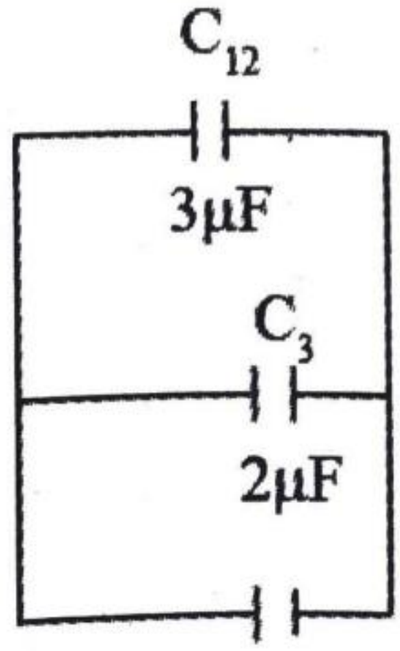
$$\frac{1}{C_{12}}=\frac{1}{C_1}+\frac{1}{C_2}=\frac{1}{4}+\frac{1}{12}=\frac{3+1}{12}=\frac{4}{12}\Rightarrow C_{12}=\frac{12}{4}=3\mu F$$

Now, the circuit modified with C_{12} and C_3 are in parallel.



$C_T = C_{12} + C_3 = 2 + 3 = 5\mu F$

b) $Q_{12} = (V)(C_{12}) = (12)(3) = 36\mu C =$

$Q_1 = Q_2$

$Q_3 = (V)(C_3) = (12)(2) = 24\mu C$

Now, $U_1 = \frac{1}{2}\frac{Q_1^2}{C_1} = \frac{1}{2}\frac{\left(36\times10^{-6}\right)}{4\times10^{-6}} = 162\mu J$

$$U_2 = \frac{1}{2}\frac{Q_2^2}{C_2} = \frac{1}{2}\frac{\left(36\times10^{-6}\right)^2}{12\times10^{-6}} = 162\mu J$$

$$U_3 = \frac{1}{2}\frac{Q_3^2}{C_3} = \frac{1}{2}\frac{\left(24\times10^{-6}\right)^2}{2\times10^{-6}} = 144\mu J$$

c) $U = \frac{1}{2}CV^2 = \frac{1}{2}\left(5\times10^{-6}\right)(12)^2 = 360\mu J$

Or $U = U_1 + U_2 + U_3 = 162 + 54 + 144 = 360\mu J$

General Secondary Education Certificate Exam

69. A boy observed that a pen rubbed on his dry hair attracted a nearby neutral piece of paper. What is the cause of the attraction?

A. Magnetic force
B. Nucleus force
C. Gravitational force
D. Electrostatic force

Answer: D

70. The electric field strength at a point is defined by the force on?

A. A source charge
B. A test charge
C. An electron
D. A proton

Answer: B

71. Suppose that the electric potential outside a living cell is higher than that of inside the cell by 0.07V. How much work is done by the electric force when a sodium ion (charge = 1.6×10^{-19}C) moves from the outside to the inside?

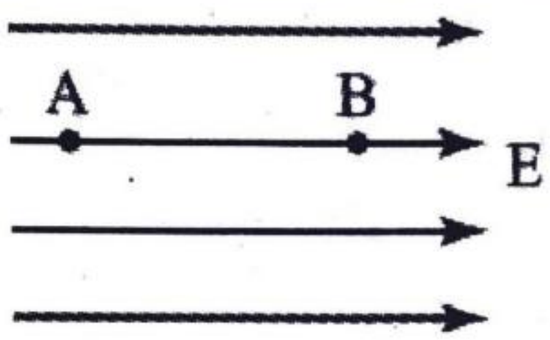
A. 7×10^{-17}J
B. 7.6×10^{-19}J
C. 1.1×10^{-20}J
D. 1.6×10^{-19}J

Solution: $V = \frac{W}{Q} \Rightarrow W = VQ = (0.07)(1.6 \times 10^{-19})$

$V = 1.1 \times 10^{20}$J

Answer: C

72. A free electron and a free proton are moved from A to B in a uniform external electric field E shown below



Which of the following is always true about the magnitude of their electrical potential energies?

A. increase for both electron and proton
B. decrease for both electron and proton
C. increase for electron and decrease for proton
D. increase for proton and decrease for electron

Explanation: To displace the electron from point A to B work is done on it, which is then stored as potential energy

Answer: C

73. Which one of the following statements is in disagreement with coulomb's law

A. The force between two charged bodies is proportional to the square of the distance between the two bodies.
B. The force between two charged bodies is proportional to the magnitude of the charge on the bodies
C. Bodies of the same charges exert repulsive force on each other
D. Bodies of different charges exert attractive force on each other.

Explanation: mathematically, coulomb's law, $F = \frac{KQ_1Q_2}{r^2}$

Answer: A

74. Which one of the following is a false statement?

A. Both gravitation and electric forces depends on the inverse of the square of the distance
B. Electrostatic force applies for two moving point charges
C. Electrostatic force is much greatest than the gravitational force
D. Charges are not created nor destroyed.

Explanation: $F_e = \frac{KQ_1Q_2}{r^2}$ and $F_g = \frac{Gm_1m_2}{r^2}$

Answer: B

75. Two electrons are 10^{-8}m apart. What is the magnitude of the force the two electrons exert on each other? $\left(\text{take } q_1 = q_2 = 1.6\times10^{-19}\text{C}, K = \frac{1}{4\pi\in_o} = 9\times10^9\right)$

A. 2.3×10^{-12}N
B. 1.4×10^{-17}N
C. 1.8×10^{-37}N
D. 0.4N

Solution:

$$F = \frac{KQ_1Q_2}{r^2} = \frac{\left(9\times10^9\right)\left(1.6\times10^{-19}\right)\left(1.6\times10^{-19}\right)}{\left(10^{-8}\right)^2} = 2.3\times10^{-12}\text{N}$$

Answer: A

76. What is the electric field strength at a distance of 20m from a point charge of 8×10^{-10}C?

A. 1.8×10^{-2}N/c
B. 0.72N/c
C. 144N/c
D. 3.5N/c

Solution:

$$E = K\frac{Q}{r^2} = \left(9\times10^9\right)\frac{\left(8\times10^{-10}\right)}{\left(20\right)^2} = 1.8\times10^{-2}\text{N/c}$$

Answer: A

77. In an experiment, a positively charged glass rod is used to charge a metal sphere by conduction. What type of charge is found on the sphere?

A. Positive
B. Negative
C. Positive or negative
D. The same if it were charged by induction

Explanation: Charging by conduction results in bodies of the same charge.

Answer: A

78. If charge Q_1 and attracts charge Q_2 and charge Q_2 repels charge Q_3. Which of the following is correct about their signs?

	Q_1	Q_2	Q_3
A.	−	+	−
B.	+	−	+
C.	−	+	+
D.	+	+	−

Answer: C

79. A charge q is located between Q_1 and Q_2 as shown q is closer to Q_1 than Q_2. If the net force on q is zero.
What would you say about Q_1 and Q_2?

A. $Q_1$and Q_2 are equal in magnitude and the same in sign
B. $Q_1$and Q_2 are equal in magnitude and opposite in sign
C. $Q_1$and Q_2 are unequal in magnitude and opposite in sign
D. $Q_1$and Q_2 are unequal in magnitude and the same in sign

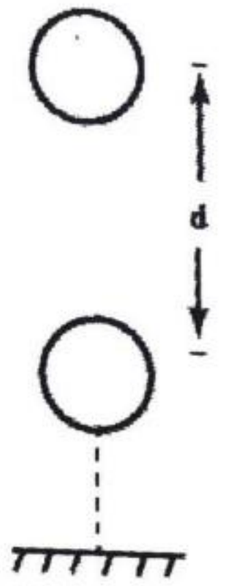
Explanation: The force on q due to $Q_1$and Q_2should be equal and opposite. For this to happen, $Q_1$and Q_2should have the same sign.

Answer: D

80. The charge on each of the pith balls shown is 20nc. What should the distance between them be for the electrostatic force to just balance the weight of the upper pith ball.

$\left(K = 9\times10^{9}\,\frac{Nm^2}{c^2},\ g = 10\ m/s^2\right)$ $m = 0.1g$

A. 1.8 mm
B. 1.8cm
C. 6mm
D. 6cm



Solution: $F = W \Rightarrow \frac{KQ^2}{d^2} = mg$

$$\Rightarrow d = \sqrt{\frac{KQ^2}{mg}}$$

$$d = \sqrt{\frac{\left(9\times10^{9}\right)\left(20\times10^{-19}\right)^2}{\left(10^{-4}\right)(10)}} = 0.06m = 6cm$$

Answer: D

81. Consider four heavily charged plastic rods. Rods 1 and 2 attract each other whereas rods 3 and 4 repel each other, which one of the following statement is correct?

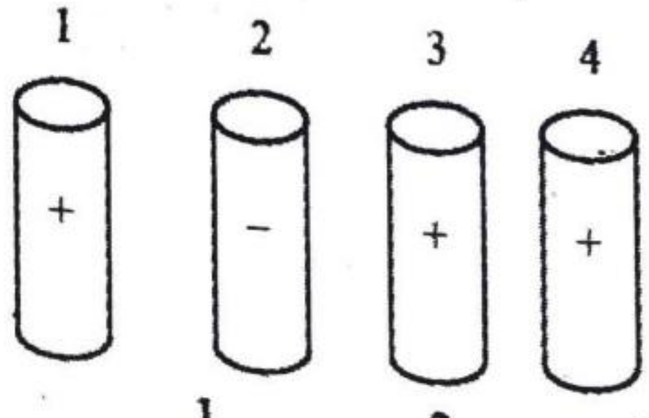
A. If rod1 repeals rod 4, rod 2 attracts rod 3
B. If rod1 repeals rod 3, rod 2 also repels rod 4
C. If rod1 attracts rod 3, rod 2 repels rod 4
D. If rod1 repeals rod 4, rod 2 repels rod 3

Explanation:

Case 1

rod 1 and 3 repel

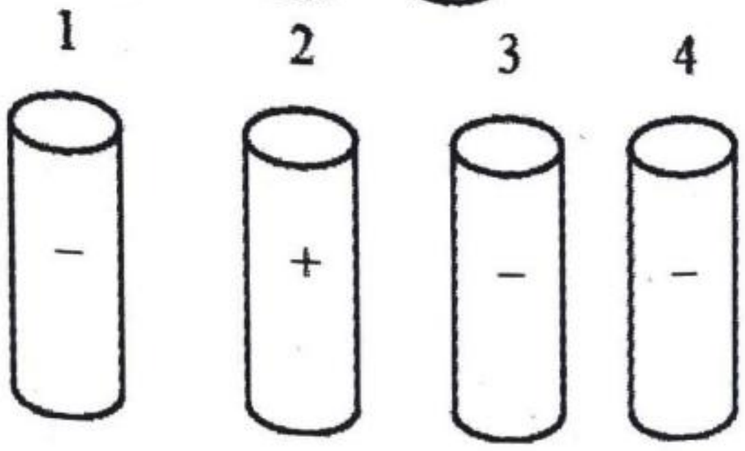
rod 2 and 4 attract



Case 2

rod 1 and rod 3 repel

rod 2 and rod 4 attract



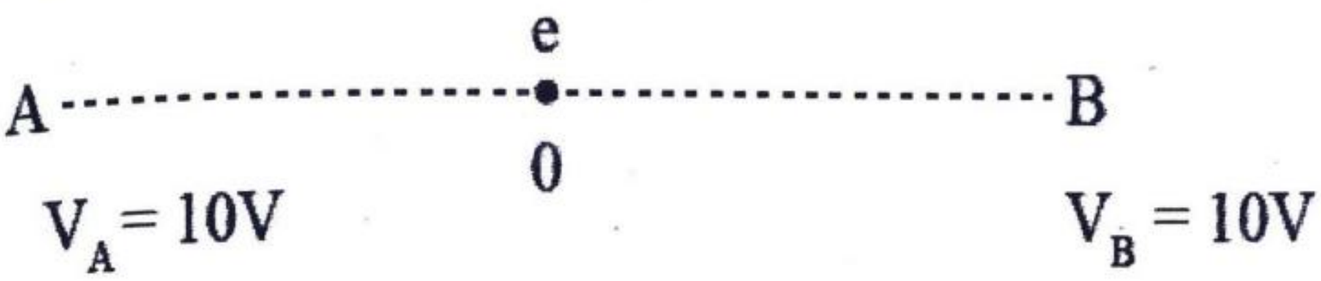
Thus, if rod 1 repel rod 4, rod

1

racts rod 3.

Answer: A

The electric potential at point A is V_A is = 10J/C at point B is V_B = 30V. which one of the following is the following is true about an electron placed at midpoint between points A and B



A. The electron will stay at rest at point O
B. The electron will start moving towards point B
C. The electron will start moving towards point A
D. The distance between A and B must be known at decide the direction of the electron motion.

Explanation: The electron moves from higher potential (B) to lower potential (A).

Answer: C

The electric potential created by a point charge at a distance r from the charge is V. What is the electric potential at a distance of 4r from the charge?

A. 4V
B. $\frac{1}{2}V$
C. $\frac{1}{4}V$
C. $\frac{1}{16}V$

Solution: $V_1 = K\frac{Q}{r_1} = V$ .....(*)

$V_2 = K\frac{Q}{r_2} = K\frac{Q}{4r_1}$ .....(**)

taking ratio, $\frac{V_2}{V_1} = \frac{KQ/4r_1}{KQ/r_1} = \frac{1}{2} \Rightarrow V_2 = \frac{1}{4}V_1 = \frac{1}{4}V$

Answer: C

84. What will happen to the electric potential at a point in the field if the distance from the charge doubles?

A. Increase two times
B. Decrease by half
C. Increase four times
D. Decrease to quarter

Solution: $V_1 = K\frac{Q}{r_1}$ .....(*)

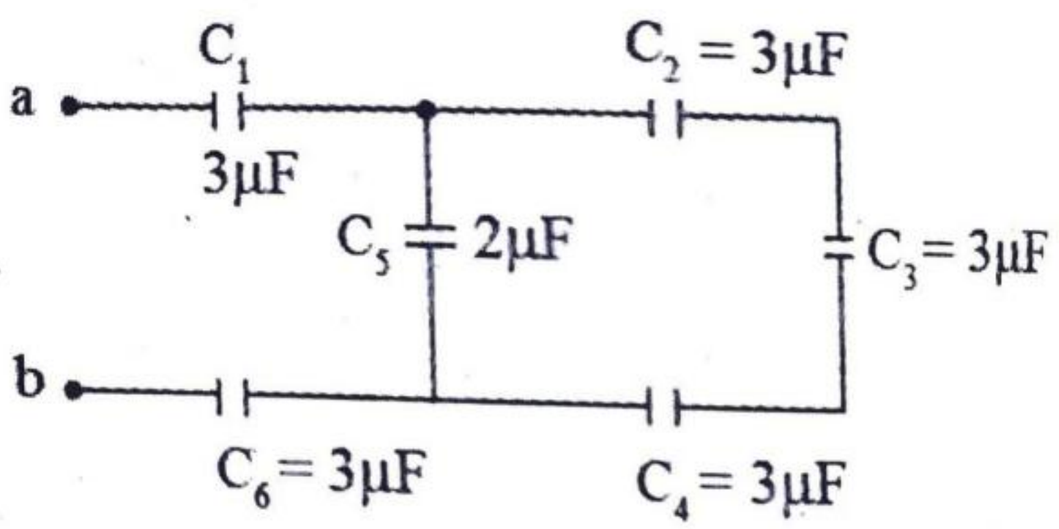
$$V_2 = K\frac{Q}{r_2} = K\frac{Q}{2r_1} \text{.....(**)}$$

taking ratio, $\frac{V_2}{V_1} = \frac{KQ/2r_1}{KQ/r_1} = \frac{1}{2} \Rightarrow V_2 = \frac{1}{2}V_1$

Answer: B

85. Six capacitors are connected as shown. The equivalent capacitance of the network is

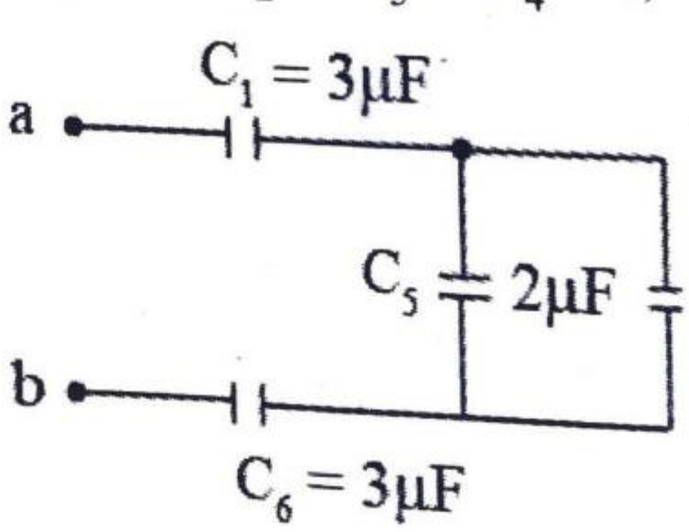
A. 17μF
B. 15μF
C. 0.5μF
D. 1μF



Solution:

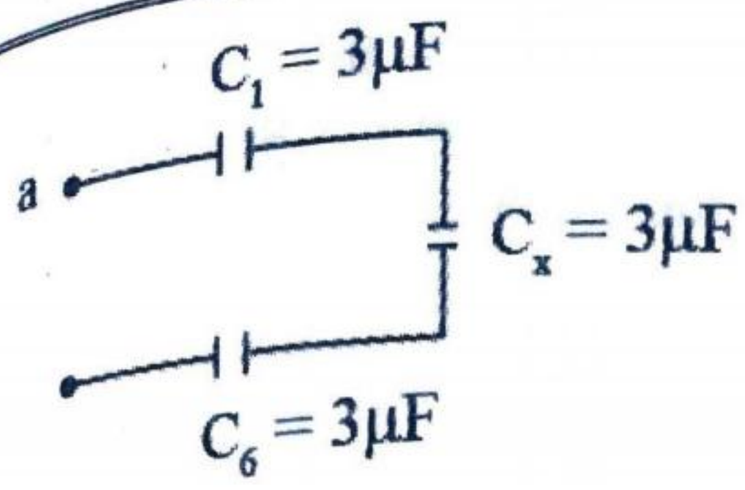
C_2,C_3, and C_4 are in series,

$$\frac{1}{C_{234}} = \frac{1}{C_2} + \frac{1}{C_3} + \frac{1}{C_4} = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = \frac{3}{3} \Rightarrow C_{234} = 1\mu F$$



$C_{234} = 1\mu F$, C_5 and C_{234} are in parallel

$C_x = C_5 + C_{234} = 1 + 2 = 3\mu F$,



$$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_6} + \frac{1}{C_x} = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = \frac{3}{3} \Rightarrow C_T = 1\mu F$$

Answer: D

6. The distance between plates of a parallels plate capacitor is 1cm. if the electric field strength is 10N/C. What is the potential difference between the plates of the capacitor.

A. 1J/C
B. 10J/C
C. 0.1J/C
D. 0.01J/C

Solution: $V = Ed = (10)(10^{-2}) = 0.1J/C.$

Answer: C

7. What is the capacitance of a capacitor?

A. It is ability to store charge per unit potential difference
B. It is the measure of its ability to resist current flow
C. It is the ratio of volume of capacitor to the charge
D. It is the ratio of the potential difference to the current

Explanation: Capacitance $= \dfrac{\text{charge stored}}{\text{potential differnece}}$

Answer: A

8. What will happen if a dielectric is inserted between the plates of a capacitors?

A. The charge stored increases
B. The capacitor discharges
C. The capacitance increases
D. The electric filed decreases.

Answer: B

9. Using two equal conducting sheets, a student made an air gap capacitor. What should he do to increase the capacitance of the capacitor?

A. Charge the capacitor to its maximum capacity
B. Increase the potential difference between the sheets
C. Increase the gap between the sheets
D. Insert sheet of paper between the sheets.

Answer: D

90. When you open an electrolyte capacitor, you usually find a wet sheet of paper. What is the reason for the paper to be wet?

A. To conduct charge from positive to negative plate
B. To lubricate the plate for charges to move easily
C. To increase the capacitance of the capacitor
D. Due to condensation of charges between plates

Answer: C

91. Two positive point charges repel each other with force of 0.16N when their separation is 1.5m. What force do they exert on each other when their separation is 1m?

A. 0.54N
B. 0.64N
C. 0.24N
D. 0.36N

Solution: $F_1 = K\frac{Q_1Q_2}{r_1^2} \Rightarrow 0.16 = K\frac{Q_1Q_2}{(1.5)^2} = KQ_1Q_2 = 0.36$

$$F_2 = K\frac{Q_1Q_2}{r_2^2} = \frac{0.36}{(1)^2} = 0.36N$$

Answer: D

92. The area of each of the plates of a parallel plate capacitor is $20cm^2$. If the plates are 0.5cm apart, what is the capacitance of the capacitor? $\left(\in_o = 8.85\times10^{-12}\right)$

A. 22.12PE
B. 11.06PF
C. 3.54PF
D. 7.07PF

Solution: $C = \frac{\in_o A}{d} = \frac{\left(8.85\times10^{-12}\right)\left(20\times10^{-4}\right)}{\left(0.5\times10^{-2}\right)} = 3.54PF$

Answer: C

93. Which one of the following is correct

A. The lines of force are longer where the electric field is strong
B. The electric fields are stronger where the field, lines are thicker
C. The electric field gets stronger where the fields lines bend
D. The line of force are denser where the electric field is stronger

Answer: D

94. A 3.2×10^{-14}kg oil drop is in an electric field of $E = 2 \times 10^{5}$N/c. if the oil drop is in equilibrium when the upward electric force (F_e) equals the gravitational force (F_g) on the drop what was the charge of the oil drop when it was in equilibrium?

A. 1.6×10^{-19}C
B. -1.6×10^{-19}C
C. 1.6×10^{-18}C
D. -1.6×10^{-18}C

Solution:

$$F_e = F_g = Eq = mg \Rightarrow q = \frac{mg}{E} = \frac{\left(3.2 \times 10^{-14}\right)(10)}{\left(2 \times 10^{5}\right)}$$

$$q = -1.6 \times 10^{-18}\text{C}$$

Answer: C

95. Two particles, with charges q_1 and q_2 are on the x-axis with q_1 at $x = a$ and q_2 at $x = -2a$. For the net force on a third charged particle at $x = 0$ to be zero q_1 and q_2 must be related by

A. $q_2 = 4q_1$
B. $q_2 = -2q_1$
C. $q_2 = -4q_1$
D. $q_2 = 2q_1$

Solution:

$$F_1 = F_2 = \frac{Kqq_1}{a^2} = \frac{Kqq_2}{(-2a)^2} \Rightarrow q_2 = 4q_1$$

Answer: A

96. A 2μF and a 1μF capacitors are connected in parallel and potential difference is applied across the combination the 2μF capacitors has

A. half the charge of the 1μF capacitors
B. twice the potential difference of the 1μF capacitors
C. twice the stored energy of the 1μF capacitors

D. half the stored energy of the 1μF capacitors

Solution: $\text{Energy} = \frac{1}{2}CV^2 \Rightarrow E_1 = \frac{1}{2}(1)(V^2) = \frac{V^2}{2}$

$$E_2 = \frac{1}{2}(2)V^2 = V^2 \Rightarrow E_2 = 2E_1$$

Answer: C

97. If a 4μF capacitor has a potential difference of 1000V, how much energy is stored in the capacitor

A. 1/2J B. 2J C. 4J D. 8J

Solution: $\text{Energy} = \frac{1}{2}CV^2 \Rightarrow = \frac{1}{2}\left(4\times10^{-6}\right)\left(1000\right)^2 = 2\text{J}$

Answer: B

98. The electric potential at point A is 10V and it is 50V at point B. how much work is done by an external force in moving a proton from point A to point B?

A. 64J
B. 6.4×10^{-18}J
C. 50J
D. 40J

Solution: $W = qV = qV_{AB} = \left(1.6\times10^{-19}\right)\left(50-10\right)^2 = 6.4\times10^{-18}\text{J}$

Answer: B

99. Which one of the following is true about absolute potential and potential difference?

A. Both quantities are arbitrary and can be taken as zero.
B. Potential difference is a location dependent quantity
C. Both quantities are measured in the same unit
D. Absolute potential is energy while potential difference is work per unit time

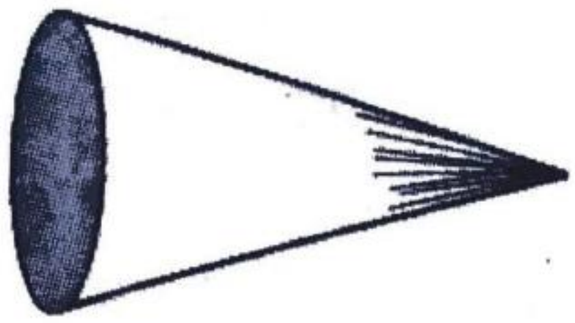
Answer: C

100. The cone shaped solid conductor is negatively charged. Which of the following statement is true?

A. The charge density is greatest at the tip
B. The charge is uniformly distributed on surface

C. The electric field is weak at the tip
D. There is an electric field inside the conductor

Answer: A

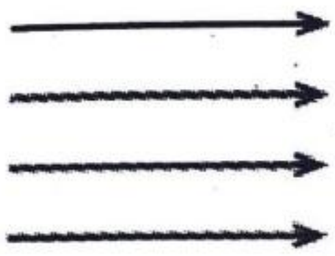


101. Tangent line to an equipotential surface and electric field of a charged body at any point on the surface must be?

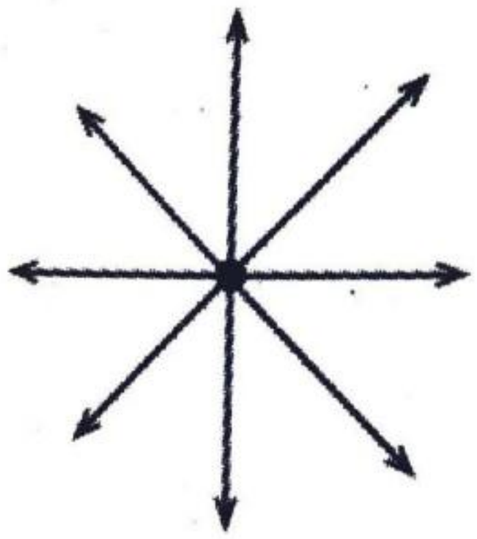
A. Parallel
B. Perpendicular
C. Opposite in direction
D. Same in direction

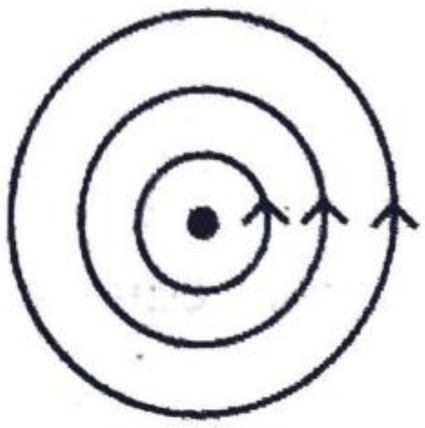
Answer: B

102. If the magnetic field in a region is uniform, which set of magnetic field lines can describe this?

A. 

C.

B. 

D. 

Answer: A

Unit Ten: Conceptual Questions

1. Why do charged objects soon lose their charge on a damp day?
2. All atoms that make up matter contain tiny charges, how is it that most objects we meet appear to be uncharged.
3. Show how the charges in the uncharged object will distribute themselves in the presence of the positively charged rod.

Charged uncharged

4. Why will a plastic comb that has been rubbed attract small pieces of paper?
5. Why do plastic objects frequently become covered with dust?
6. Explain the effect of inserting a dielectric in the gap between the plates of a parallel plate capacitor
7. Why is the inside of a car a safe place to shelter in a storm?
8. Two capacitors of 5μF and 10μF could be arranged either in series or in parallel which arrangement gives the highest capacitance?
9. If two capacitors A and B, have the same area and dielectric constant, but the distance between the plates of A is twice the distance between the plates of B, which has higher capacitance?
10. Why is it possible to discharge a positive charge by connecting it to the ground since positive charges do not move.
11. Which one will experience increasing electric potential as it moves in electric field. Proton or electron
12. How much work is required to move a charge from one location to another on equipotential line.

Unit Ten: Problems

Coulomb's law

13. Two point charges, 3C and 2C are placed 5m apart. Calculate the force between them.
14. How far apart should two electrons be if the electric force each exerts on the other equals the weight of an electron

Electric field

15. How big a force will a charge of 2C experience if placed in an electric field of 18N/c?
16. What is the electric field strength 0.4m from a 7×10^{-5}C
17. Two charges 4×10^{-6}C and 8×10^{-6}C are 2m apart.
    a) What is the electric field midway between them
    b) At what distance the field is zero

18. A body whose mass is 10^{-6}kg carries a charge of 10^{-6}C. What is the magnitude of the electric field to balance the gravitational force on the body?

Electric potential energy and potential difference

19. Calculate the potential difference between two points if 2J of work is done to move a 4×10^{-3}C charge.

20. The potential difference between two parallel metal plates is 6V and the separation of the plate sis 2mm. calculate the electric field between the plates?

21. A proton is released in an electric field of 3000v/m that is along the +ve x axis. What will be its speed as it passes x = 50cm?

22. An electric travels at a speed of 5×10^{6}m/s between two plates 100mm long and 20mm apart to which a period of 30V is applied.
    a) What size force will the electron experience
    b) What acceleration will this given to the electron
    c) If the positive plate is the lower one, in which direction will the electron accelerate.
    d) How long will it take it pass through the plates.

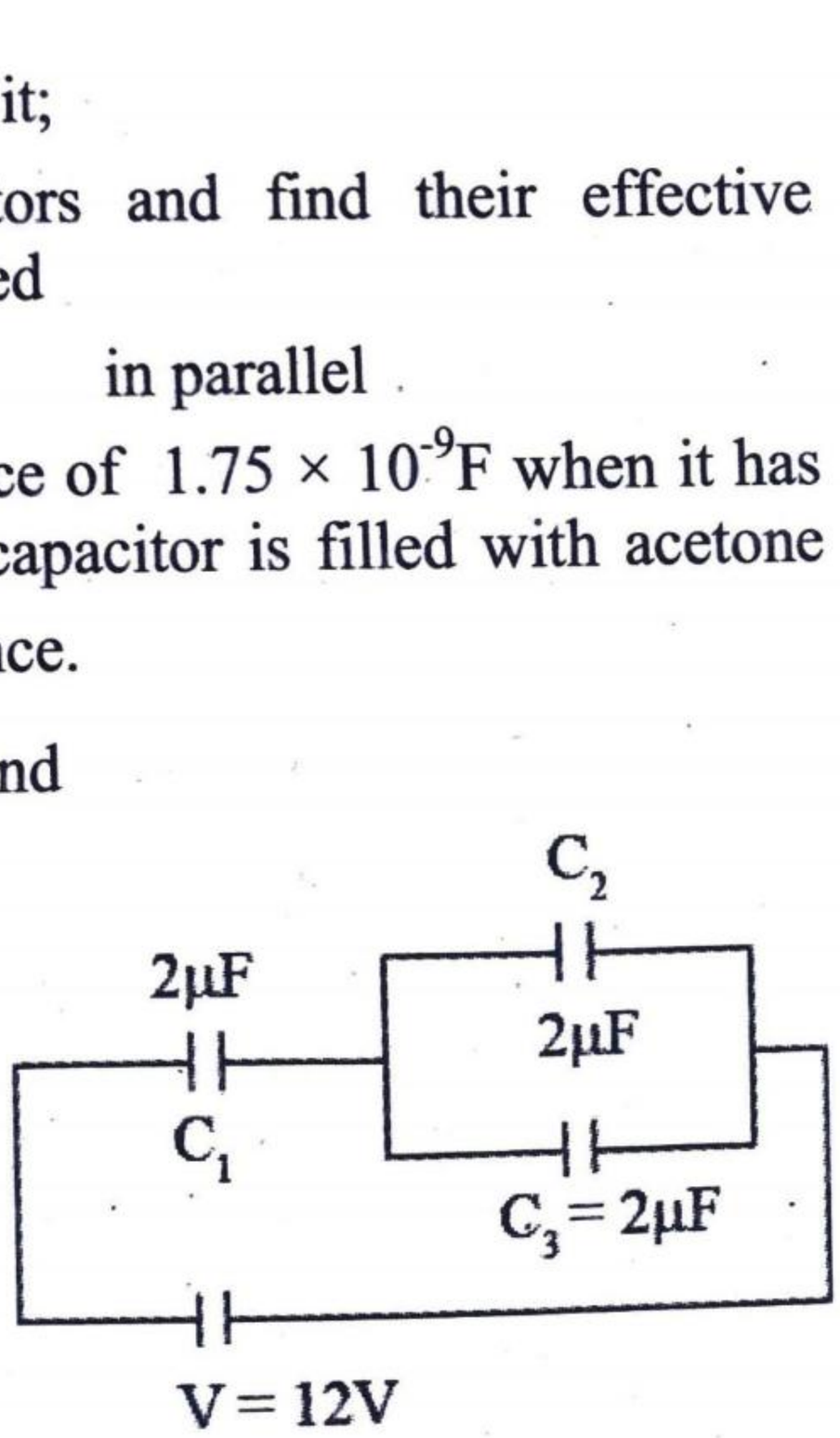
Capacitance and energy stored in a capacitor

23. A 2μF capacitor is charged form a 6V batter
    a) How much charge will it hold;
    b) How much energy is stored on it;

24. Take a 2μF and a 5μF capacitors and find their effective capacitance when they are connected
    a) In series b) in parallel

25. A certain capacitor has a capacitance of 1.75×10^{-9}F when it has air between its plates. When the capacitor is filled with acetone ($\epsilon_o = 27$) what will be its capacitance.

26. For the capacitor network shown, find
    a) The equivalent capacitance
    b) The potential drop in each capacitor
    c) The charge stored in each capacitor



d) The energy stored in each capacitor
e) The total energy stored in the system

Unit Ten: Explanation to Questions

1. Because of the leak off charge via the atmosphere
2. Because there are equal number of protons and electrons in an atom and hence it is neutral under normal condition.
3. There exists polarization.

4. Because of the induced charge, the comb attract charges of opposite sign towards itself like the above case.
5. Because static charges are stored on them since they are insulators and hence can easily attract small particles due to induction (at a distance).
6. It reduces neutralization of plates of a capacitor and hence increase its capacitance
7. It is safe because the metal shell deflects charges
8. When they are connected in parallel ($C_T = 15\mu F$)
9. Capacitor B. because capacitance $C = \epsilon \frac{A}{d}$, it decrease as distance increases.
10. Because, electrons come from the earth and neutralize it. Protons do not move readily!
11. The electron, since it is moving opposite to the field
12. Zero. Because, $W = \Delta VQ$, here, $\Delta V = 0$

Unit Ten: Solution to problems

13. $$F = \frac{KQ_1Q_2}{r^2} = \frac{(9\times10^9)(3)(2)}{(5)^2} = 2.16\times10^9 m$$

14. $$F = \frac{KQ_1Q_2}{r^2} \Rightarrow mg = \frac{Ke^2}{r^2},\ r^2\frac{Ke^2}{mg} = r\sqrt{\frac{Ke^2}{mg}}$$

$$r = e\sqrt{\frac{K}{mg}} = 1.6\times10^{-19}\sqrt{\frac{9\times10^{9}}{\left(9.1\times10^{-31}\right)(10)}} = 0.5\times10^{1} = 5\text{m}$$

15. $E = {}^{F}\!/_{Q} \Rightarrow F = EQ = (18)(2) = 36\text{N}$

16. $E = \dfrac{KQ}{r^{2}} = \dfrac{\left(9\times10^{9}\right)\left(7\times10^{-5}\right)}{(0.4)^{2}} = 3.9\times10^{-3}\text{N/c}$

17. a) $E_{net} = E_1 - E_2 = \dfrac{KQ_1}{r_1^{\,2}} - \dfrac{KQ_2}{r_2^{\,2}}$

$$E_{net} = \frac{\left(9\times10^{9}\right)\left(4\times10^{-6}\right)}{(0.4)^{2}} - \frac{\left(9\times10^{9}\right)\left(8\times10^{-6}\right)}{(1)^{2}} = 3.6\times10^{4}\text{N/c}$$

t

owards the 4μC charge.

b) $\dfrac{KQ_1}{x^{2}} = \dfrac{KQ_2}{(2-x)^{2}} = \dfrac{1}{x^{2}} = \dfrac{2}{(2-x)^{2}}$

$$\Rightarrow 2x^{2} = (2-x)^{2} = 4 - 4x + x^{2}$$

$\Rightarrow x^{2} + 4x - 4 = 0$. Solving for x, x ≈0.8 from the 4μC

18. $EQ = mg = E = {}^{mg}\!/_{Q} = \dfrac{\left(10^{-6}\right)(10)}{10^{-6}} = 10\text{N/c}$

19. $V = {}^{W}\!/_{Q} = {}^{2}\!/_{\left(4\times10^{-3}\right)} = 0.5\times10^{3}\text{V}$

20. $V = Ed = E = {}^{V}\!/_{d} = {}^{6}\!/_{\left(2\times10^{-3}\right)} = 3\times10^{3}\text{V/m}$

21. $EQd = \dfrac{1}{2}mv^{2} = V = \sqrt{\dfrac{2EQd}{m}} = \sqrt{\dfrac{2VQ}{m}}$, $V = Ed$

$$V = \sqrt{\frac{(2)(3000)(1.6\times10^{-19})(0.5)}{(1.67\times10^{-27})}} = 0.4 \times 10^{6}\,\text{m/s}$$

22. a) $F = EQ = \left(\frac{V}{d}\right)Q = \left(\frac{30}{0.1}\right)(1.6\times10^{-19}) = 2.4\times10^{-16}\text{N}$

b) $EQ = ma \Rightarrow a = \frac{EQ}{m} = \frac{2.4\times10^{-16}}{9.1\times10^{-31}} = 0.24\times10^{15}\,\text{m/s}^2$

c) downward

d) $S = Vt \Rightarrow t = S/V = \frac{0.1}{5\times10^{6}} = 2\times10^{-4}\text{sec}$

23. a) $Q = CV = (2\times10^{-6})(6) = 12\mu\text{C}$

b) $U = \frac{1}{2}CV^2 = 36\mu\text{J}$

24. a) $\frac{1}{C_T} = \frac{1}{2} + \frac{1}{5} \Rightarrow \frac{1}{C_T} = \frac{7}{10} \Rightarrow C_T = \frac{10}{7}\mu\text{F} = 1.4\mu\text{F}$

b) $C_T = 2 + 5 = 7\mu\text{F}$

25. $C_1 = \frac{\epsilon_o A}{d}$ and $C_2 = \frac{\epsilon_r \epsilon_o A}{d} = \epsilon_r (C_1) \Rightarrow (27)(1.75\times10^{-9}\text{F})$

$C_2 = 4.7\times10^{-8}\text{F}$

26. a) $C_T = \frac{4}{3}\,\mu\text{F}$

b) $V_1 = 8\text{V}, V_2 = V_3 = 12 - 8 = 4\text{V}$

c) $Q_1 = 16\mu\text{C}, Q_2 = 8\mu\text{C}, Q_3 = 8\mu\text{C}$

d) $U_1 = 64\mu\text{J}, U_2 = 16\mu\text{J}, U_3 = 16\mu\text{J}$

e) $U = 96\mu\text{J}$

Unit Eleven

Current Electricity

Light bulb, stove, heater, frige, Tv, computer, phone, machines big or small, all depend on electricity to work. To realize just how big an impact electricity has on our daily lives, just think about what happens when there is a power failure for a day or two. Nowadays, life without electricity seems almost impossible.

11.1 Electric Current.

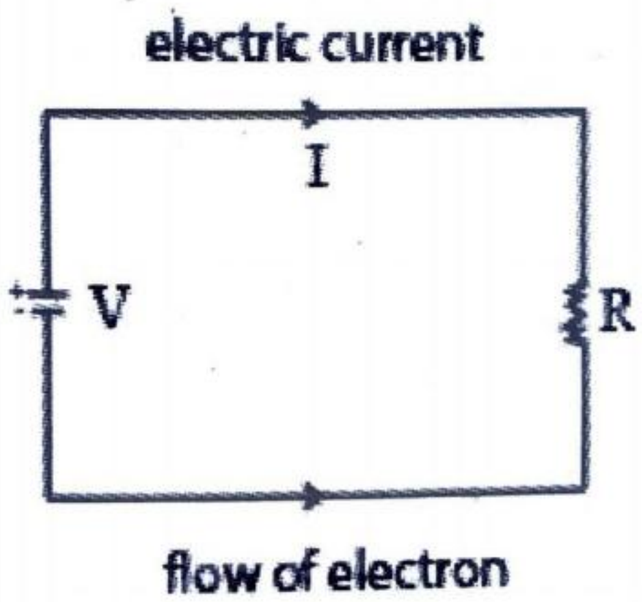
The flow of charged particles (electrons) though a wire just like the flow of water molecules through a pipe is known as *electric current*.

Electric current: Is the flow of charge per unit time across a given cross sectional area.

Mathematically, $\text{Current} = \dfrac{\text{charge}}{\text{time}} \Rightarrow \boxed{I = \dfrac{Q}{t}}$

The SI unit of electric current is Ampere (A).

Thus, one $\text{Ampere} = \dfrac{\text{one coulomb}}{\text{one second}} \Rightarrow \boxed{1A = \dfrac{1C}{S}}$



☞ **Remember:** in actual sense, electrons are the one which move in the wire, but for practical purpose, we conventionally assume positive charges to form what we call electric current. So, when we say current,we mean the flow of positive charges which is always opposite in direction to the flow of electrons.

Conceptual Example

1. If electrons keep going though a wire, will the wire get thinner and thinner as electrons move from it. Explain?

Explanation: No. the wire thickness will not be affected because, the electrons circulate through it and do not 'escape' from it.

Illustrative Examples

2. How long will it take for a charge of 600C to pass through a conductor carrying a current of 3A?

Solution: $I = \frac{Q}{t} \Rightarrow t = \frac{Q}{I} = \frac{600}{3} = 200\text{sec}$

3. If a current of 15A flows through a wire for 1hr. Calculate.
   a) The amount of charge
   b) The number of electrons that cross the wire

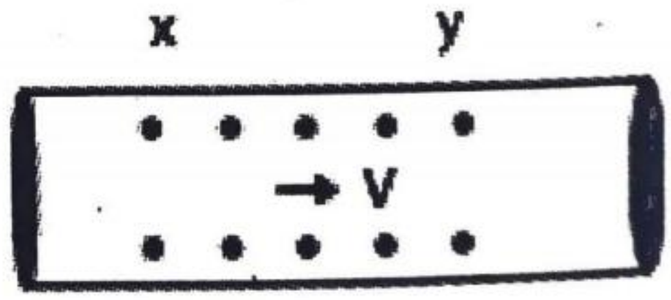
Solution:

a) $I = \frac{Q}{t} \Rightarrow Q = It = (15)(1)(60)(60) = 54000\text{C}$

b) $Q = ne \Rightarrow n = \frac{Q}{e} = \frac{54000}{1.6\times10^{-19}} = 3.375\times10^{23}$ electrons

Drift Speed

If (n) free electrons per volume of wire, each electron carrying charge (e) and is moving at average speed (v) takes a time (t) to move from point x to y through a wire of area (A), then



→ The number of electrons between x and y is, nAvt

→ The total charge passing point y in time t is, Q = nAevt finally, the current is $I = \frac{Q}{t} = \frac{nAevt}{t}$

$\Rightarrow \boxed{I = nAev}$

Drift speed: Is the average speed that an electron attains in an electric field.

Conceptual Example

4. If the cross sectional area of a wire gets thinner and thinner along its length, what will happen to the drift speed of the electrons

Explanation: from the conservation of charge, the current in the wire remains constant hence, from $I = nAev$ $v = I/nAe$, thus as area decreases, speed increases.

Illustrative Examples

5. A copper wire of cross-sectional area $3\times 10^{-6}m^2$ carries a current of 10A and 8.48×10^{28} electrons per meter cube.Find the drift speed of the electrons in the wire.

Solution:

$$I = nAev \Rightarrow v = \frac{I}{nAe} = \frac{10}{\left(8.48\times10^{28}\right)\left(1.6\times10^{-19}\right)\left(3\times10^{-6}\right)}$$

$$v = 2.46\times10^{-6}\,m/s$$

6. A current of 3A is flowing through a 5mm diameter wire. If the drift speed of the electrons in the wire is 2×10^{-6}m/s. Calculate the number of electrons passing a point in a second.

Solution: $I = nAev \Rightarrow n = \frac{I}{Aev}$

but, $e = 1.6\times10^{-19}C$ and $A = \pi\frac{d^2}{4} = (3.14)\frac{\left(5\times10^{-3}\right)}{4} = 2\times10^{-5}$

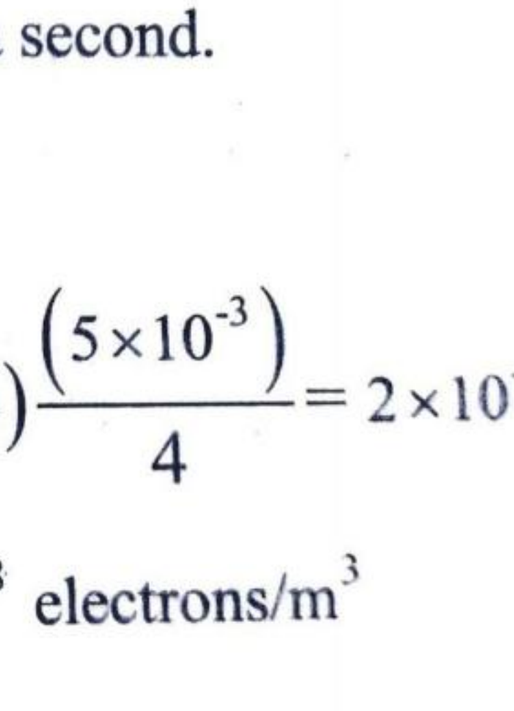
$$\Rightarrow n = \frac{3}{\left(2\times10^{-5}\right)\left(1.6\times10^{-19}\right)\left(2\times10^{-5}\right)} = 5\times10^{28}\ \text{electrons/m}^3$$

Source of electricity

Electric energy can be obtained from *chemicals* and from *heat*.

Electric energy from chemicals

If plates of zinc and copper are immersed in dilute sulphuric acid (ammonium chloride) and their terminals joined by a wire, an electric current flows. The source of the electrical



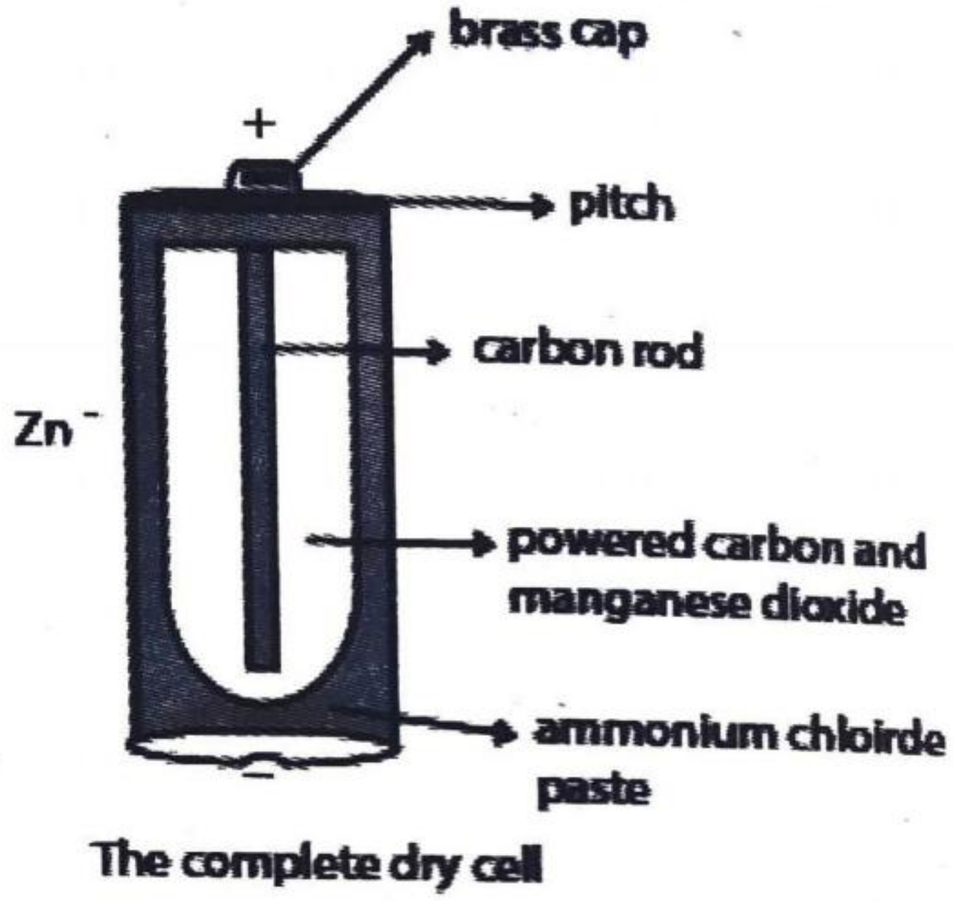
The simple voltaic cell

energy is the chemical reaction. This reaction may be represented by the equation

$$Zn + H_2SO_4 \rightarrow H_2 + ZnSO_4 + \text{energy.}$$

The chemical action maintains a p.d of about 1V between the plates.

The dry cell, which is used extensively in torches, radios and calculators do have a negative zinc plate forming the external part of the cell, and a positive carbon rod that runs down the center with a brass cap on the top to prevent the carbon from damage. The positive plate is both the carbon rod and the thick layer of the complete dry cell powdered carbon that surrounds it.



The complete dry cell

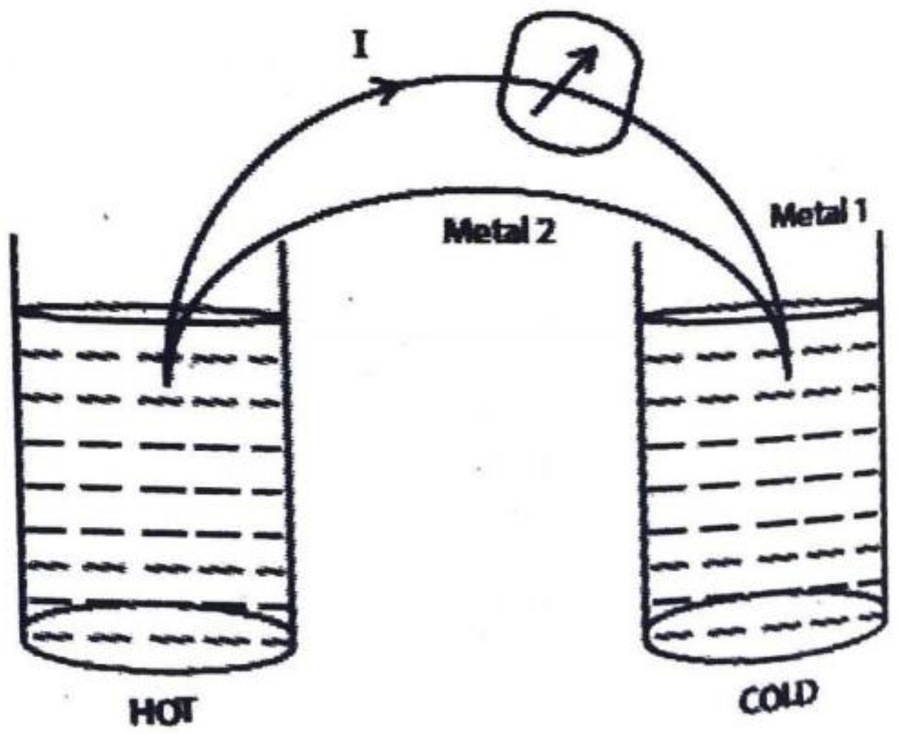
The powdered carbon is mixed with manganese dioxide so as to convert hydrogen gas formed on the carbon plate to water. And a paste of ammonium chloride is used to keep the manganese dioxide in place.

In the dry cell, a tightly packed manganese dioxide and carbon conducts well and the cell has a very low internal resistance. The p.d between its terminals is about 1.5V.

Electric energy form heat

The electricity generated from heat is known as ***thermoelectricity***.

If two wires of different materials are joined at their ends and one end is maintained hotter than the other, a small voltage difference occurs between the two junctions and this effect we call the *seebeck* effect. The *seebeck* effect is used in **thermocouple thermometer** to measure temperature.



thermocuople thermometer.

Illustrative Example

7. When one junction of a thermocouple thermometer is immersed in ice and the other in boiling water, a current of 15mA is recorded. But, when one junction is immersed in ice and the other in a liquid of unknown temperature, a current of 5mA is recorded. What is the temperature of the liquid.

Solution: In the first case, $\Delta T_1 = I_1 \Rightarrow 100 - 0 = 15 \ldots (*)$

In the second case, $\Delta T_2 = I_2 \Rightarrow 100 - 0 = 5 \ldots (**)$

taking ratio of equations (*) and (**)

$$\frac{100}{T} = \frac{15}{5} \Rightarrow T \simeq 33°C$$

Or using simply criss - crossing

$100°C = 15mA$

$\times$

$T = 15mA$, after criss cross

$\Rightarrow T = 33°C$

11.2 Ohm's law and electrical resistance

Ohm's law

For a metal wire at a constant temperature, the current that flows through it is proportional to the potential difference between its ends. This is what we call **ohm's law**.

Ohm's law: states that:

"The current that flows through a conductor at constant temperature is proportional the potential difference between its ends."

That is, voltage ~ current

$$\Rightarrow \boxed{V = IR} \leftarrow \text{Ohm's law}$$

Where: V is potential difference (voltage drop)

I is electric current, and

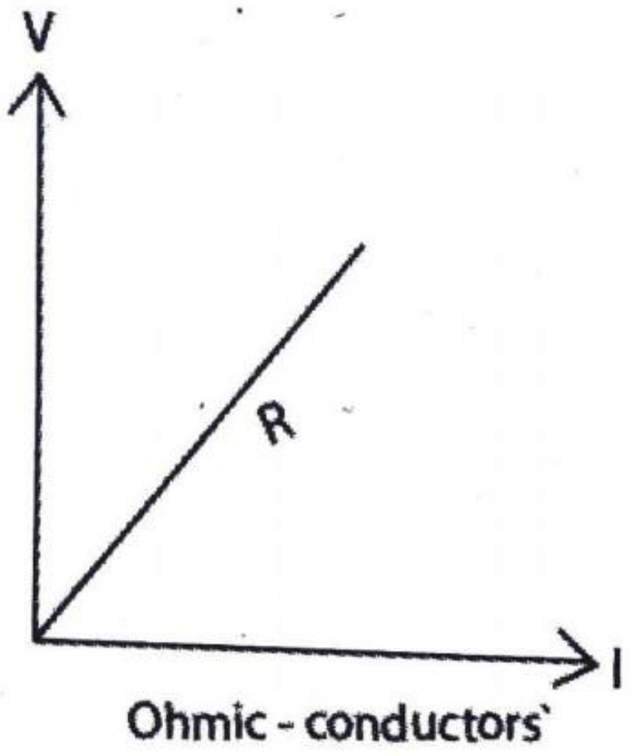
R is resistance of the conductor, its SI unit is ohm (Ω)

Resistance: Is the opposition to a flow of current in an electrical circuit.

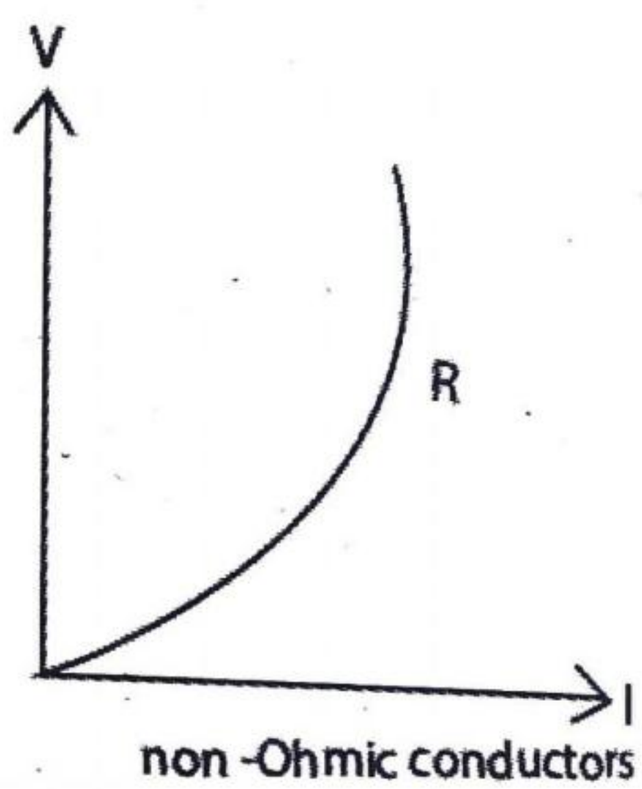
Conceptual Examples

8. Ohm's law is said to be valid usually at constant temperature. Why? Explain

Explanation:This is because, the resistance of some conductors change as their temperature changes. These type of conductors are called **non-ohmic** conductors, because they do not obey ohm's law.



Ohmic - conductors`



non -Ohmic conductors

Illustrative Examples

9. What potential difference is required to pass 8A current through a resistor of resistance 40Ω.

Solution: $V = IR = (8)(40) = 320V$

10. What is the current through a 6Ω bulb when it is operating on a 240V line?

Solution: $V = IR \Rightarrow I = \frac{V}{R} = \frac{240}{6} = 40A$

11. The potential difference across a 12Ω resistor is 3V. How long does it take for 60C of charge to pass through the resistor?

Solution: $V = IR \Rightarrow I = \frac{V}{R} = \frac{3}{12} = 0.25A$, and

$I = \frac{Q}{t} \Rightarrow t = \frac{Q}{I} = \frac{60}{0.25} = 240$ second

12. When a potential difference of 27V is applied on a resistor, a charge of 5μC passes in half hour. Calculate the resistance of the resistor.

Solution: $I = \frac{Q}{t} = \frac{5\times10^{-6}}{(30)(60)} = 0.27\times10^{-9}\,A$, and

$$V = IR \Rightarrow R = \frac{V}{I} = \frac{27}{0.27\times10^{-9}} = 10^{11}\Omega$$

Factors affecting the resistance of a conductor

Experiments show that*temperature, length, cross-sectional area* and *the material the wire made* from all affect the resistance of a conductor.

Temperature (ΔT)

→ As temperature increases, the resistance of the wire increases. Thus the increase in resistance (ΔR) due to an increase in temperature (ΔT) is given by;

$$\boxed{\Delta R = R_o \alpha \Delta T}$$

Where, α (Alpha) is a constant that varies from metal to metal,

R_o, is the resistance of the wire at 0°C

The total resistance of the wire is then,

$$R = R_o + \Delta R \Rightarrow R = R_o + R_o \alpha \Delta T$$

$$\boxed{R = R_o(1 + \alpha \Delta T)}$$

The resistance of a metal wire at a given temperature is determined by three factors:

- Its length (ℓ): as length increases, resistance increases
- Its cross-sectional area (A): as area increase, resistance decreases
- The material from which the wire is made (ρ), copper, for example is a better conductor than iron

The resistance R, of the wire can then be expressed in the form;

$$\boxed{R = \rho\frac{\ell}{A}}$$

Where, ρ(rho) is a constant called *resistivity* of the conductor and the value of which depends on the material from which the wire is made, the unit of resistivity is ohm meter (Ωm).

Conceptual Examples

13. For metals, resistance increases as the temperature of the metal is increased and vice versa. Why?

Explanation: this is because, as the temperature of the metal increases, the kinetic energy of particles of the metal increases and hence they move in random motion, as a result collisions between particles increase. Thus more collision means more resistance.

14. As length of a metal wire increases, its resistance also increase. Why?

Explanation: Because, as the wire is made longer and longer keeping the cross sectional area constant, the charges flowing in the wire faces many more collisions.

15. If the length of the wire is kept fixed and the cross sectional area is made larger, the resistance decreases. Why?

Explanation: Because when the wire is made thicker, the charges get more freedom to move through and the number of collisions decreases and so the resistance.

Illustrative Examples

16. At 0°C, the resistance of a wire is 20Ω. If the temperature coefficient of resistance is 0.0025Ω/°C. What is its resistance at 50°C?

Solution: $R = R_o + \Delta R = R_o + R_o \alpha \Delta T$

$\Rightarrow R = 20 + (20)(0.0025)(50 - 0) = 22.5\Omega$

17. A coil of wire has resistance of 45Ω at 30°C and a resistance of 60Ω at 45°C. What is the temperature coefficient of the wire?

Solution: $R = R_o + R_o \alpha \Delta T \Rightarrow R - R_o = R_o \alpha \Delta T$

$$\Rightarrow \alpha = \frac{R - R_o}{R_o \Delta T} = \frac{60 - 45}{(45)(45 - 30)} = 0.022\Omega / °C$$

18. A wire made from aluminum of resistivity $2.8 \times 10^{-8}\Omega m$ has a length of 30m and cross sectional area of $0.45 cm^2$. Find the resistance of the wire?

Solution: $R = \rho\frac{\ell}{A} = \left(2.8\times10^{-8}\right)\frac{30}{\left(0.45\times10^{-4}\right)} = 1.86\times10^{-4}\Omega$

19. What is the resistance of copper wire 30m long and 0.8cm in radius $\left(\rho_{Cu} = 1.7\times10^{-8}\Omega m\right)$.

Solution: $R = \rho\frac{\ell}{A}, A = \pi r^2 = \pi\left(0.8\times10^{-2}\right)^2 = 20.1\times10^{-8}m^2$

$$\Rightarrow R = \left(1.7\times10^{-8}\right)\frac{(30)}{\left(201\times10^{-8}\right)} = 0.25\Omega$$

20. What length of wire of resistivity $8\times10^{-7}\Omega m$ and cross sectional area $0.2mm^2$ is required to get a resistance of 10Ω?

Solution: $R = \rho\frac{\ell}{A} \Rightarrow \ell = \frac{RA}{\rho} = \frac{(10)\left(0.2\times10^{-6}\right)}{0.8\times10^{-7}} = 25m$

21. Find the resistance of a 10m length of wire which has a cross sectional area of $0.05mm^2$ and is at a temperature change of $30°C$.

$\left(\text{take } \rho = 40\mu mm,\ \sigma = 0.005\frac{\Omega}{°C}\right)$

Solution: $R_o = \frac{\rho L}{A} = \frac{\left(40\times10^{-9}\right)(10)}{0.05\times10^{-6}} = 8\Omega$

and, $R = R_o + \Delta R = R_o + R_o\alpha\Delta T$

$R = 8+(8)(0.005)(30) = 9.2\Omega$

22. A metal rod of 1mm diameter has a resistance of 0.01Ω. What should be the resistance of a rod of the same metal and length if the diameter were doubled?

Solution: $R_1 = \rho\frac{L_1}{A_1}$ .....(*) , and

$$R_2 = \rho\frac{L_2}{A_2} \quad \text{.....(**)}$$

taking ratio, $\frac{R_2}{R_1} = \rho\frac{L_2}{A_2} \Big/ \rho\frac{L_1}{A_1}$, but $L_1 = L_2$

$$\Rightarrow \frac{R_2}{R_1} = \frac{A_1}{A_2} = \pi\frac{d_1^2}{4} \Big/ \pi\frac{d_2^2}{4} = \frac{d_1^2}{d_2^2}, \text{ but } d_2 = 2d_1$$

$$\frac{R_2}{R_1} = \frac{d_1^2}{(2d_1)^2} = \frac{d_1^2}{4d_1^2} = \frac{1}{4}$$

$$\Rightarrow R_2 = \frac{1}{4}R_1 = \frac{1}{4}(0.01) = 0.0025\Omega$$

23. A potential difference of 5.1V is applied across a 2m long copper wire of cross sectional area $0.01mm^2$ if resistivity of copper at 20°C is $1.7 \times 10^{-8}\Omega m$, what is the current in the wire?

Solution: $V = IR \Rightarrow R = \frac{V}{I}$, and

$$R = \rho\frac{\ell}{A} = \frac{(1.7\times10^{-8})(2)}{10^{-8}} = 3.4\Omega$$

$$\Rightarrow I = \frac{V}{R} = \frac{5.1}{3.4}$$

$$I = 5.1A$$

24. When a potential difference of 6V is applied across a brass wire of cross sectional area $0.02cm^2$, a current of 12A flows through it. Calculate the length of the wire. The resistivity of brass is $6.6 \times 10^2\Omega m$.

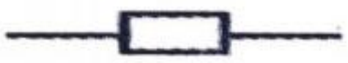
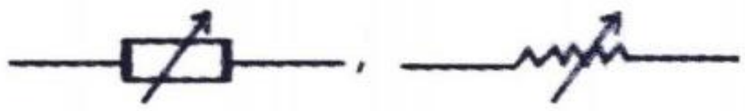
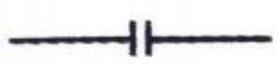
Solution: $R = \frac{V}{I} = \frac{6}{12} = 0.5\Omega$, and

$$R = \rho\frac{\ell}{A} \Rightarrow \ell = \frac{RA}{\rho} = \frac{(0.5)\left(0.02\times10^{-4}\right)}{6.6\times10^{-8}} = 15.2\text{m}$$

Electric Circuit: is a closed path along which an electric current flows.

Some common elements which can be found in electrical circuits include *light bulbs*, *batteries*, *switchesresistors*, *voltmeters* and *ammeters*.

The circuit symbols for the circuit elements we often use are the following.

Component	Symbol
Bulb	
Cell	
Battery	
Resistor	
Variable resistor	
Switch	
Ammeter	
Voltmeter	
Capacitor	

A simple electric circuit comprises battery, bulb, switch, ammeter or voltmeter.

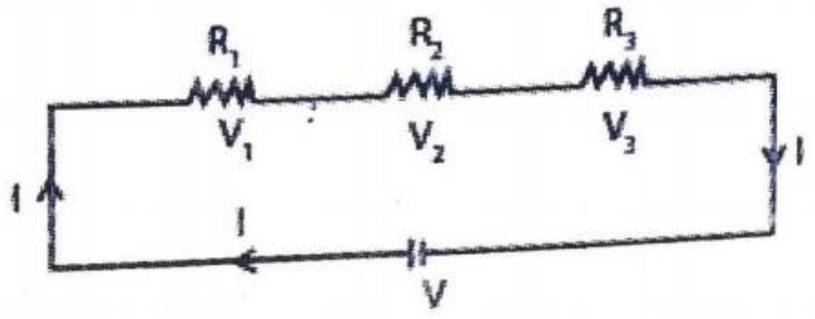
A simple Circuit

11.3 Combination of Resistors

In a circuit, resistors can be connected in series, in parallel or in combination of the two as required.

Resistors in series

In series connection, resistors are connected one after the other, and the



current has only one single path and hence it is constant at every spot.

→ Current is constant

$$\boxed{I_1 = I_2 = I_n = I_T = I}$$

→ The totalpotential drop in the system is the sum of individual potential drops across each resistors

$$V_T = V_1 + V_2 + \ldots + V_n$$

$$I_T R_T = I_1 R_1 + I_2 R_2 + \ldots + R_n V_n$$

But, $I_1 = I_2 = I_T = I_n = I$ = constant

$$IR_T = IR_1 + IR_2 + \ldots . IR_n$$

$$\Rightarrow \boxed{R_T = R_1 + R_2 + \ldots . + R_n}$$

Resistors in parallel

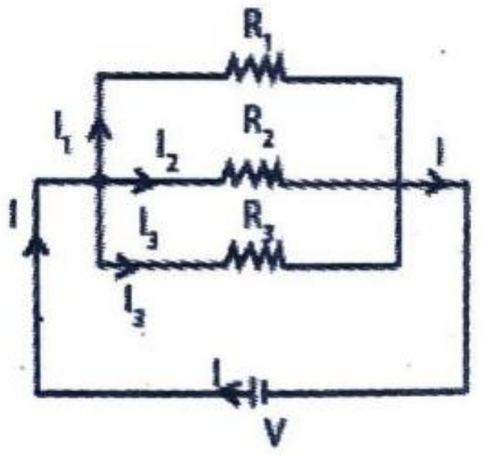
In parallel connection, resistors are connected face to face and the current do have many path ways, branches to go through. And since the polarity of each p.d on each resistor is the same, the potential drop in the system is equal to the potential drop across each resistors.

→ Potential drop is constant

$V_1 = V_2 = V_n = V_T = V$

→ The total current flowing in the circuit is the sum of individual currents flowing

$I_T = I_1 + I_2 + \ldots \ldots . I_n$



$$\frac{V_T}{R_T} = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \ldots . + \frac{V_n}{R_n}$$

But, $v_1 = v_2 = v_T = v_n = v$ = constant

$$\frac{v}{R_T} = \frac{v}{R_1} + \frac{v}{R_2} + \ldots + \frac{v}{R_n}$$

$$\Rightarrow \boxed{\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots . + \frac{1}{R_n}}$$

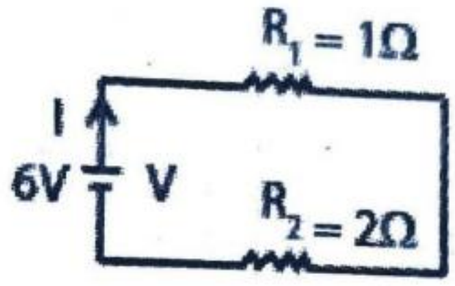
Conceptual Examples

25. Is it the series or the parallel combination of resistors that gives a larger equivalent resistance? Explain

Explanation:Series connection of resistors gives a larger equivalent resistance. For two reasons, the first reason is that, in series connection, the total resistance is the algebraic sum of each resistance, $R_T = R_1 + R_2 + \ldots + R_n$. And the second reason is that, in series connection, the length of the resistor increases since the resistors are connected one next to the other

Illustrative Examples

26. For the circuit shown, calculate



a) The current in the circuit

b) The potential drop across the 2Ω resistor.

Solution: a) $R = R_1 + R_2 = 1 + 2 = 3\Omega$

$$V = IR \Rightarrow I = \frac{V}{R} = \frac{6}{3} = 2A$$

b) $V_2 = IR_2 = (2)(2) = 4V$

27. For the circuit shown, find

a) The total resistance

b) The current flowing through R_1 and R_2

c) The total current in the circuit.

Solution: a) $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{8} + \frac{1}{24} = \frac{4}{24} = \frac{1}{6}$

$\Rightarrow R_T = 6\Omega$

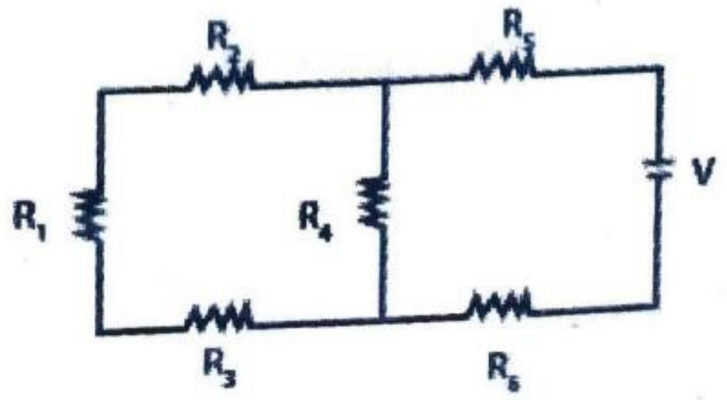
b) $I_1 = \frac{V_1}{R_1} = \frac{V}{R_1} = \frac{48}{8} = 6A$

$I_2 = \frac{V_2}{R_2} = \frac{V}{R_2} = \frac{48}{24} = 2A$

c) $I = I_1 + I_2 = 6 + 2 = 8A$

Note: since R_1 and R_2 are in parallel, $V_1 = V_2 = V = 48V$

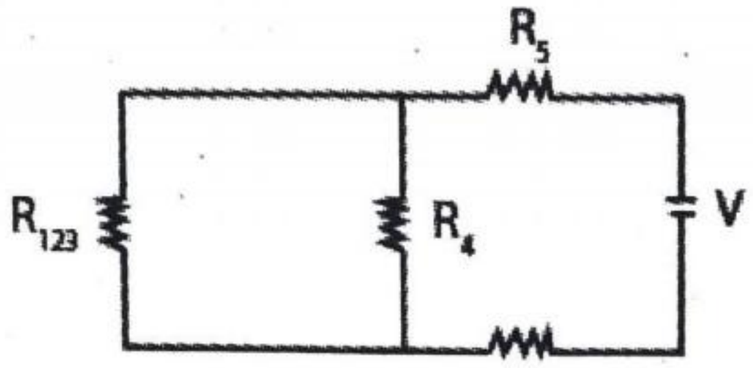
28. Six identical reistors of each 4Ω are connected in a circuit shown below. Determine the total resistance of the circuit.



Solution:R_1, R_2andR_3 are in series,

$R_{123} = R_1 + R_2 + R_3 = 4 + 4 + 4 = 12\Omega$

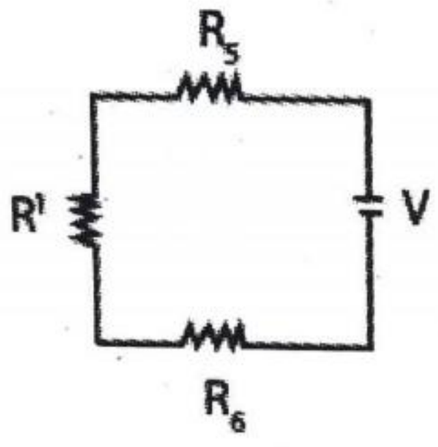
The circuit is modified to,



now $R_{123}$and $R_4$are in parallel

$$\frac{1}{R^1} = \frac{1}{R_{123}} + \frac{1}{R_4} = \frac{1}{12} + \frac{1}{4} = \frac{4}{12} = \frac{1}{3} \Rightarrow R^1 = 3\Omega$$

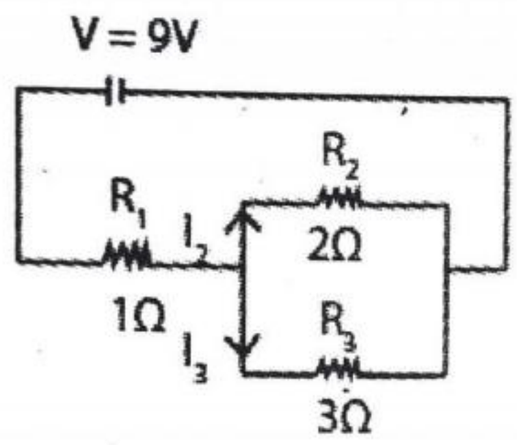
Now, the circuit modified to,



, here, R^1, R_5 and R_6 are in series

$\Rightarrow R_T = R^1 + R_5 + R_6 = 3 + 4 + 4 = 11\Omega$

29. For the circuit shown, find



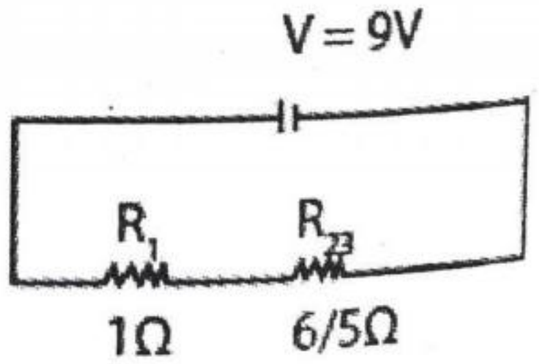
a) The total resistnace
b) The total current
c) The current I_1, I_2 and I_3
d) The potential drop acorss each resistor.

Solution:

a) R_2 and $R_3$are in parallel

$$\frac{1}{R_{23}} = \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{2} + \frac{1}{3} = \frac{5}{6} \Rightarrow R_{23} = \frac{6}{5}\Omega$$

Now, the circuit modifies to;



$R_1$and

R_{23} are in series,

$$R_T = R_1 + R_{23} = 1 + \frac{6}{5} = \frac{11}{5}\Omega$$

b) $I = \frac{V}{R} = \frac{9}{(11/5)} = \frac{(9)(5)}{11} = \frac{45}{11} A$

c) $I_1 = \frac{45}{11} A$. The p.d across R_1 is, $V_1 = R_1 I_1 = (1)\left(\frac{45}{11}\right)$

$V_1 = \frac{45}{11} V = 4.1V$ and $V - V_1 = 9 - 4.1 = 4.9V$

Since R_2 and R_3 are in parallel, V_2 and V_3 are equal

Hence, $I_2 = \frac{V_2}{R_2} = \frac{4.9}{2} = 2.45A$, and

$I_3 = \frac{V_3}{R_3} = \frac{4.9}{3} = 1.63A$

d) $V_1 = 4.1V$, $V_2 = 4.9V$ and $V_3 = 4.9\ V$

30. Three resistors are connected in parallel, find the current in each resistors.

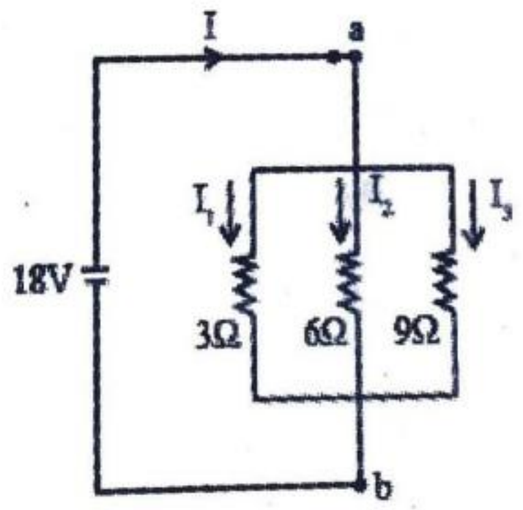
Solution:The resistors are in parallel, and the potential difference across each is the same and is equals to 18V.

Applying ohm's law to each resistor,

$I_1 = \frac{V}{R_1} = \frac{18}{3} = 6A$

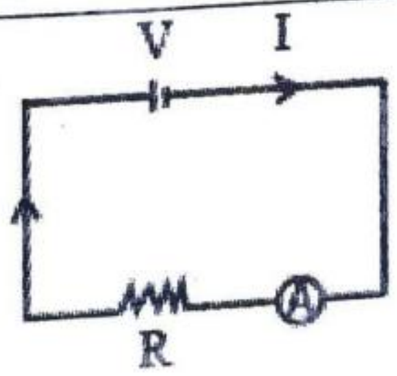
$I_2 = \frac{V}{R_2} = \frac{18}{6} = 3A$

$I_3 = \frac{V}{R_3} = \frac{18}{9} = 2A$



Ammeters and Voltmeters

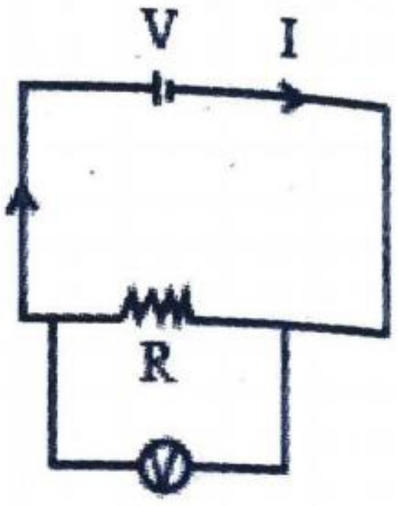
Ammeters: Is a device used to measure the electric current in a circuit.



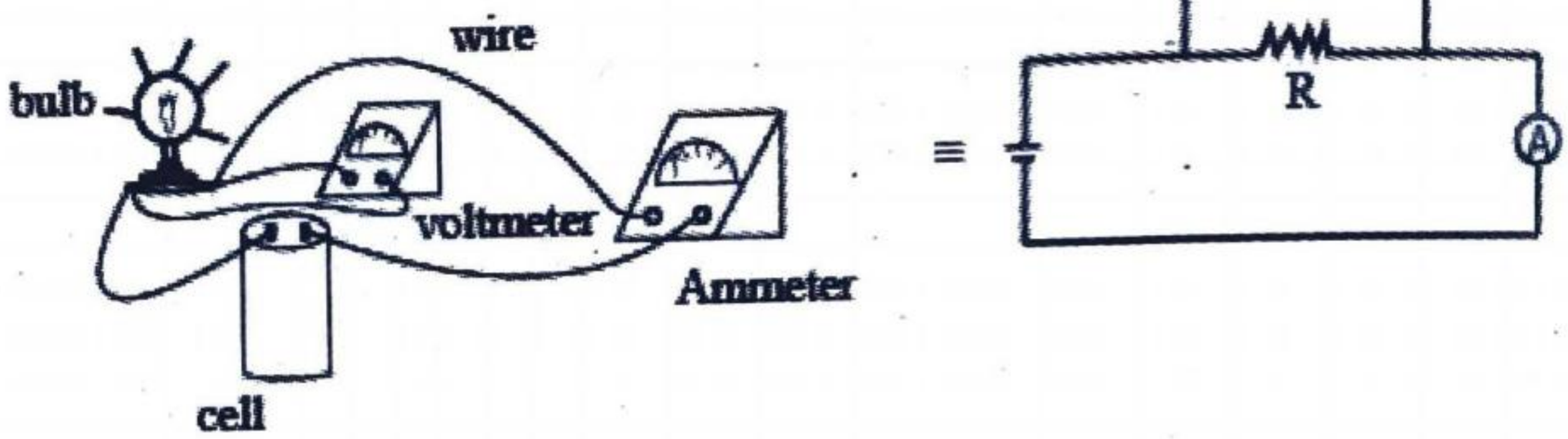
An ammeter should have a very small resistance and its correct position in a circuit is in *series* with the resistor.

Voltmeter: Is a device used to measure potential difference in a circuit.

A voltmeter should have a very large resistance and its correct position in a circuit is in *parallel* with the resistor.



In a given circuit, we can connect both ammeter and voltmeter at the same time as shown



Conceptual Examples

31. Why do we want an Ammeter to have a small resistance and a voltmeter to have a large resistance? Explain

Explanation: If the ammeter has any resistance at all, when we connect it in a circuit, it will increase the total circuit resistance and so reduce the circuit current. Ideally therefore, an ammeter should have zero resistance. On the other hand, the ideal voltmeter should have infinitely large resistance so that no current flows through it. Any current flowing through a voltmeter means an increase in the total circuit current, which will affect the voltmeter reading.

11.4 E.M.F and Internal resistance of a cell

Although electrons move freely in metal wires, they do not flow unless the wire is connected to a source of electric energy such as a battery so as to pump the charges around the circuit.

Water flows downward, from higher level to lower level. In the same manner electric current flows from higher potential (+) to the lower potential (-) ends (terminals) of a battery.

Electromotive force (E.M.F) or ε: is the total potential difference produced by the action in a source of electrical energy like a cell, generator or dynamo.

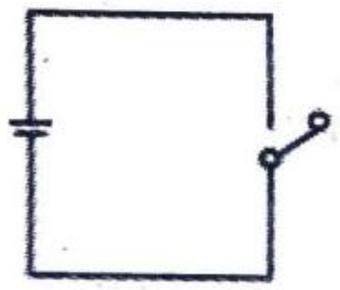
Conceptual Examples

32. From its name, we can easily understand that electromotive force is a force. Is it true or false?

Explanation: False, electromotive force is not a force at all. Instead, the emf determines the amount of work a battery does to move certain amount of charge around a circuit

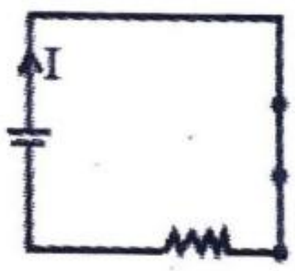
The potential difference across the ends (terminals) of a cell is known as *terminal potential difference*.

When the circuit is 'open circuit', that is when there is no current,flowing in the circuit, the terminal potential difference is equal to the electromotive force of the cell.



Open Circuit

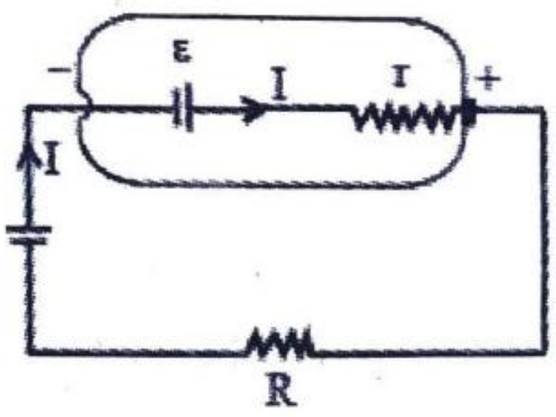
However, when the circuit is 'closed circuit' that is when there is flow of current in the circuit, the terminal potential difference is less than the electromotive force of the cell.



Closed Circuit

Now consider a cell of electromotive force ε, internal resistance (r) and external (load) resistance (R).

The electromotive force is dropped as internal potential difference (V_{int}) and external (terminal) potential difference.



$$\Rightarrow \boxed{\varepsilon = V_{int} + V_{ext}}, \text{ where}$$

$V_{int} = Ir$ and $V_{ext} = IR$

$\Rightarrow \varepsilon = Ir + IR$

$$\boxed{\varepsilon = I = (r + R)}$$

☞ **Note that:Internal resistance (r):** is the resistance inside the cell itself

Internal potential difference (V_{int}): is the potential difference needed to move charges in the internal resistance

External potential difference (V_{ext}):is the potential difference needed to move charges in the external resistance.

Illustrative Examples

33. A battery of emf = 12V and internal resistance r = 1Ω is connected to a 3Ω external resistance. Find
    a) the current in the circuit
    b) the external (terminal) potential difference

Solution: a) $\varepsilon = I(r + R) \Rightarrow I = \dfrac{E}{r + R} = \dfrac{12}{1+2} = 3A$

b) $V_{ext} = IR = (3)(3) = 9V$

34. A car battery with an emf of 12V is used to light a car's head lights. If the terminal potential differenc is 11.4V. Calculate
    a) The potential difference used to drive current through the battery
    b) The internal resistance of the battery if the current in the circuit is 3A.

Solution: a) $\varepsilon = V_{int} + V_{ext}$

$V_{int} = \varepsilon - V_{ext} = 12 - 11.4 = 0.6V$

b) $V_{int} = IR \Rightarrow r = \dfrac{V_{int}}{I} = \dfrac{0.6}{3} = 0.2\Omega$

Combination of cells in series and in parallel

cells in series.

When cells are connected in series,

→ The total emf is the sum of the emf of each cells.

$\varepsilon_T = \varepsilon_1 + \varepsilon_2 + ... + \varepsilon_n$

→ The total internal resistance is the sum of the internal resistance of each cell.

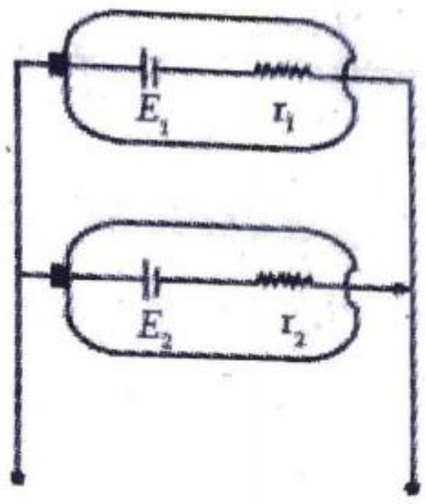
$$r_T = r_1 + r_2 + \ldots\ldots + r_n$$

Cells in parallel

When cells are connected in parallel.

→ The total emf is equal to the emf of each cell.

→ The total internal resistance is the reciprocal sum of each internal resistance

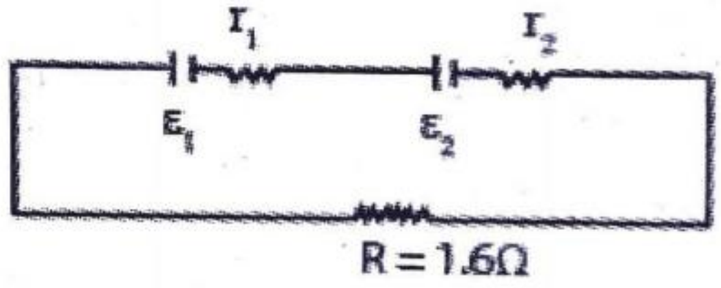


$$\frac{1}{r_T} = \frac{1}{r_1} + \frac{1}{r_2} + \ldots\ldots + \frac{1}{r_n}$$

Illustrative Examples

35. Two cells of emf 1.5V and interanl resistance 0.4Ω each are connected in series with a 1.6Ω external resistor. Calculate

a) the total emf of the battery
b) the total internal resistance of the battery
c) the current in the circuit



Solution: a) $\varepsilon = \varepsilon_1 + \varepsilon_2 = 1.5 + 1.5 = 3V$

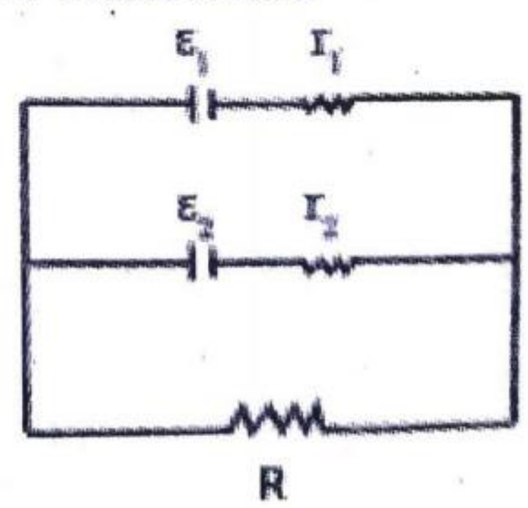
b) $r = r_1 + r_2 = 0.4 + 0.4 = 0.8\Omega$

$R_{total} = r + R = 0.8 + 1.6 = 2.4\Omega$

c) $\varepsilon = I(r + R) \Rightarrow I = \dfrac{\varepsilon}{r + R} = \dfrac{3}{0.8 + 1.6} = 1.25A$

36. Two cells of emf 1.5V and interanl resistance 0.4Ω each are connected in series with a 1.6Ω external resistor. Calculate

a) the total emf of the battery
b) the total internal resistance of the battery
c) the current in the circuit



Solution:

a) $\varepsilon = \varepsilon_1 + \varepsilon_2 = 1.5V$

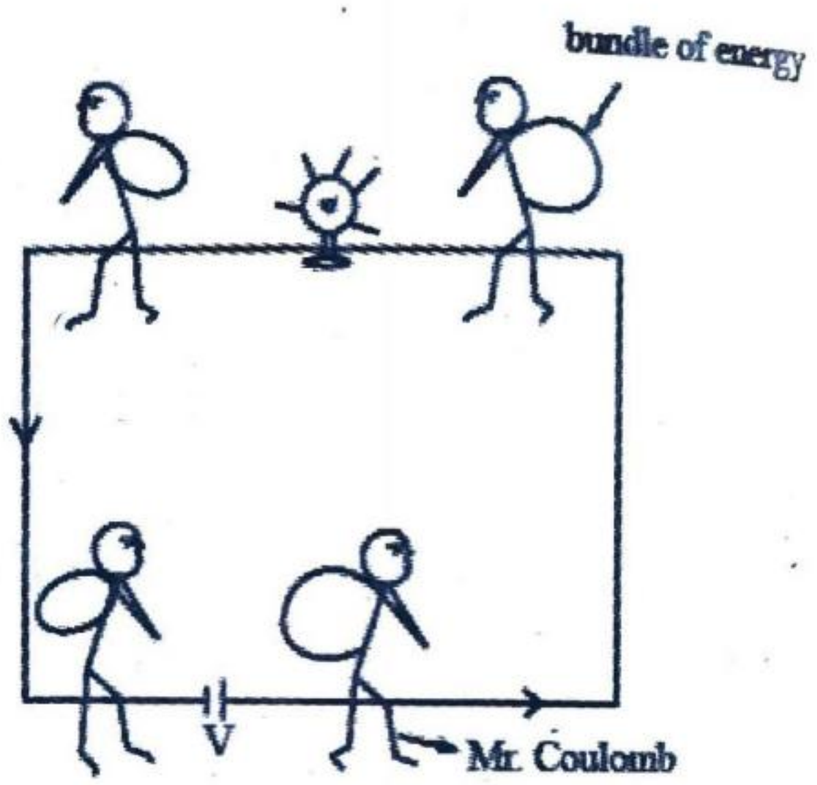
b) $\frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2} = \frac{1}{0.4} + \frac{1}{0.4} = \frac{2}{0.4} = \frac{1}{0.2} \Rightarrow r = 0.2\Omega$

c) $\varepsilon = I(r + R) \Rightarrow I = \frac{\varepsilon}{r + R}$

$= \frac{1.5}{0.2 + 1.6} = \frac{1.5}{0.8} = 0.83A$

11.5 Electrical Energy and Power

Electric energy is converted into other forms of energy such as light and heat. Mr. coulomb, for example travels around the circuit carrying "bundle" of energy, and unloads most of it in the bulb where the electrical energy is converted into heat and light Mr. Coulomb then goes around the circuit and receives a fresh 'bundle' of energy as he passes through the battery. And the cycle repeats.



The electrical energy produced by a current (I) flowing through a potential difference (V) for a time t is given by

$$\boxed{\text{Energy} = VIT}$$

The SI unit of electric energy is joule (J)

Power: Is the rate at which energy is being supplied or converted.

$$\text{Power} = \frac{\text{Energy}}{\text{time}} = \frac{VIT}{t}$$

$$\boxed{\text{Power} = VI}$$

The SI unit of power is watt (w)

$$1 \text{ watt} = \frac{1 \text{ Joule}}{1 \text{ second}} \Rightarrow \boxed{1W = \frac{1\ J}{s}}$$

Cost of electrical energy

Electric energy consumed is paid for each kilowatt hour (kwh).

Kilowatthour (kwh): Is the energy used by a 1kw appliance working for 1 hour.

The amount of payment (cost) for the electric consumed is obtained by

$$\text{Cost} = (\text{Energy})\,(\text{rate of payment})$$

Where, *energy* = (power) (time) = VIt and should always be converted into kwh.

Rate of payment: is a fixed payment value for each kilowatthour consumption.

Illustrative Examples

37. A current of 0.3A flows through a 300Ω resistor. How much power is lost in the resistor?

Solution: $P = IV = I^2R = (0.3)^2\,(300) = 27W$

38. What is the power out put when a current of 0.4A delivered due to a 1.5V battery?

Solution: $P = IV = (0.4)\,(1.5) = 0.6W$

39. An electric lamp is marked 110W and 220V. What is its

a) Current
b) Resistance

Solution:a) $P = IV \Rightarrow I = \frac{P}{V} = \frac{110}{220} = 0.5A$

b) $P = IV = \frac{V^2}{R} \Rightarrow R = \frac{V^2}{P} \frac{(220)^2}{110} = 440\Omega$

40. An electric light has a resistance of 110Ω with a current of 0.5A calculate the energy used by the light in one hour.

Solution: $\text{Energy} = VIt = I^2Rt = (0.5)^2(110)(3600)$

$= 99000J = 99kJ$

41. A 1.8KW electric fire is accidentally left on over night for 8 hours. If the cost of a kwh electrical energy is 0.345 Birr, how much money has been wasted?

Solution: Energy = (power) (time) = (1.8) (8) = 14.4 kwh

Cost = (Energy) (rate) = (14.4) (0.345) = 4.986 Birr

42. A woman uses two 60w light bulbs at her home. The two bulbs remain switched on for 6 hours every day. Find the cost of electrical energy in one month. (rate = 27 cents/kwh)?

Solution: Energy = (power) (time)

$= (2 \times 60)(6 \times 30) = 21600\text{wh} = 21.6\text{kwh}$

Cost = (Energy) (rate) = (21.6) (27) = 583.2 cents

Cost = 5.83 birr

43. Find the cost of using five 60w lamps and four 100w lamps for 8 hours, assuming the cost of electrical energy is 30 cents per unit.

Solution:

Energy = (power) (time)

$= (5 \times 60 + 4 \times 100)(8) = (5.6)(30) = 5600\text{wh} = 5.6\text{ kwh}$

Cost = (Energy) (rate)

= (5.6) (30) = 168 cents

= 1.68 birr

44. In a certain family, a 1kw stove is on for 1 hour every day and four 60w bulbs are on for 3 hours every day. Caculate the monthly electric cost if rate is 27 cents/wh.

Solution: Energy for stove = (power) (time) = (1) (1×30) = 3kwh

Energy for bulb = (power) (time)

$= (4\times60)(3\times30) = 21600\text{kh} = 21.6\text{ kwh}$

Total energy = 3 + 21.6 = 24.6kwh

Cost = (energy) (rate) = (24.6) (27) = 664.2 cents

Cost = 6.642 Birr.

General Secondary Education Certificate Exam

45. A current in a torch bulb is 0.1A. How long will it take for one coulomb of charge to pass through the bulb?

A. 5 sec
B. 10 sec
C. 0.1 sec
D. 1 sec

Solution: $I = \frac{Q}{t} \Rightarrow t = \frac{Q}{I} = \frac{1}{0.1} = 10\text{sec.}$ **Answer: B**

46. A 100 watt light bulb is used for 10 hrs every day. What is the cost of its operation for one month at a fixed rate of 30 cents per kwh?

A. 15 birr C. 9 birr
B. 3 birr D. 6 birr

Solution: Cost = (Energy) (rate), Energy = (power) (time),

Energy = (100) (10)(30) = 30,000wh = 30kwh

Cost = (Energy) (rate)= (30)(30) = 900 cents = 9 birr

Answer: C

47. A current of one Ampere corresponds to the flow of one coulomb of charge

A. In one second
B. Across a one ohm resistor
C. In a uniform electric field
D. Across one volt p.d

Explanation: $\text{current} = \frac{\text{charge}}{\text{time}} \Rightarrow 1A = \frac{1C.}{1\text{ sec}}$ **Answer: A**

48. For an electric current in a wire.

A. The direction of current and direction of flow of electrons in the wire are opposite.
B. Electric current in the wire is due to the random motion of electrons in the wire.
C. Electric current in the wire is due to the perpendicular motion of electrons in the wire
D. Electrons flow in the wire in the direction direction of conventional current

Explanation: conventional current is opposite to the direction of flow of electrons in a wire. **Answer: A**

49. A wire of 1m length and 2mm diameter has a resistance of 6Ω. What will be the resistance of the same type of wire with the same diameter if its length is 5m?

A. 6Ω B. 5Ω C. 30Ω D. 60Ω

Solution: $R = \frac{\rho\ell}{A} = R_1 = \frac{\rho\ell_1}{A} = 6.....(*)$

$$R_2 = \frac{\rho\ell_2}{A} = ...(**), \text{ taking ratio}$$

$$\frac{R_2}{R_1} = \frac{\ell_2}{\ell_1} \Rightarrow R_2 = \left(\frac{\ell_2}{\ell_1}\right)(R_1) = \frac{5}{1} \times 6 = 30\Omega$$

Answer: C

50. A resistor wire of length (ℓ) and cross sectional area (A) is connected in a simple electric circuit. This wire is then folded into two and connected in the same circuit. What happens to the circuit resistance?

A. doubled
B. quadrupled
C. halved
D. remains the same

Explanation: $R_1 = \frac{\rho\ell_1}{A_1}.....(*)$

$R_2 = \frac{\rho\ell_2}{A_2}$, $\ell_2 = \frac{1}{2}\ell_1$ and $A_2 = 2A_1$

$\Rightarrow R_2 \frac{\rho\ell_1}{4A_1}.....(**)$, taking ratio

$$\frac{R_2}{R_1} = \frac{\rho\ell_1}{4A_1 / \rho\ell_1 / A_1} \Rightarrow R_2 = \frac{1}{4}R_1$$

Answer: B

51. suppose a resistor is connected to a constant voltage supply. When will the current in the resistor decrease.

A. When length of the resistor is increased
B. When length of the resistor is reduced
C. When cross sectional area of the resistor increased
D. When the resistance is decreased

Answer: A

52. Consider a simple circuit with a cell emf of 4.5V and a resistor of 40Ω. The current in the circuit is 0.1A what will be the current in the circuit if additional cell of 4.5V is connected to the previous cell in series?

A. 0.18A
B. 0.05A
C. 0.2A
D. 0.21A

Solution: $\varepsilon = I(R+r) \Rightarrow R + r = \frac{\varepsilon}{I} = \frac{4.5}{0.1} = 45\Omega$

$$R + r = 45 \Rightarrow r = 45 - R = 45 - 40 = 5\Omega$$

Next, connection, $V = V_1 + V_2 = 4.5 + 4.5 = 9V \Rightarrow \varepsilon = 9V$

$$\varepsilon = I\left(R + r + r\right) \Rightarrow I = \frac{\varepsilon}{R + r + r}$$

$$I = \frac{9}{40 + 5 + 5} = \frac{9}{50} = 0.18A$$

Answer: A

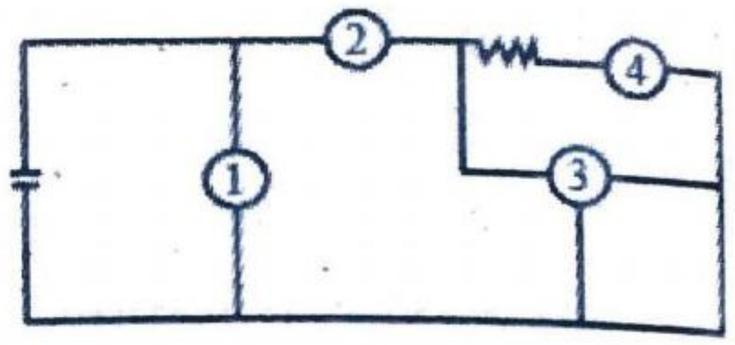
53. The correct way of connecting an ammeter and a voltmeter in a circuit

A. Both connected in series with the resistor
B. Both connected in parallel with the resistor
C. Ammeter in series and voltmeter in parallel with the resistor
D. Ammeter in parallel and voltmeter in series with the resistor

Answer: C

54. In a circuit diagram below, which one is a voltmeter?

A. $\frac{1}{3}\Omega$
B. 1.5Ω
C. 1 and 3
D. 1, 3 and 5



Explanation: voltmeter should be in parallel with the resistor.

Answer: C

55. You are given three 1Ω resistors. All of the following equivalent resistance can be produced using all three except

A. $\frac{1}{3}\Omega$
B. 1.5Ω
C. 1Ω
D. 3Ω

Answer: A

56. How is a galvanometer used as a voltage measuring device?

A. A large multiplier resistor must be attached to it in series
B. A large shunt resistor must be connected to it in series
C. A small shunt resistor must be connected to it in series
D. It can be used directly since it is basically a voltage measuring device

Answer: A

57. To protect damage by large current, a galvanometer is modified by connecting

A. Multiplier resistor in parallel with it to be a voltmeter
B. Multiplier resistor in series with it to be an ohm meter
C. Shunt resistor in series with it to be an ammeter
D. Shunt resistor in parallel with it to be an ammeter

Answer: D

58. If a 15V dry cell is connected to a resistor of 144Ω, a current of 0.1A flows through the resistor. What is the internal resistance of the battery?

A. 300Ω
B. 150Ω
C. 10Ω
D. 6Ω

Solution: $\varepsilon = I\left(r + R\right) \Rightarrow R + r = \frac{\varepsilon}{I} = \frac{15}{0.1} = 150\Omega$

$$R + r = 150 \Rightarrow r = 150 - R = 150 - 144 = 6\Omega$$

Answer: D

59. A battery of emf 5V and internal resistance 10Ωis connected in series with an external resistor of resistance 240Ω. What is the voltage drop on the external resistor.

A. 0.2V
B. 5.22V
C. 4.8V
D. 5V

Solution: $\varepsilon = I\left(r+R\right) \Rightarrow I = \dfrac{\varepsilon}{r+R} = \dfrac{5}{10+240} = \dfrac{5}{250\Omega} = 0.02A$

$V_{ext} = IR = (0.02)\,(240) = 4.8V$

Answer: C

60. An 8Ω speaker is connected across the output terminals of a power amplifier that delivers 64W. What is the current supplied to the speaker?

A. 8A
B. $\sqrt{2}A$
C. 4A
D. $2\sqrt{2}A$

Solution: $P = I^2R \Rightarrow I = \sqrt{\dfrac{P}{R}} = \sqrt{\dfrac{64}{8}} = \sqrt{8} = 2\sqrt{2}A$

Answer: D

61. You wish to triple the rate of electrical energy dissipation in a heating device. To do this you should triple,

A. The p.d keeping the resistance the same
B. The resistance keeping the current the same
C. The resistance keeping the p.d the same
D. The current keeping the resistance the same.

Explanation: $P = \dfrac{V^2}{R} \Rightarrow P_1 = \dfrac{V^2}{R_1} \ldots\ldots(*)$

$3P_1 = P_2 = \dfrac{V^2}{R_2} \ldots\ldots\ldots\ldots(**)$, taking ratio

$$\frac{P_2}{P_1} = \frac{V^2/R_2}{V^2/R_1} = \frac{R_2}{R_1} = \frac{3P_1}{P_1} = \frac{R_2}{R_1} \Rightarrow R_2 = 3R_1$$

Answer: B

62. The p.d between two points is 100V. A particle with a charge of 2C is transported from one point to the other. What is the magntidue of the work done?

A. 200J
B. 100J
C. 50J
D. 2J

Solution: W = QV = (2) (100) = 200J. **Answer: A**

63. In an experiment four resistors, 3Ω, 4Ω, 12Ω and 16Ω are required at different time. But the lab technician told you to use only two resistors. Which two resistors do you choose to do the experiment.

A. 3Ω and 6Ω
B. 3Ω and 4Ω
C. 4Ω and 12Ω
D. 4Ω and 16Ω

Explanation: It is clear that the connection is parallel. Hence, the resultant of the four when connected in parallel is about 4Ω and the 3Ω and 4Ωwhen connected in parallel gives a resultant of about 1.7Ω which is somewhat close to 1.4Ω

Answer: B

64. A student wanted to use a fuse to safeguard her radio that has a 5V transformer supply. If the fuse operates at a maximum of 6W, how should it be connected?

A. In series with the radio before the mains input
B. In series the radio next to the transformer
C. In series the radio before the mains
D. In series with the radio next to the transformer.

Answer: A

65. The terminal p.d of a battery is maximum when

A. the emf is the minimum
B. the battery is being discharged
C. there is no current flowing
D. there is a steady current flowing **Answer: C**

66. a lightning protection provided to a building is

A. an overhead grid of insulators extended to the ground
B. an overhead grid of wire conductors extended to the ground
C. a system of lightning rod and conductors extended to the building
D. a stem of lighting rods connected to a building is steel from

Answer: B

67. A transformer on a pole near a factory steps the voltage down from 3600V to 220V.What is the current in the primary coil when the transformer delivers 900KV to the factory?

A. 250A B. 0.25A C. 4.1A D. 220A

Solution: for an ideal transformer, power in = power out

$$\Rightarrow P_{in} = P_{out} \Rightarrow I_P V_P = P_{out} \Rightarrow I_P = \frac{P}{V_P} = \frac{900000}{3600}$$

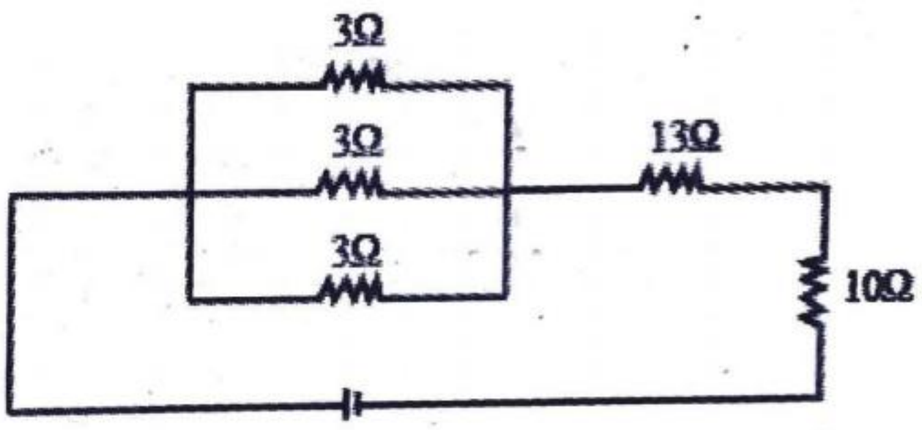
$I_P = 250A$

Answer: A

68. What is the p.d across the 10Ω resistor?

A. 1V C. 5V
B. 12V D. 10V

Solution: The three 3Ω resistors are in parallel and their resultant is 1Ω. Then the total resistance is 24Ω.



$I = \frac{V}{R} = \frac{12}{24} = 0.5A$ and the potential drop at the 10Ω is, $V = (I)(R) = (0.5)(10) = 5V$

Answer: C

69. A stove that draws 10A current from 220v line is used for 5 hours. If the electrical company demands 20 cents per kilowatt hour, what is the cost of the electrical energy consumed?

A. 2 Birr and 20 cents
B. 55 cents
C. 1 Birr and 10 cents
D. 1 Birr

Solution: cost = (energy) (rate)

energy = (power) (time) = VIt

energy = (10) (220) (5) = 11000wh = 11kwh

cost = (11) (20) = 220 cents = 2 Birr &20 cents

Answer: A

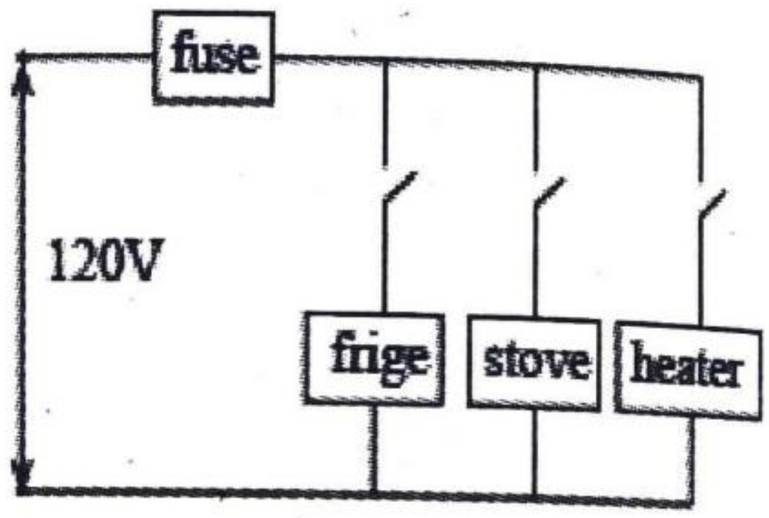
70. The potential difference between the terminals of a battery when the battery is isolated electrically is

A. electric field
B. electric force
C. electromotive force
D. terminal voltage

Answer: C

71. A 360W refrigerator, a 1200W stove, a 720W heater and a 15A fuse are connected to 120V main. Under what condition will the fuse melt?

A. When the refrigerator and the stove and switched on
B. When the refrigerator and the heater are switched on
C. When the heater and the stove are switched on
D. When any of the devices is switched on



Solution: the fuse melts when the current through it is greater than 15A. When the stove and heater are on.

$$I_{stove} = \frac{P_{stove}}{V_{stove}} = \frac{1200}{120} = 10A,\ I_{heater} = \frac{P_{heater}}{V_{heater}} = \frac{720}{120} = 6A$$

$$I = I_{stove} + I_{heater} = 10 + 6 = 16A > 15A$$

Answer: C

Additional Examples

72. A cell supplies a current of 0.9A through a 2Ωresistor and a current of 0.3A through a 7Ω resistors. The internal resistance of the cell is

A. 2Ω B.Ω1 C. 1.2Ω D. 0.5Ω

Solution: $\varepsilon_1 = \varepsilon_2 = I_1(R_1 + r) = I_2(R_2 + r)$

$$\Rightarrow 0.8(2 + r) = 0.3(7 + r)$$

$$6 + 3r = 7 + r \Rightarrow r = 0.5\Omega$$

Answer: D

73. A current flows in a metallic conductor of varying cross section. The quantity that remains constant along the length of the conductor is

A. drift speed
B. Electric field and drift speed
C. Current
D. Current and drift speed

Answer: C

74. Potential difference of a cell in an open circuit is 6V but it falls to 4V when a current of 2A is flowing through it. The internal resistance of the cell is

A. 1Ω B. 2Ω C. 3Ω D. 4Ω

Solution: $\varepsilon = V_{int} + V_{ext} \Rightarrow 6 = V_{int} + 4 \Rightarrow V_{int} = 2V$

$V_{int} = Ir \Rightarrow 2 = (2)(r) \Rightarrow r = 1\Omega$

Answer: A

75. Ten identical resistances, each having a resistance of 1Ω are joined in parallel. The combination has the resistance of

A. 10Ω
B. 1Ω
C. 0.1Ω
D. 0.01Ω

Solution:

$$\frac{1}{R_T} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \frac{1}{R} = \frac{10}{R}$$

$$\frac{1}{R_T} = \frac{10}{R} \Rightarrow R_T = \frac{R}{10} = \frac{1}{10} = 0.1\Omega$$

Answer: C

76. A wire of resistance R is cut into n equal parts. These parts are then connected in parallel. The equivalent resistance of the combination is

A. nR
B. R/n
C. n/R
D. R/n^2

Solution: $\frac{1}{R_T} = \frac{1}{R/n} + \frac{1}{R/n} + \ldots\ldots\ldots + \frac{1}{R/n}$

$\Rightarrow \frac{1}{R_T} = \frac{n}{R/n} \Rightarrow \frac{1}{R_T} = \frac{n^2}{R}$ or $R_T = \frac{R}{n^2}$

Answer: D

77. A 50V battery is connected across a 10Ω resistor. The current is 4.5A the internal resistance of the battery is

A. zero B. 1.1Ω C. 0.5Ω D. 5Ω

Solution: $\varepsilon = I(r + R) \Rightarrow 50 = 4.5\ (10 + r) \Rightarrow r = 1.1\Omega$

Answer: B

78. When both length and cross sectional area of a wire are doubled, then its resistance

A. Will be come half
B. Will be doubled
C. Will remains the same
D. Will become four times

Answer: C

79. The resistance of a conductor carrying a current of 2A when the potential difference across it is 20V is

A. 0.1Ω B. 1Ω C. 10Ω D. 40Ω

Solution: $V = IR \Rightarrow R = \frac{V}{I} = \frac{20}{2} = 10\Omega$

Answer: C

80. Three equal resistors are given. How many different combinations can be made from them?

A. three B. four C. five D. six

Explanation:
- all in series
- all in parallel
- two in series and parallel with the third
- two in parallel and series with the third

Answer: B

81. A total charge of 5C passes through a conductor in 10ms. The current during this time is

A. 50A B. 125A C. 500A D. 250A

Solution: $I = \frac{Q}{t} = \frac{5}{10\times10^{-3}} = 500A$

Answer: C

82. The resistance of a wire of uniform diameter d and length ℓ is R. the resistance of another wire of the same material but diameter 2d and length 4ℓ will be

A. 2R C. $R/2$

B. R D. $R/4$

Solution: $R = \frac{\rho\ell}{A} = \frac{\rho\ell}{\frac{\pi d^2}{4}} = \frac{4\rho\ell}{\pi d^2} \ldots (*)$

$R^1 = \frac{\rho\ell^1}{A^1}$, but $\ell^1 = 4\ell$ and $d^1 = 2d$

$$R^1 = \frac{\rho(4\ell)}{\frac{\pi d^2}{4}} = \frac{16\rho\ell}{\pi(2d)^2} = \frac{16\rho\ell}{4\pi d^2} \ldots (**)$$

Taking ratio of equation (**) to equation (*)

$$\frac{R^1}{R} = \frac{4\rho\ell}{\frac{4\rho\ell}{\pi d^2}} = 1 \Rightarrow R^1 = R$$

Answer: B

83. An electric bulb rated for 500W at 100V is used in a circuit having a 200V supply. The resistance R that must be put in series with the bulb, so that the bulb draws 500W is

A. 18Ω C. 40Ω

B. 20Ω D. 700Ω

Solution: $P = VI \Rightarrow I = \frac{P}{V} = \frac{500}{100} = 5A$, and

$$V = IR \Rightarrow 100 = 5R \Rightarrow R = 20\Omega$$

Answer: B

84. The time required by 1kw heater to raise the temperature of 10 litre of water through 10°C is

A. 42 sec C. 420 sec
B. 210 sec D. 840 sec

Solution: Energy = Heat

$$Pt = mC\Delta T \Rightarrow t = \frac{mC\Delta T}{p} = \frac{(10)(4200)(10)}{1000}$$

$$t = 420 \text{ sec}$$

Answer: C

85. How many 2.1V cells with internal resistance 0.2Ω each are to be connected in series to obtain a battery so that a conductor of resistance 6Ω would carry a current of 1.5A?

A. 4 B. 5 C. 6 D. 7

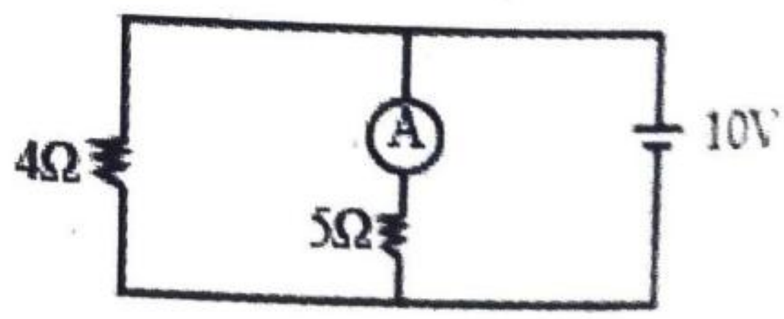
Solution: $n\varepsilon = I(nr + R) \Rightarrow 2.1n = 1.5\ (0.2n + 6)$

$\Rightarrow 2.1n = 0.3n + 9 \Rightarrow 1.8n = 9 \Rightarrow n = 5$. **Answer: B**

86. In the circuit, the reading of the ammeter is,

A. 40A B. 10A C. 5A D. 2A

Solution: The resistors are in parallel hence, the potential difference is the same for then.

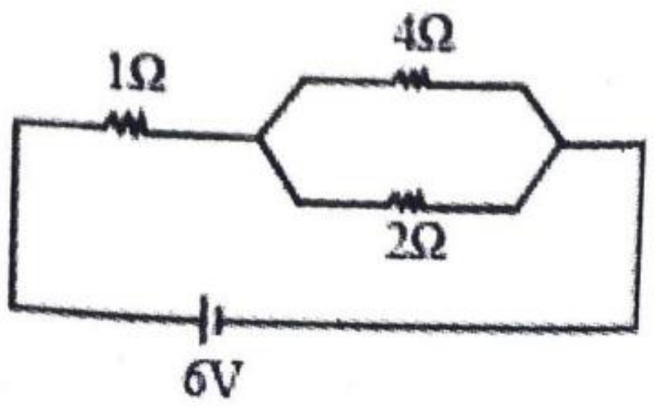


$$V = IR \Rightarrow I = \frac{V}{R} = \frac{10}{5} = 2A$$

Answer: D

87. What is the current flowing through 2Ω resistance shown below

A. $\frac{12}{7}$A C. 12A



B. 7A D. 2A

Solution: The 4Ω and 2Ω are in parallel, thus

$$\frac{1}{R^1}=\frac{1}{2}+\frac{1}{4}\Rightarrow\frac{1}{R^1}=3/4\Rightarrow R^1=4/3\,\Omega$$

$R_T=1+\frac{1}{4}=7/3\,\Omega$. The p.d at the 1Ω is

$V=IR=\left(\frac{6}{7/3}\right)(1)=\frac{18}{7}$V, the p.d across the 4Ω and the 2Ω is the same, and is equal to, $V=6-\frac{18}{7}=\frac{42-18}{7}=\frac{24}{7}$V. then the current in the 2Ω resistor is, $I=V/R=\frac{\left(\frac{24}{7}\right)}{(2)}=\frac{24}{14}=\frac{12}{7}$A. **Answer: A**

88. Two resistors of 15Ω and 30Ω are connected in parallel. What should be the value of R to be connected in series with the two so that the net resistance will be 20Ω?

A. 5Ω
B. 10Ω
C. 15Ω
D. 120Ω

Solution:

$$\frac{1}{R^1}=\frac{1}{15}+\frac{1}{30}=\frac{3}{30}\Rightarrow R^1=10.\ R_T=R^1+R=20\Rightarrow R=10\Omega.$$

Answer: B

89. What is the current in the circuit shown

A. 8.31A
B. 6.82A
C. 4.92A
D. 2A

Solution: the two 6Ω resistors are in series, $R^1=6+6=12\Omega$. Then, R^1 is in parallel with the 3Ω resistor, and the total resistance is,

B. 20Ω D. 700Ω

Solution: $P = VI \Rightarrow I = \frac{P}{V} = \frac{500}{100} = 5A$, and

$$V = IR \Rightarrow 100 = 5R \Rightarrow R = 20\Omega$$

Answer: B

84. The time required by 1kw heater to raise the temperature of 10 litre of water through 10°C is

A. 42 sec C. 420 sec
B. 210 sec D. 840 sec

Solution: Energy = Heat

$$Pt = mC\Delta T \Rightarrow t = \frac{mC\Delta T}{p} = \frac{(10)(4200)(10)}{1000}$$

$$t = 420 \text{ sec}$$

Answer: C

85. How many 2.1V cells with internal resistance 0.2Ω each are to be connected in series to obtain a battery so that a conductor of resistance 6Ω would carry a current of 1.5A?

A. 4 B. 5 C. 6 D. 7

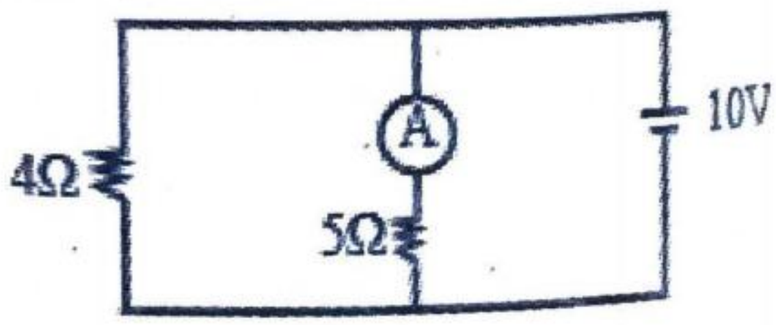
Solution: $n\varepsilon = I(nr + R) \Rightarrow 2.1n = 1.5\ (0.2n + 6)$

$\Rightarrow 2.1n = 0.3n + 9 \Rightarrow 1.8n = 9 \Rightarrow n = 5$. **Answer: B**

86. In the circuit, the reading of the ammeter is,

A. 40A B. 10A C. 5A D. 2A

Solution: The resistors are in parallel hence, the potential difference is the same for then.

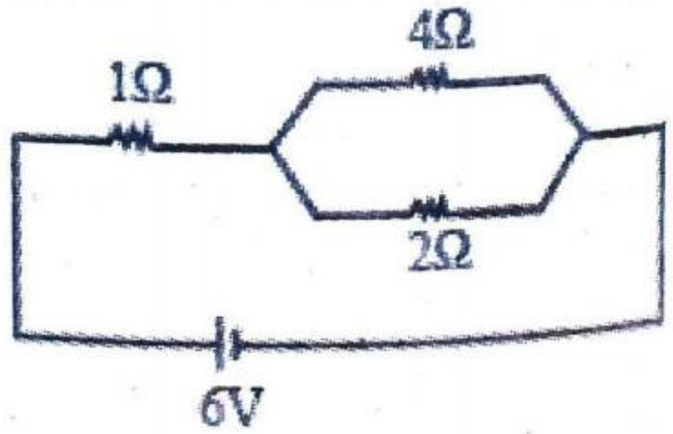


$$V = IR \Rightarrow I = \frac{V}{R} = \frac{10}{5} = 2A$$

Answer: D

87. What is the current flowing through 2Ω resistance shown below

A. $\frac{12}{7}A$ C. 12A



B. 7A D. 2A

Solution: The 4Ω and 2Ω are in parallel, thus

$$\frac{1}{R^1}=\frac{1}{2}+\frac{1}{4}\Rightarrow\frac{1}{R^1}=3/4\Rightarrow R^1=4/3\,\Omega$$

$R_T=1+\frac{1}{4}=7/3\,\Omega$. The p.d at the 1Ω is

$V=IR=\left(\frac{6}{7/3}\right)(1)=\frac{18}{7}$V, the p.d across the 4Ω and the 2Ω is the same, and is equal to, $V=6-\frac{18}{7}=\frac{42-18}{7}=\frac{24}{7}$V. then the current in the 2Ω resistor is, $I=V/R=\frac{\left(\frac{24}{7}\right)}{(2)}=\frac{24}{14}=\frac{12}{7}$A. **Answer: A**

88. Two resistors of 15Ω and 30Ω are connected in parallel. What should be the value of R to be connected in series with the two so that the net resistance will be 20Ω?

A. 5Ω C. 15Ω
B. 10Ω D. 120Ω

Solution:

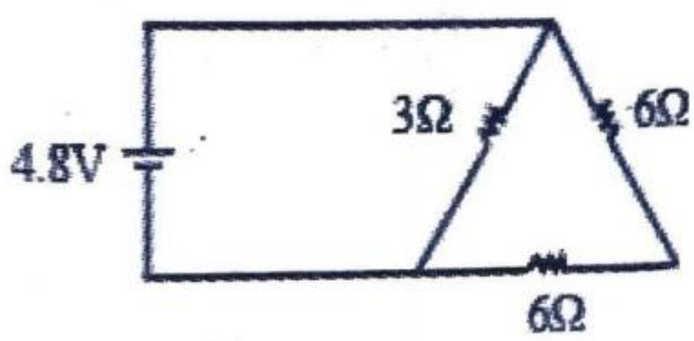
$$\frac{1}{R^1}=\frac{1}{15}+\frac{1}{30}=\frac{3}{30}\Rightarrow R^1=10.\ R_T=R^1+R=20\Rightarrow R=10\Omega.$$

Answer: B

89. What is the current in the circuit shown

A. 8.31A C. 4.92A
B. 6.82A D. 2A

Solution: the two 6Ω resistors are in series, $R^1=6+6=12\Omega$. Then, R^1 is in parallel with the 3Ω resistor, and the total resistance is,



$$\frac{1}{R_T}=\frac{1}{12}+\frac{1}{3}=\frac{5}{12}\Rightarrow R_T=2.4\Omega.$$

and $V=IR\Rightarrow I=\frac{V}{R}=\frac{4.8}{2.4}=2A.$

Answer: D

90. The resistance of a conductor is 5Ω at 50°C and 6Ω at 100°C. its resistance at 0°C is

A. 3Ω B. 4Ω C. 1Ω D. 2Ω

Solution: $5=R_O(1+50\alpha)$ and

$6=R_O(1+100\alpha)$

Taking ratio

$$\frac{5}{6}=\frac{1+50\alpha}{1+100\alpha}\Rightarrow 6+300\alpha=5+500\alpha\Rightarrow\alpha=\frac{1}{200}$$

then, substituting into one of the equations,

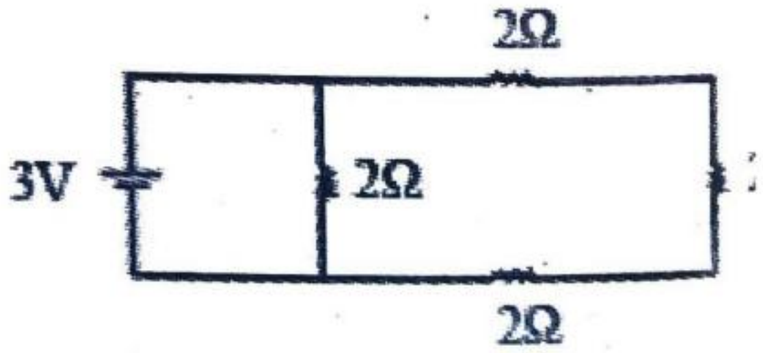
$$5=R_o\left(1+50\left(\frac{1}{200}\right)\right)\Rightarrow R_o=4\Omega$$

Answer: B

91. The current I in the circuit shown is

A. 1A C. 1.2A
B. 0.5A D. 2A

Solution: $R^1=2+2+2=6\Omega$, then



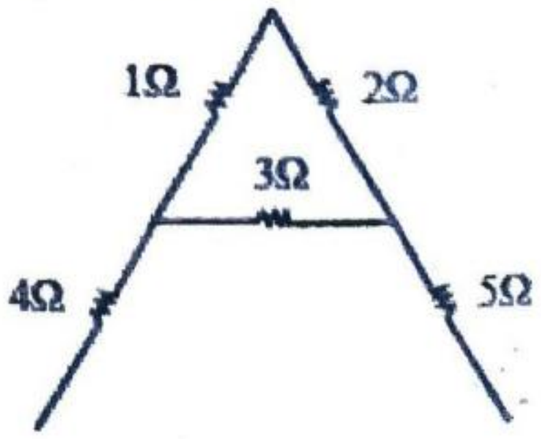
$$\frac{1}{R_T}=\frac{1}{2}+\frac{1}{6}=\frac{4}{6}\Rightarrow R_T=\frac{6}{4}\Omega$$

$$I=\frac{V}{R}=\frac{3}{\left(\frac{6}{4}\right)}=\frac{12}{6}=2A.$$

Answer: D

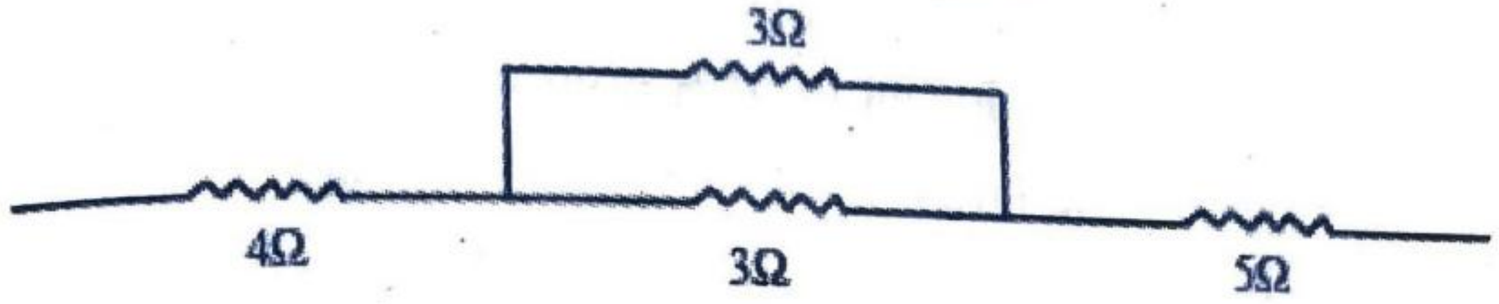
92. Find the equivalent resistance of the circuit shown.

A. 5Ω C. 1Ω
B. 10.5Ω D. 3Ω



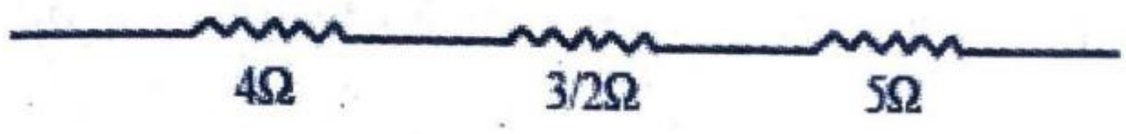
Solution: The 1 and 2Ω are in series,

$R^1 = 1 + 2 = 3\Omega$, then the circuit modified to



the two 3Ω resistors are in parallel,

$\frac{1}{R^{11}} = \frac{1}{3} + \frac{1}{3} = \frac{2}{3} \Rightarrow R^{11} = \frac{3}{2}\Omega$, the circuit is then

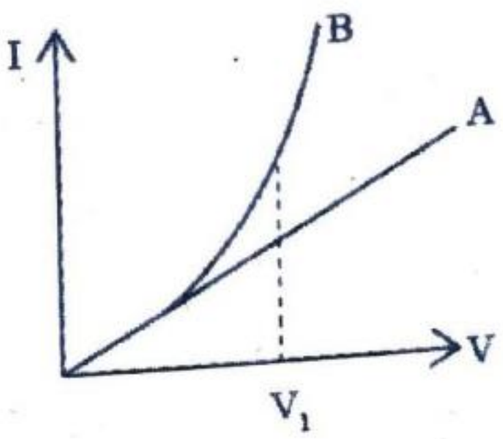


$$R_T = 4 + \frac{3}{2} + 5 = 10.5\Omega$$

Answer: B

Unit Eleven: Conceptual Questions

1. Is there any physical quantity whose unit is 'Ampere second'?
2. What is the difference between a primary and a secondary cell?
3. Explain how it is possible for a current to flow through a metal wire.
4. Is there any difference between a cell and a battery
5. Explain why it is wrong to write the unit of resistivity as Ω/m?
6. What will happen to the resistance of a wire if only the length of the wire doubled.
7. Why is an ammeter connected in series while a voltmeter in parallel with a resistor?
8. Is electric energy measurable directly?
9. A given car battery is rated as 250 amp-hours. What does this rating indicate?
10. The resistivity of tungsten increases with temperature when a light containing a tungsten filament heat up, what will its power consumption be
11. Which metal has a greater resistance when the voltage is at V_1

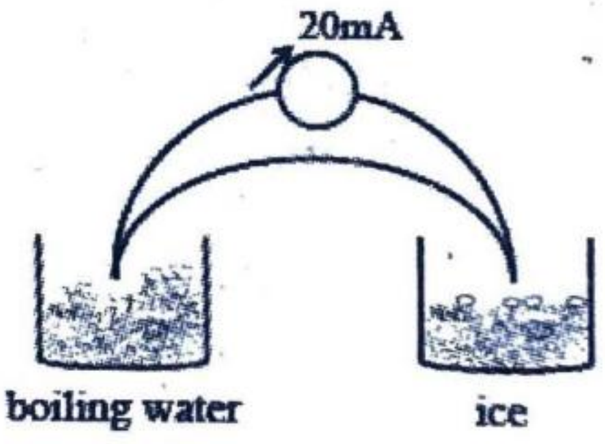


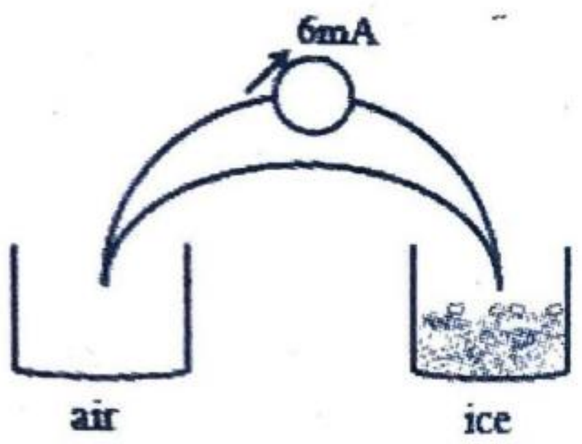
12. Is it possible to connect a group of resistors of value R in such a way that the equivalent resistance is less than R.
13. Light bulb A has twice the power rating of light bulb B. How does the resistance of bulb A compare with that of bulb B.
14. Two cells one small and the other large are labeled 1.5V each. If they have the same internal resistance which one will drive more charge across a conductor.

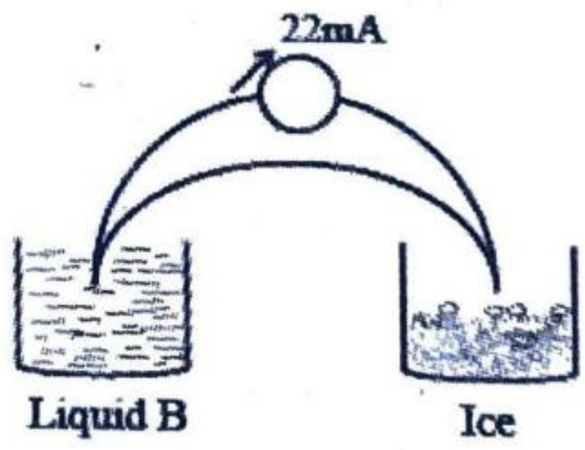
Unit Eleven: Problems

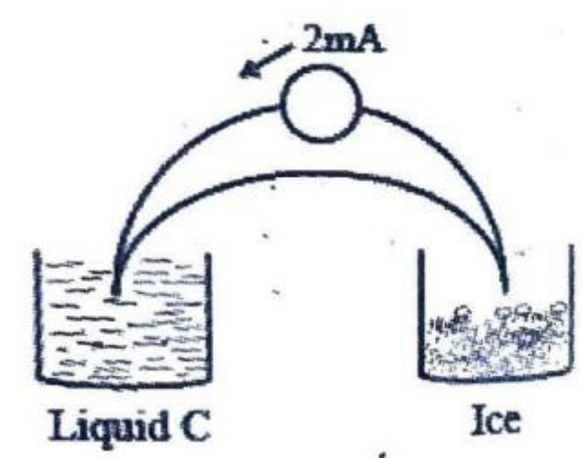
Electric Current

15. Charge flows along a wire at a rate of 3A. how many coulombs of charge will pass a given point in 2 minutes
16. A very sensitive ammeter records a current of 30μA. How long will it take before 6μC of charge has flowed.
17. What temperature does the thermo couple indicate for
    a) Air
    b) Liquid B
    c) Liquid C







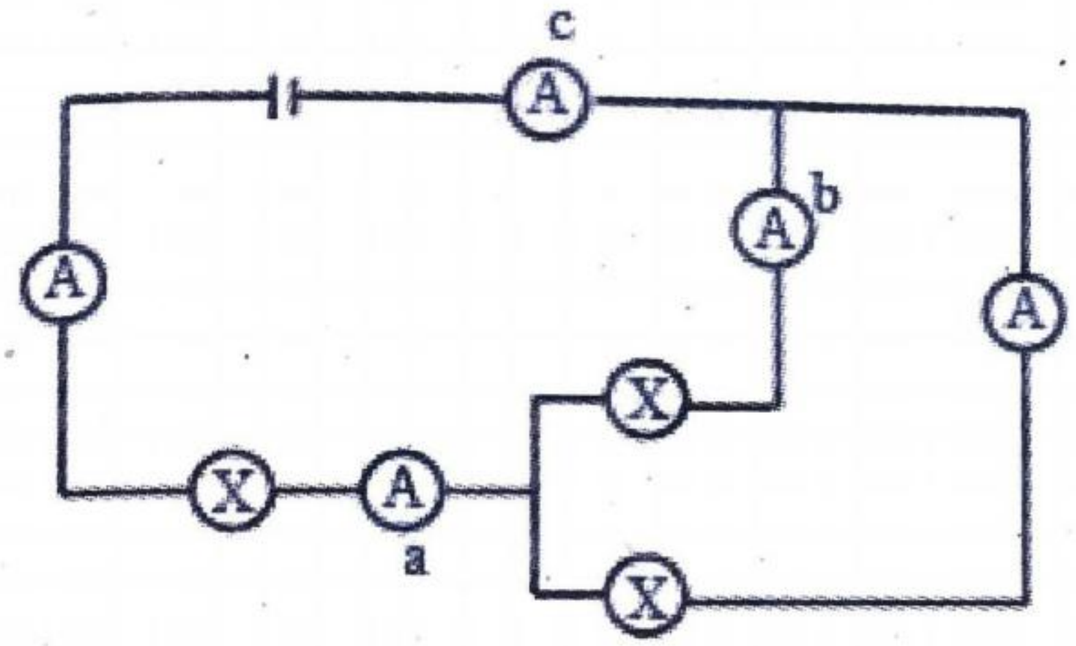


Ohm's law and electrical resistance

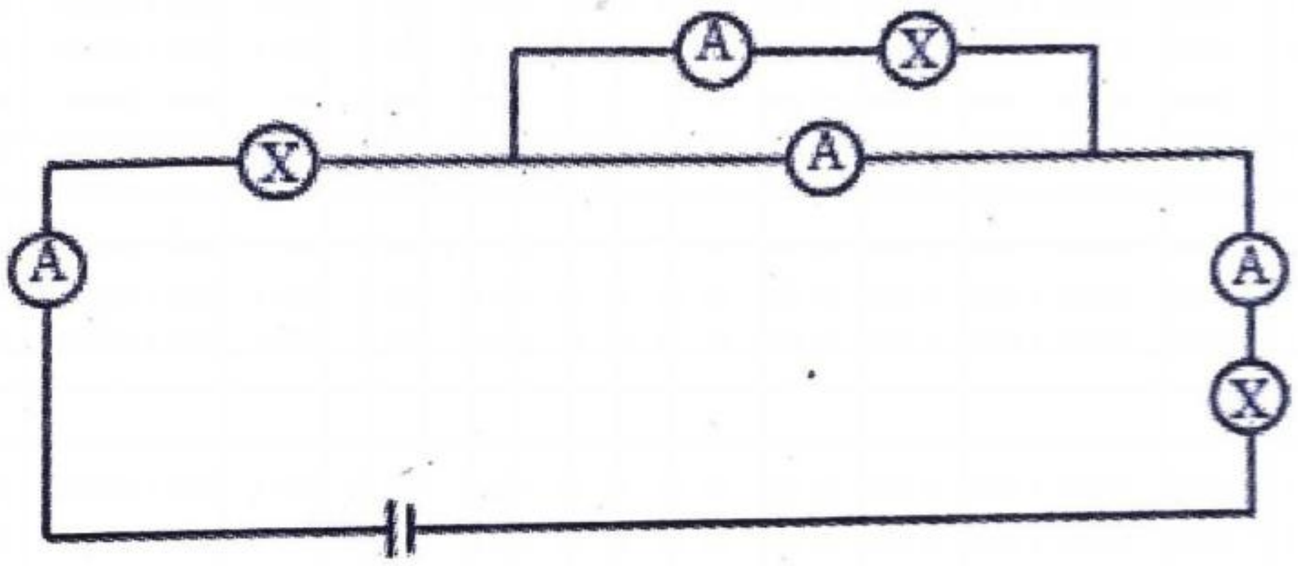
18. A battery of emf 6V is connected to a circuit with a total resistance of 12Ω. What current would you expect
19. The resistivity of iron is $10^{-7}\Omega$? Find the resistance of a 12km length of a railway line with across sectional area of $200cm^2$.
20. A tungesten resistor ($\alpha = 5.8 \times 10^{-3}K^{-1}$) with a value of 10Ω at 0°C is warmed form that temperature up to 150°C work out its resistance when hot.

Combination of resistors

21. What should ammeters (a), (b) and (c) read, and which bulb is the brightest?

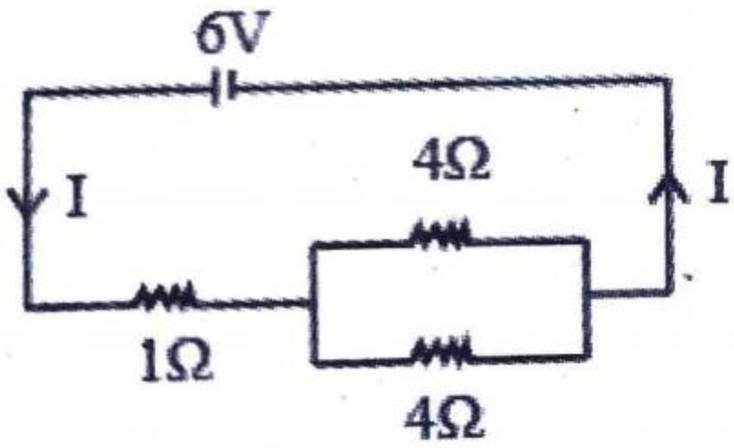


22. Give the missing ammeter readings C and d. Assume all the bulbs are identical ones



23. For the circuit shown below, what is

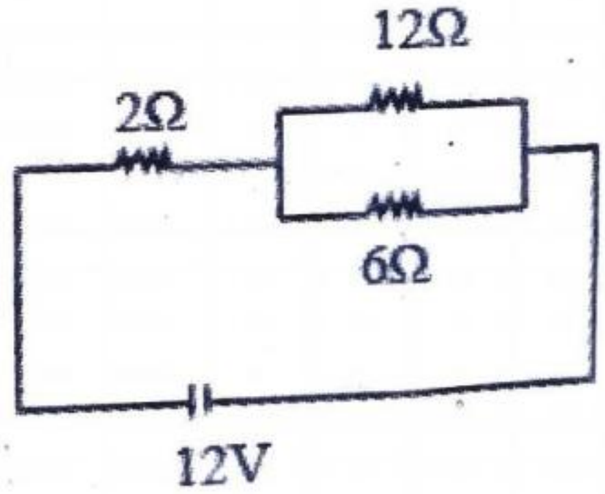
a) The effective resistance
b) The current



24. Calculate the effective resistance of a 25Ω and a 100Ω resistors connected in parallel. What current will be drawn from a 50V supply connected to the circuit

25. What is the

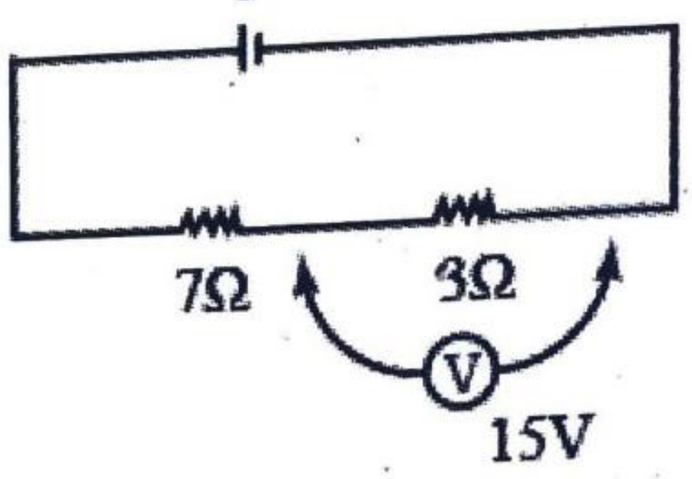
a) current through the 6Ω resistor
b) potential difference across the 2Ω



e.m.f and interval resistance

26. A battery has an emf of 3V and an interval resistance of 1Ω. What current will it give if it is connected to a circuit of 4Ω resistor.
    a) How much charge will it hold;
    b) How much energy is stored on it;

27. a) What current is flowing through the 3Ω resistor
    b) If its internal resistance is 2Ω, what is the emf of the battery.



Unit Eleven: Explanation to Questions

1. Yes, it is electric charge
2. Secondary cell can be charged but not primary cell
3. It is due to a potential difference between its ends
4. Yes, a battery is a combination of two or more cells
5. The unit of resistivity is Ωm and not Ω/m
6. The resistance also doubles
7. In series, current is constant, hence the current in the ammeter is the same as the current in the circuit. And in parallel, voltage is constant, hence the voltage across the resistor is the same as the voltage across the voltmeter.
8. Yes, electric meter can measure it in kwh
9. It indicates the charge
10. As temperature increases, resistance increases which causes the power to decrease, because $P = V^2/R$
11. Wire A
12. Yes, when they are connected in parallel, the total resistance is less than each of the individual resistances

13. Assuming the same voltage supply V for both, then $P_A = \frac{V^2}{R_A}$ and $P_B = \frac{V^2}{R_B}$, But $P_A = 2P_B$, now taking ratio

$$\frac{P_A}{P_B} = \frac{V^2/R_A}{V^2/R_B} = \frac{R_B}{R_A} \Rightarrow \frac{2P_B}{P_B} = \frac{R_B}{R_A} \Rightarrow R_A = \frac{1}{2}R_B$$

14. The smaller and the larger cell, do have the same efficiency

Unit Eleven: Solution to problems

15. $I = \frac{Q}{t} \Rightarrow Q = It = (3)(2 \times 60) = 360C$

16. $I = \frac{Q}{t} \Rightarrow t = \frac{Q}{I} = \frac{6}{30} = 0.2 \text{ sec}$

17. a) $100°C = 20mA$ b) $100°C = 20mA$ c) $100°C = 20mA$

$T°C = 6mA$ $T_B = 22mA$ $T_C = -2mA$

Criss-crossing, $T = 30°C$ $T_B = 110°C$ $T_C = -10°C$

18. $V = IR \Rightarrow I = V/R = 6/12 = 0.5A.$

19. $R = \rho\ell/A = (10^{-7})\left(\frac{12 \times 1000}{200 \times 10^{-4}}\right) = 6 \times 10^{-2}\ \Omega.$

20. $\Delta R = R_o \alpha \Delta T = (10)(5.8 \times 10^{-3})\ (150) = 8.7\Omega$

$R_T = R_o + \Delta R = 10 + 8.7 = 18.7\Omega$

21. $a = 0.7A$, $b = 0.35A$ $c = 0.7A$

22. $c = 2.5A$, $d = 5A$

23. a) the two 4Ω resistors are in parallel, $\frac{1}{R^1} = \frac{1}{4} + \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$

$R^1 = 2\Omega.\ R_T = 1 + 2 = 3\Omega$

b) $I = V/R = 6/3 = 2A.$

24. $\frac{1}{R}=\frac{1}{25}+\frac{1}{100}=\frac{5}{100}\Rightarrow R=20\Omega$. and $I=\frac{V}{R}=\frac{50}{20}=2.5A$

25. The total resistance is $\frac{1}{R^1}=\frac{1}{6}+\frac{1}{12}=\frac{3}{12}\Rightarrow R^1=4\Omega$

and $R_T=2+R^1=2+4=6\Omega$

a) $I=\frac{V}{R}=\frac{12}{6}=2A.$

b) $V_2=(I_2)(R_2)=(2)(2)=4V$

26. $\varepsilon=I(r+R)\Rightarrow 3=I(1+4)\Rightarrow I=\frac{3}{5}=0.6A$

27. a). $I=\frac{V}{R}=\frac{15}{3}=5A.$

b) $\varepsilon=I(r+R)=5(2+(7+3))=60V$

Unit Twelve

Electromagnetism

The operation of electric motors, generators, transformers, video and audio tape recorders, TV, radio, computer disks, compass, electric bell and electric meters such as Ammeter, voltmeter, etc would be impossible without magnetism. Earth's magnetism helps us find our way round, and protect us from harmful radiation. Birds, sea turtles, and some bacteria use earth's magnetism for navigation. Without magnetism we would not even see, for light itself originates in an interaction between magnetism and electricity.

12.1 Magnetism

The first magnetic phenomena is observed from a natural magnet, called ***lodestone***, it was found near the ancient city of *magnesia*, and hence the name *magnet*.

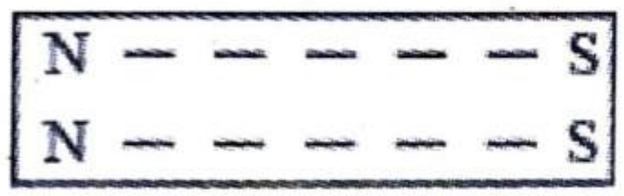
Iron, cobalt, nickel, steel and alloys of these materials are strongly attracted by a magnet and are called ***Ferromagnetic***, which means they can be made to act as a magnet.

The atoms in a magnetic material are grouped into microscopic magnetic regions called *domains*.

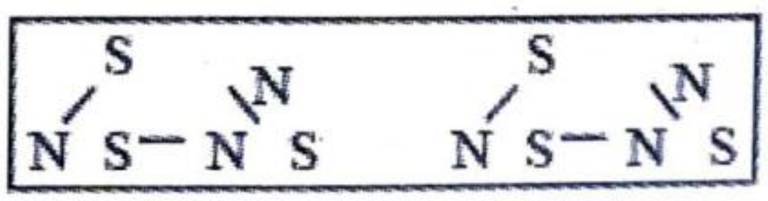
Magnetic Domains: Are group of atoms that act as tiny magnets in a magnetic material.

When the domains are oriented, they form poles and the material become magnet.

When the domains point in random direction, the material loses its magnetism.



magnetised



unmagnetised

Magnetic Shielding

Sometimes you may not want a piece of equipment to become magnetised as it could be damaged as a result. A magnetic shield stops the equipment from being affected by a magnet.

> **Magnetic Shielding:** Is the process of limiting the penetration of magnetic fields using a barrier of conductive material such as *metalsheet*, *metal foam* and *plasma* (ionized gas).

For example, a compass placed between two magnets doesn't show the actual direction since it is prevented form deflection by the magnets.

But, when a tin can cover is placed over the compass, the effect of the magnets is prevented and hence the compass can deflect.

Conceptual Examples

1. Why do temporary magnets lose their magnetic property after a while? Explain

   Explanation: Because, in the case of temporary magnet, after the external field is removed, the magnetic domains return to their random arrangement.

2. Why do permanent magnets called permanent magnet are alloyed with other substances that keep the domains well arranged.

   Explanation: Because the domains of such magnets remain ordered after the removal of the external field. The domains remained aligned because, the iron, cobalt or nickel that made the permanent magnet are alloyed with other substances that keep the domains well arranged.

3. What is the possible cause for the magnetic effect of a permanent magnet? Explain

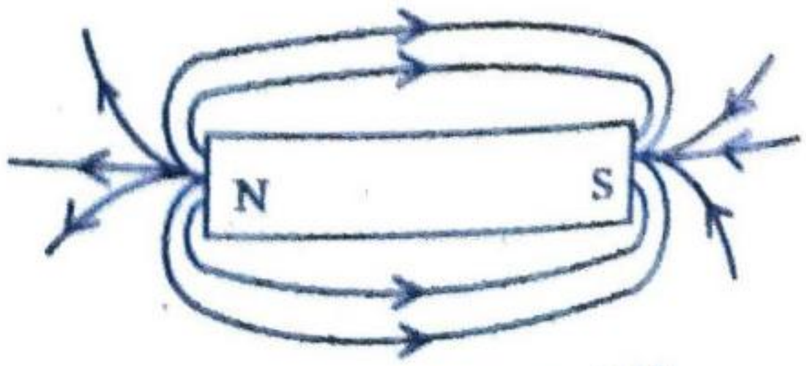
   Explanation: The magnetic effect of a permanent magnet results from the spinning of electrons on their axis as they move in atoms.

12.2 Concepts of magnetic field

Magnetic field represents the effect a magnet has on its surrounding.

> **Magnetic field:** Is a region around a magnet where a magnetic force may be felt.

The imaginary lines that are used to visualize the magnetic field are called magnetic field lines.

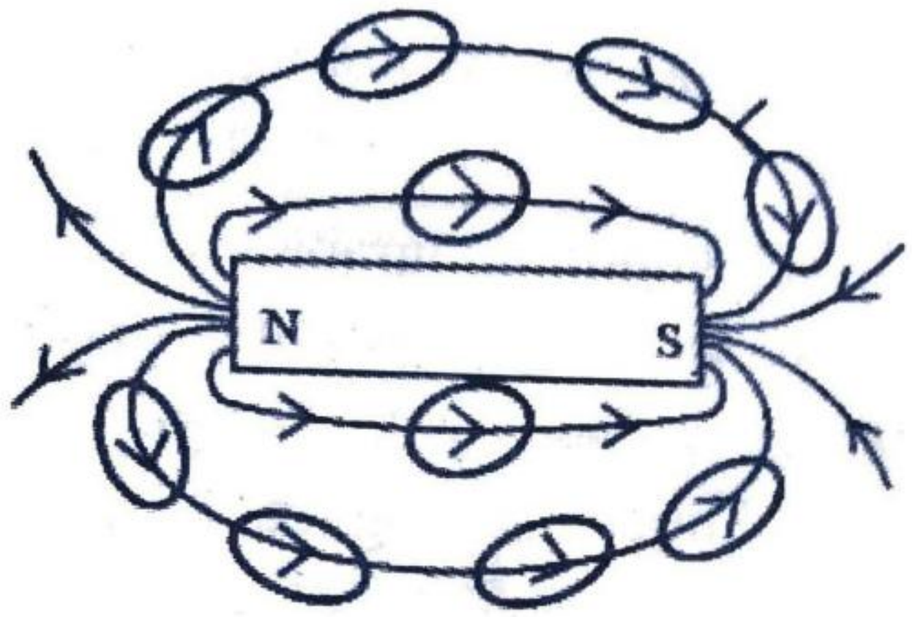


Direction of mangeic field

☞ The magnetic field lines originate from the North pole and terminate at the South pole.

The magnetic field of a bar magnet can be traced with the aid of a compass.

☞ The direction of the magnetic field at a given location is the direction in which the north pole (→) of a compass points when placed at that location.



small compass can be used to trace the magnetic field lines of a bar magnet

Conceptual Examples

4. Is it possible to separate poles of a magnet by cutting the magnet into pieces? Explain.

   Explanation: North and south poles always exist together. They can not be isolated by cutting the magnet in to pieces.

5. If the field lines are observed to be crowded together at some region, what can you say about the nature of the magnetic field at that location?

   Explanation: The magnetic field (force) is strong in the region where the field lines are closer together.

The force between magnets

Two magnets may attract or repel each other when brought closer together. This is explained by the basic law of magnetism.

> **Law of magnetism,** states that
> "like poles repel each other and unlike poles attract each other".

That is, two North poles or two South poles repel each other while a north pole and a south pole attract each other.

Note: The word "pole" is to mean ends of a magnet.

Earth's Magnetic Field

When a bar magnet is suspended from a string, it come to rest in a north – south direction. Similarly a compass needle points north – south from these observations we can conclude that the earth is acting like a magnet, since it interact with the magnet or the compass and that earth's magnetic field is like that of a giant bar magnet.

Since the north pole of a compass needle points toward the north magnetic pole of the earth, and since opposites attract, if follows that.

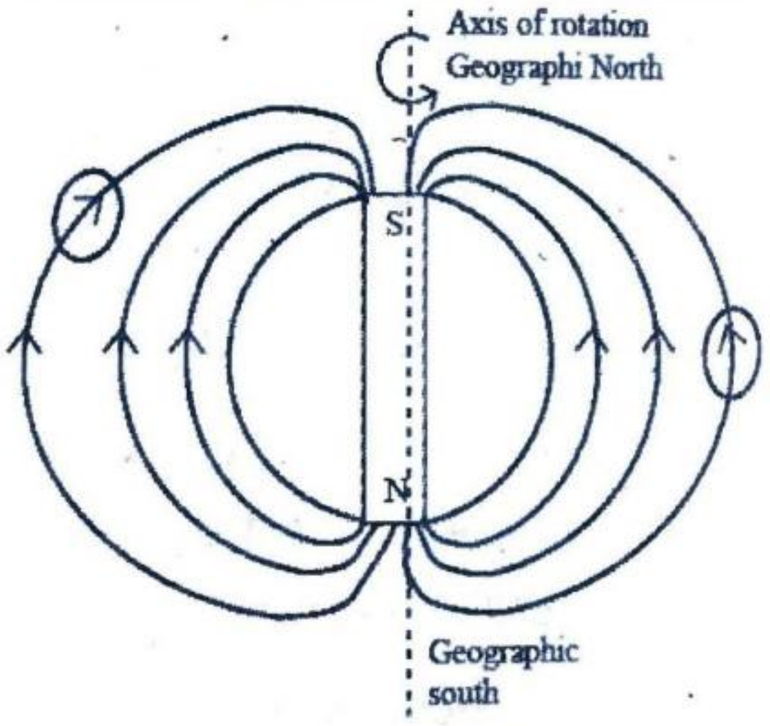
> The north geographic pole of the Earth is actually near the south pole of the Earth's magnetic field.

Although the Earth's magnetic field is similar in many respects to that of a huge bar magnet, recent theories indicate that there is no such permanent magnet inside the earth. However, according to the **dynamotheory** of magnetism;

> The flowing currents of molten materials in the Earth's core are the primary causes of earth's magnetic field.

The Earth's magnetic field is similar to that of a gaint bar magnet tilted slightly from the rotational axis.

Notice that the north geographic pole is actually near the south pole of the Earth's magnetic field.



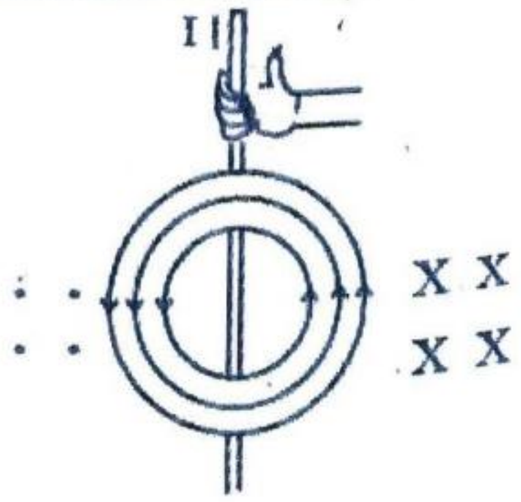
magentic field
of the Earth

Magnetic field lines around a current carrying wire

Magnets are not the only sources of magnetic field. An electric current also gives rise to a magnetic field. The direction of the magnetic field produced due to a current carrying wire is determines using the *right hand rule*.

> **Right Hand Rule:** If you grasp the wire with your right hand in such a way that your thumb is pointing in the direction of the current, then your fingers will curl in the direction of the magnetic field.

For example, if we have a wire carrying current upwards, then the right hand rule can be applied



as shown to determine the direction of the magnetic field.

Where, the dot (.) indicates the magnetic field is *out of* the page, and

the cross (×) indicates the magnetic field is *into* the page.

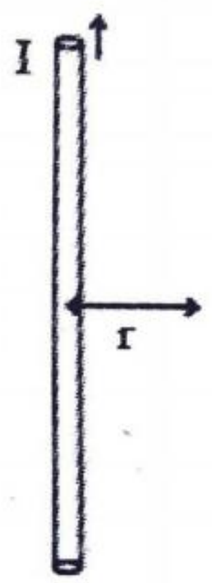
☞ **Note that;** magnetic field lines around a wire are represented as a series of concentric circles.

Magnetic field strength at a point due to a current carrying wire.

The magnetic field strength (B) at a point a distance (r) from the current carrying wire is given by

$$\boxed{B = \frac{\mu_o I}{2\pi r}}, \text{ where } \mu_o = 4\pi \times 10^{-7}\ \text{Tm/A}, \text{ and is}$$

called *permeability of free space.*



The SI unit of magnetic field strength (B) is Tesla (T)

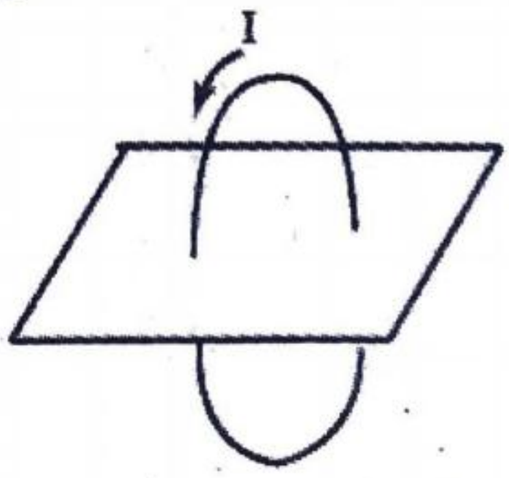
Illustrative Examples

6. Determine the magnetic field strength at a point 5cm from a wire carrying a current of 10A.

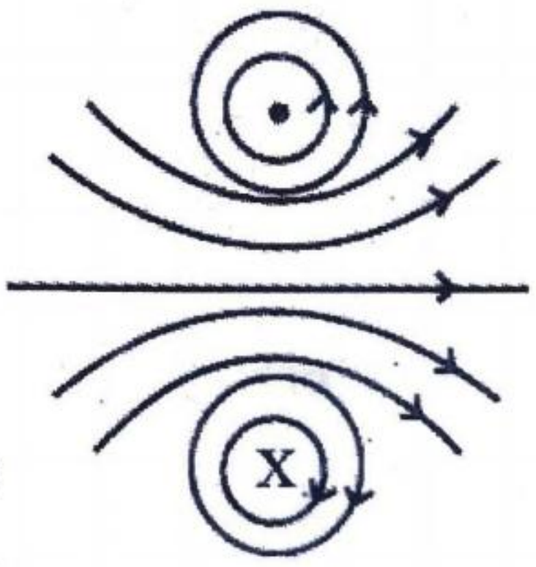
Solution: $B = \frac{\mu_o I}{2\pi r} = \frac{\left(4\pi \times 10^{-7}\right)(10)}{(2\pi)\left(5 \times 10^{-2}\right)} = 4 \times 10^{-5}\ \text{T}$

Magnetic field of a circular loop

Consider a circular loop carrying current as shown. Then the direction of the magnetic field due to the loop is again determined using the **right hand rule**



current carrying loop



Its magnetic field

Here, dot (.) indicates current out of the page, and

(×) indicates current into the page.

The magnetic field strength at the center of a circular loop

The magnetic field strength at the center of a current loop of N turns, radius R and carrying a current I is obtained by:

$$B = \frac{N\mu_o I}{2R}$$

Illustrative Examples

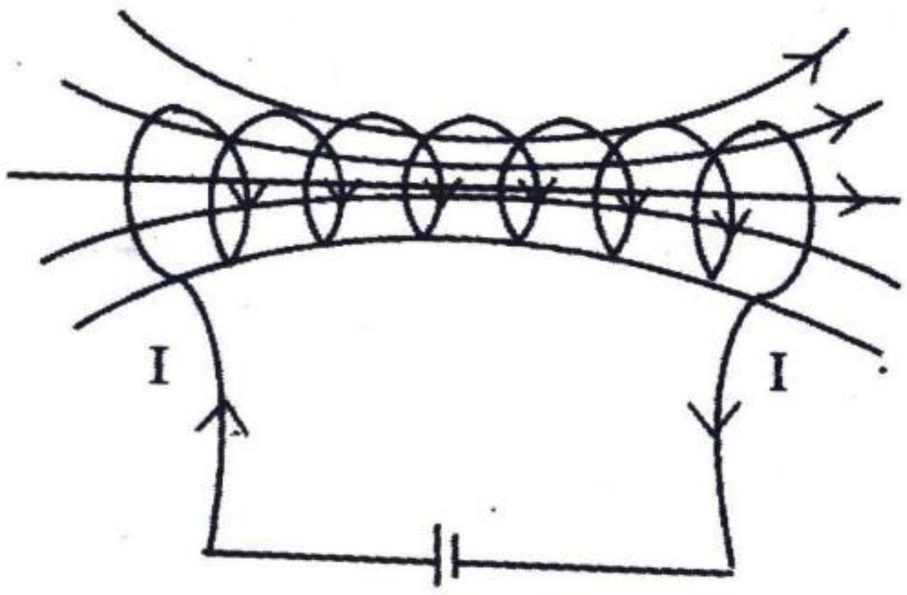
7. A circular coil of wire 0.05m in radius, with 30 turns carries a current of 5A .what is the magnetic field strength at its center?

Solution: $B = \frac{N\mu_o I}{2R} = \frac{(30)(4\pi \times 10^{-7})(5)}{2(0.05)} = 1.9 \times 10^{-3} T$

Magnetic field of a solenoid

Solenoid: Is a coil of many turns carrying current.

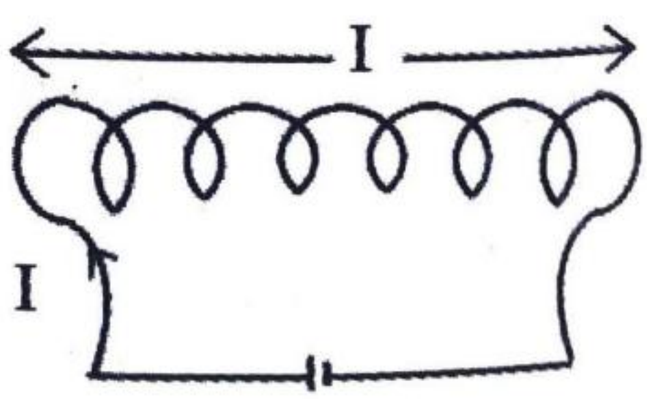
The direction of the magnetic lines of a solenoid can be found by applying the right hand rule. That is, if you grasp part of the solenoid with your right in such a way that the fingers encircling in the direction of the current, then the thumb points in the direction of the North pole of the solenoid.



Current direction Finger curl in the current direction

Magnetic field strength in side a solenoid

The magnetic field strength in side a solenoid length ℓ and N turns carrying a curren given by



$$\boxed{B = \mu_o \frac{N}{\ell} I = \mu_o nI}$$

Where $n = \frac{N}{\ell}$, number of turns per length

Illustrative Examples

8. A solenoid that is 75cm long and 92,000 turns carries a current of 8.4A. What is the magnetic field produced inside the solenoid?

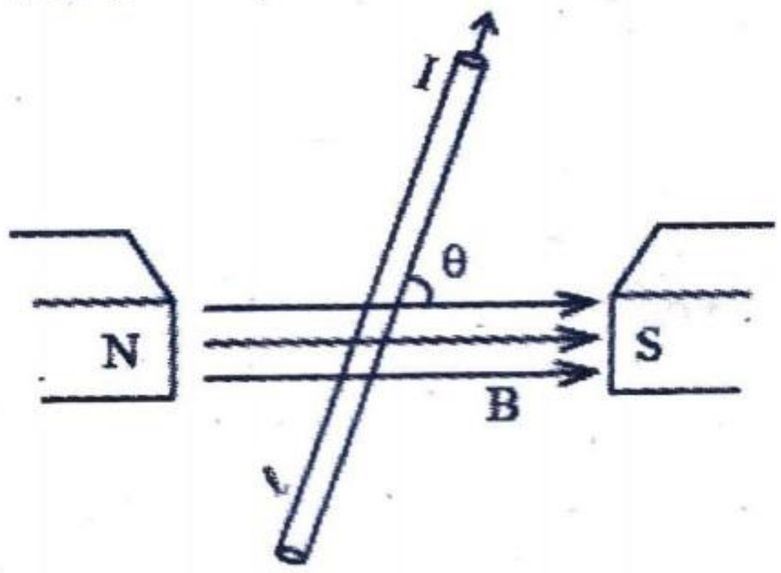
Solution: $B = \mu_o \frac{N}{\ell} I = \left(4\pi \times 10^{-7}\right) \frac{(92{,}000)}{(0.75)} (8.4) = 1.3\text{T}$

☞ **Remember;** in all the three cases, the straight wire, the circular loop and the solenoid, the field lines are closed loops.

12.3 Magnetic forces on a current carrying wire

The magnetic force on a wire of length ℓ carrying current I and placed at an angle θ to an external magnetic field B is given by

$$\boxed{F = BI\ell \sin\theta}$$



☞ **Note that**

→ the force is maximum when the current and the magnetic field are perpendicular (at right angle $\theta = 90°$) to each other, $F = BI\ell$

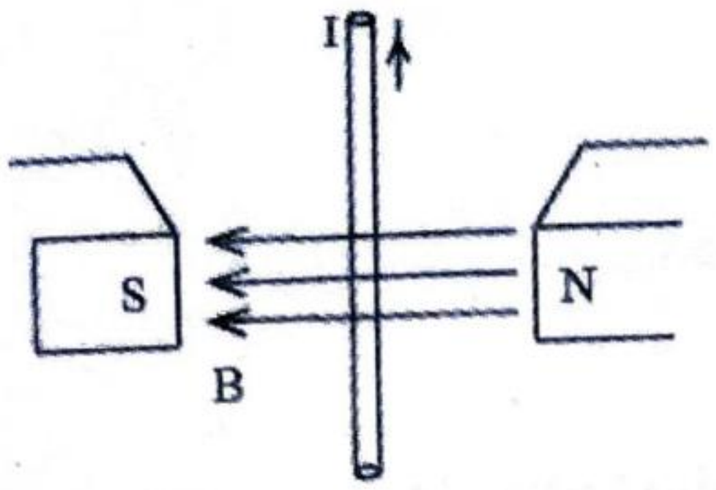
→ the force is minimum ($F = 0$) when the current and the magnetic field are parallel or antiparallel ($\theta = 0$ or $\theta = 180°$) to each other

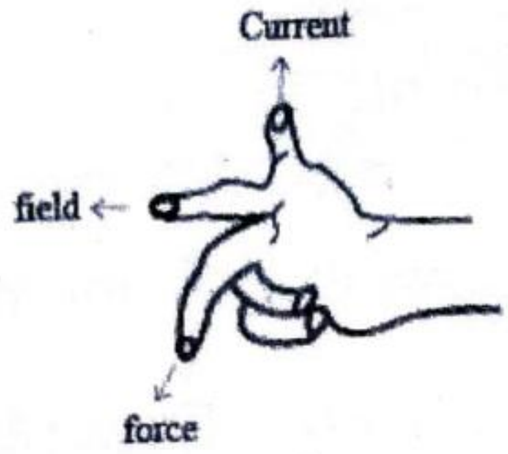
Direction of the magnetic force on a current carrying wire

The direction of the magnetic force on a current carrying wire is determined using the *right hand rule*.

Right Hand Rule: If you point the first finger of your right hand in the direction of the magnetic field and the thumb in the direction of the current, then the second finger will point in the direction of the magnetic force.

That is, for the arrangements shown,



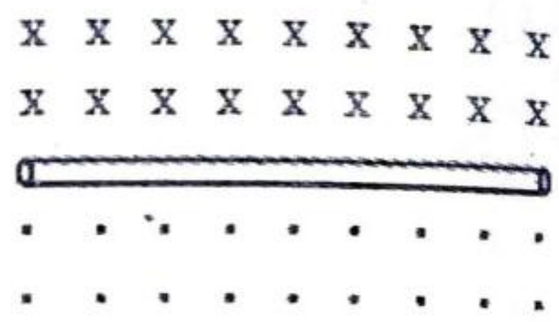


The right hand rule
(The first finger, the second finger and the thumb are mutually perpendicular to each other)

Since the current is upward (thumb direction), the field to the left (first finger direction), then the force is out of the page (second finger direction).

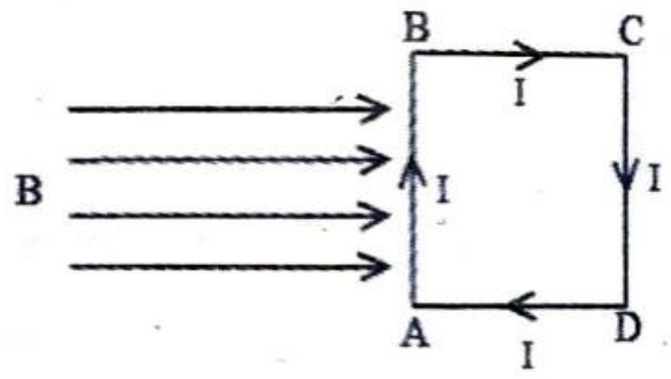
Conceptual Examples

9. The magnetic field due to the horizontal current carrying wire is shown. Does the current in the wire flow to the left or to the right?



Explanation:If you point the thumb of your right hand along the wire to the left, your fingers curl into the page above the wire. Thus, the current flows to the left.

10. Determine the magnetic force on each segment of a square of length ℓ and current I.



Explanation: Using the relation, $F = BI\ell\sin\theta$ and the right hand rule, we have,

$\rightarrow$ for segment AB, $\theta = 90° \Rightarrow F = BI\ell$, into the page

$\rightarrow$ for segment BC, $\theta = 0° \Rightarrow F = 0$

$\rightarrow$ for segment CD, $\theta = 90° \Rightarrow F = BI\ell$, out of the page

$\rightarrow$ for segment DA, $\theta = 180° \Rightarrow F = 0$

$\rightarrow$

Illustrative Examples

11. A current carrying coductor of legnth 40cm and current 8A is placed in a magnetic field of 5T at an angle of 53°. Calculte the force on the wire.

Solution: $F = BI\ell\sin\theta = (5)\,(8)\,(40 \times 10^{-2})(\sin 53)°$, $\sin 53° = 0.8$

$F = 1.28N$

12. A magentic field produces a force of 1N on a wire of 25cm long carrying a current of 5A perpendicular to the field. What is the magnitude of the magnetic field?

Solution: $F = BI\ell \Rightarrow B = \frac{F}{I\ell} = \frac{1}{(5)(25\times10^{-2})} = 0.8T$

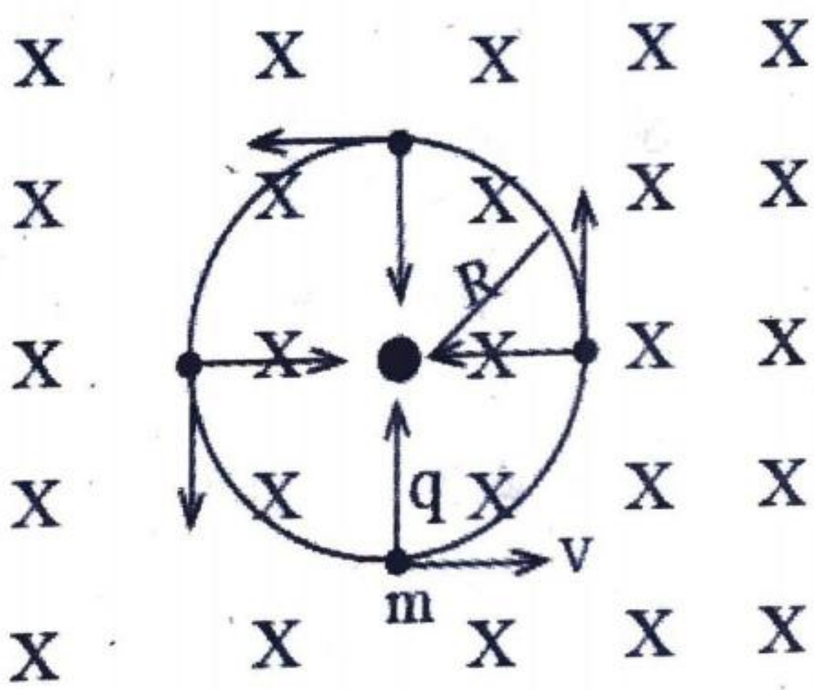
Magnetic force on a charged particle in Magnetic field

A single charged particle carrying charge (q) and moving at a velocity (V) perpendicular to an external magnetic field (B), experiences a force given by.

$$\boxed{F = BqV}$$

If a charged particle moves perpendicular to a uniform magnetic field it experiences a force of constant magnitude, that always points towards the same spot. This force is what we call *centripetal force*.

Now, consider a positive charged particle of mass (m) and charge (q) moving at a velocity (V) in a uniform magnetic field (B) pointing into the page. The particle then follows a circular path of radius (R), which you can check using the right hand rule with the thumb of your right hand pointing in the direction of the velocity of the charged particle. So we can have



Centripetal force = magnetic force

$$\frac{mV^2}{R} = BqV$$

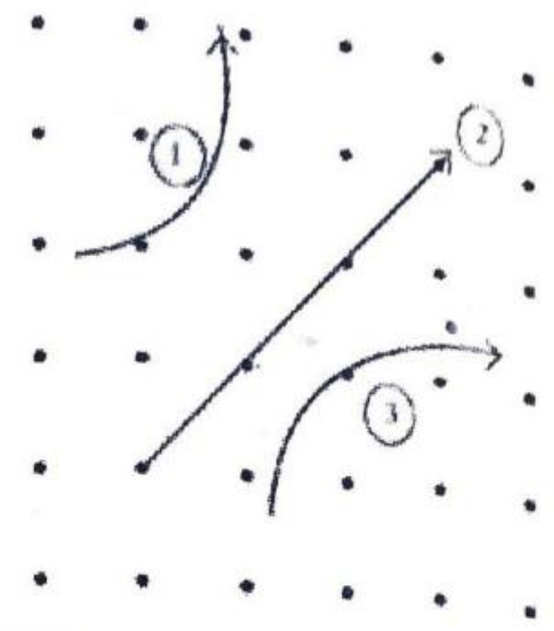
$$\Rightarrow \boxed{\frac{mV}{R} = Bq}$$

Conceptual Examples

13. Does a constant magnetic field do work on a charged particle? Explain.

Explanation: No. Because, the force exerted by a constant magnetic field on a moving charged particle is perpendicular to the the velocity of the charged particle as a result, the magnetic force does no work on the particle.

14. Three particles travel through a region of space where the magnetic field is out of the page. For each of the three particles, state whether the particle's charge is positive, negative or zero.



Explanation: Using the RHR, particle 3 must have a positive charge, particle 1 must be negatively charged and since particle 2 is undeflected its charge must be zero.

Illustrative Examples

15. An electron moving perpendicular to a magnetic field of 4.6×10^{-3}T follows a circular path of radius 2.8mm. What is the electron's speed? ($m_e = 9.11 \times 10^{-31}$kg).

Solution:

$$\frac{mV}{R} = Bq \Rightarrow V = \frac{BqR}{m} = \frac{\left(4.6\times10^{-3}\right)\left(1.6\times10^{-19}\right)\left(2.3\times10^{-3}\right)}{9.11\times10^{-31}}$$

$$V = 2.26\times10^{6}\,m/s$$

16. A prtoton traveling at a speed of 5×10^{5}m/s enters a magnetic field of 0.01T. If the velocity of the proton is perpendicular to the field, what is the radius of the circular path that the proton will move in? ($m_p = 1.67 \times 10^{-27}$kg).

Solution: $$\frac{mV}{R} = Bq \Rightarrow R = \frac{mV}{Bq} = \frac{\left(1.67\times10^{27}\right)\left(5\times10^{5}\right)}{\left(0.01\right)\left(6.1\times10^{-19}\right)}$$

$R = 0.52$m.

17. A proton moves at a speed of 10^{6}m/s perpendicular to a magnetic field in a circular path of radius 4cm. what is the magnitude of the magnetic field?

Solution: $\frac{mV}{R} = Bq \Rightarrow B = \frac{mV}{Rq} = \frac{\left(1.67\times10^{-27}\right)\left(10^{6}\right)}{\left(4\times10^{-2}\right)\left(1.6\times10^{-19}\right)} = 0.26T$

Conceptual Examples

18. Two charged particles move at right angles to a magnetic field and deflect in opposite directions. Can one conclude that the particles have opposite charges? Explain

Explanation:No. The particles may have the same sign charge but move in opposite directions along the same line.

19. An electron moves with constant velocity through a region of space. Can one conclude that the magnetic field is zero in this region? Explain

Explanation: No, if the electron moves in the same or opposite direction to the field, the magnetic force on it is zero and hence the electron moves with constant velocity.

20. An electron moving in the positive x direction at right angle to a magnetic field experiences a magnetic force in the positive y direction. What is the direction of the magnetic field?

Explanation: Using the right hand rule, the magnetic field is in the positive z direction

21. A charged particle moves in a circular path in the presence of a constant magnetic field applied perpendicular to the particle's velocity. Does the particle gain energy from the magnetic field? Explain

Explanation: No. Because, the magnetic force is always perpendicular to the displacement of the particle hence no work is done on the particle and hence no energy gain.

Important differences between electric and magnetic forces on charged particles.

- The electric force is always parallel or antiparallel to the direction of the electric field, whereas, the magnetic force is perpendicular to the magnetic field.
- The electric force acts on a charged particle independent of the particle's velocity, whereas, magnetic force acts on a charged particle only when the particle is in motion.

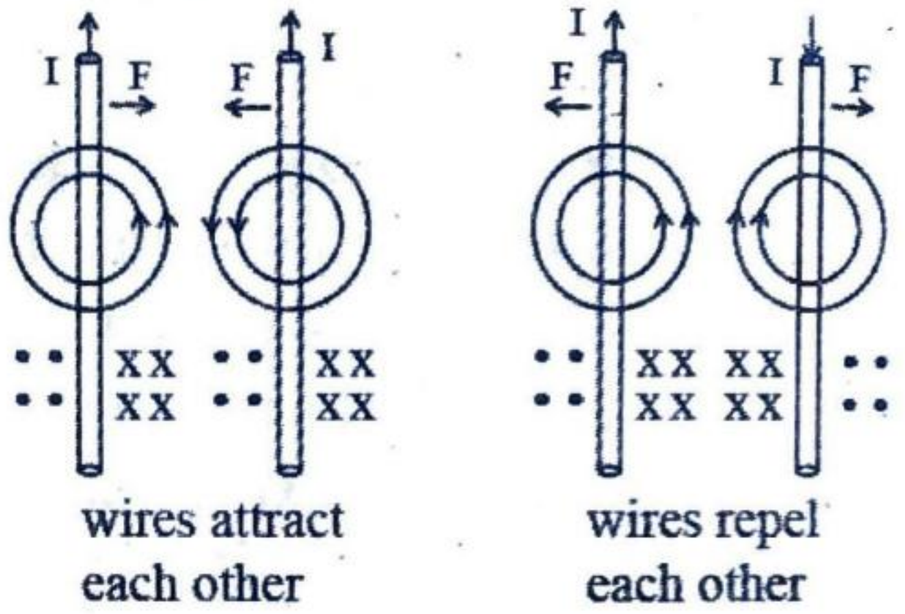
→ The electric force does work in displacing a charged particle, whereas the magnetic force associated with a steady magnetic field does no work when a charged particle is displaced.

Magnetic force between two parallel current carrying wires

Two parallel wires each carrying currents interact with each other. This is so because, each wire produced a magnetic field at the site of the other wire and both wires exert magnetic forces on each other. The forces can be either attractive or repulsive.

☞ If the currents in the two wires are in the *same direction*, the wires *attract* each other. On the other hand, if the wires carry currents in the *opposite* direction, they *repel* each other.

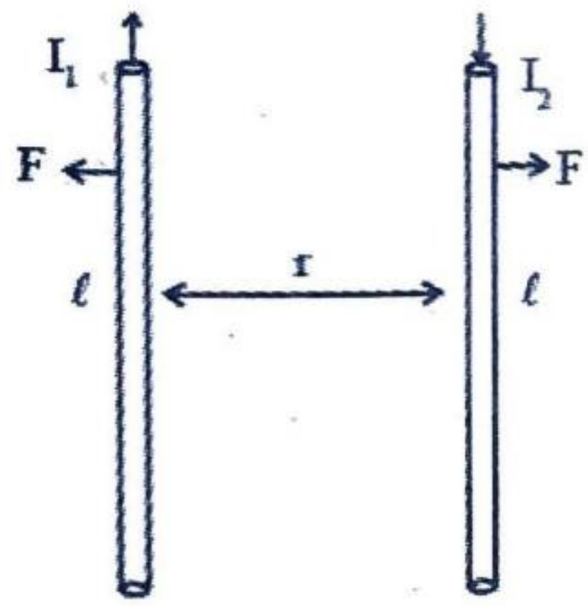
Applying the right hand rule for each wire, the magnetic field between the wires is opposite when the currents are in the same direction and it is the same when the currents are in opposite directions.



The magnitude of the force with which two wires that are a distance (r) apart, each of length (ℓ) and carrying currents of (I_1) and (I_2) is given by,

$$F = \frac{\mu_o I_1 I_2 \ell}{2\pi r}$$

Where, $\mu_o = 4\pi \times 10^{-7} N/A^2$



Conceptual Examples

22. Why do two parallel wires carrying currents in the same direction attract each other? Why do they repel each other when they carry currents in opposite directions? Explain

Explanation: Using the right hand rule, for each wire, the direction of the magnetic field between the wires is opposite when they carry currents in the same direction, hence the wires attract.

However, the direction of the field between the wires is the same when they carry currents in opposite direction, as a result the wires repel each other.

Illustrative Examples

23. Two main lines, each 250m length, run parallel to each other with a separation of 35cm. if the lines carry parrallel currents of 120A, what is the magntidue and direction of the force between them?

Solution: $F = \frac{\mu_o I_1 I_2 \ell}{2\pi r} = \frac{(4\pi \times 10^{-7})(120)(120)(250)}{(2\pi)(0.35)}$

$F = 2.1N$ attractive

Defining the Ampere

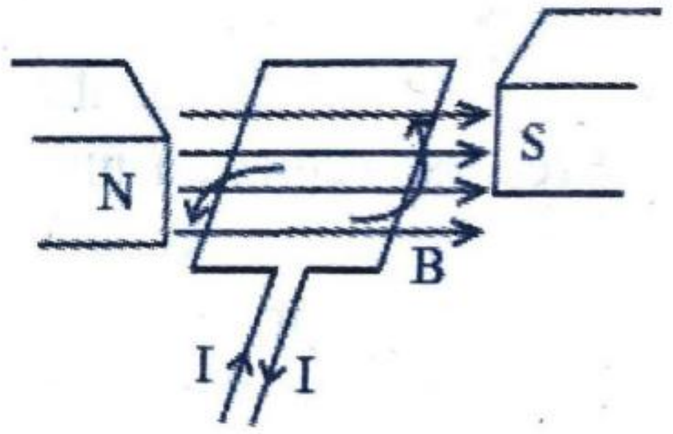
One Ampere: Is the amount of current when flowing through each of two parallel wires that are 1m apart in vacuum, they exert a force of $2 \times 10^{-7}N$ on each other on every meter length.

Torque on a rectangular current carrying wire.

The turning effect of a force is called *Torque* (τ). The torque on a rectangular loop of area A having N turns, carrying a current I and making an angle θ with the magnetic field is given by

$$\tau = NIABsin\theta$$

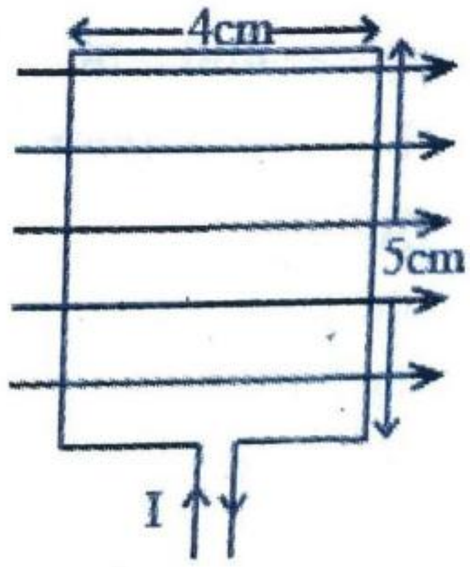
The maximum torque is attained when $\theta = 90° \Rightarrow \tau = NIAB$



A rectangular coil rotating in the field.

Illustrative Examples

24. A rectangular coil with 200 turns is 5 cm high and 4cm wide when the coil is placed in magnetic field of 0.35T, its maximum torque is 0.22 N.m. what is the current in the coil?



Solution:

$\Rightarrow \tau = NIAB \quad , \quad A = hw$

$$I = \frac{\tau}{NAB} = (5 \times 10^{-2})(4 \times 10^{-2})$$

$$= \frac{0.22}{(200)(2\times10^{-3})(0.35)} \quad = 2\times10^{-3}\,m^2$$

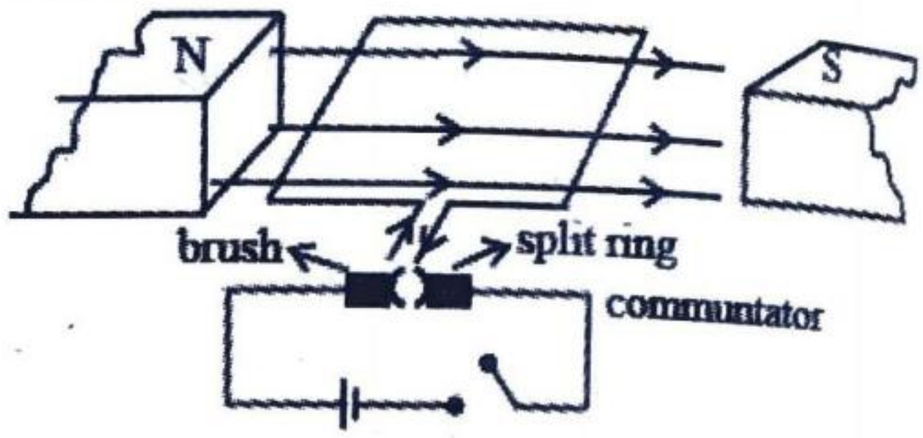
$$= 1.6A$$

The Electric Motor

Motor: Is a device that converts electrical energy into mechanical energy.

The basic principle behind the operation of an electric motor is rotation of a current carrying loop in a magnetic field.

The main components of a typical electric motor are magnet, rotating coil (armature), split ring commutator, brush, source of electric energy and switch.



An electric Motor

☞ **Remember,** A simple loop of wire in a magnetic field will not rotate more than 180°. For the loop to rotate 360° in the field, the current flowing in the loop must reverse direction just as the loop reaches the vertical position. For this purpose, a *split ring commutator* is used. The split ring commutator conducts current into the loop by rubbing against the brushes. As the current is reversed, the force direction also changes and makes the loop to spin in one direction through out.

Conceptual Example

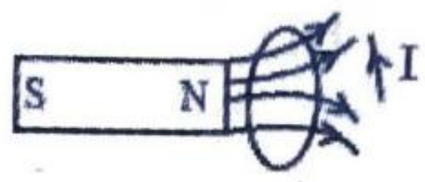
25. In an electric motor, a current carrying rectangular coil is placed in a uniform magnetic field. In which orientation, the coil will not tend to rotate?

Explanation: $\tau = NIAB\sin\theta$, No rotation means $\tau = 0$, it is so when the magnetic field is parallel to the area in this case $\theta = 0°$, and $\sin0° = 0 \Rightarrow \tau = 0 \Rightarrow$ No rotation.

12.4 Electromagnetic Induction

A changing magnetic field across a conductor can produce an electric current just like a batter. Effects of this nature have led to a host of applications in everyday life, from electric motors and generators to microphones, speakers and computer disk drives.

Electromagnetic Induction: is the process of generating a current by the relative motion between a wire and a magnetic field.



The electromagnetic induction is also known as the *dynamo effect*.

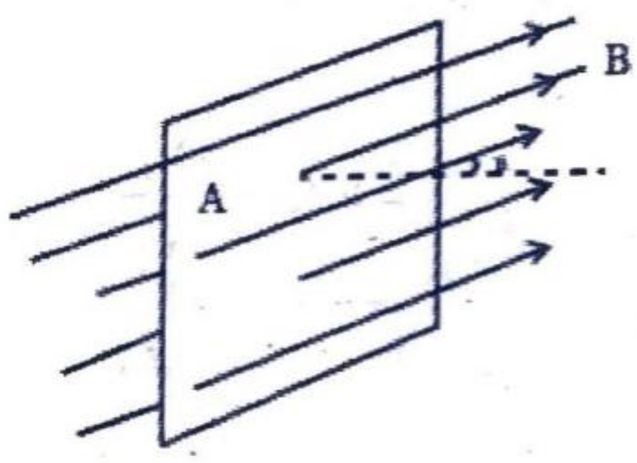
The changing magnetic field across a conductor results in an induced electromotive force that acts in driving changes through the conductor.

Induced emf: Is the voltage produced by electromagnetic induction

The induced emf is produced only when there is a rate of change of magnetic flux..

Magnetic flux (ϕ): Is a measure of the number of field lines that cross a given area.

Suppose a magnetic field B crosses a surface area A at angle θ that is perpendicular (normal) to the plane of the area, then the magnetic flux is given by:



$$\phi = BA\cos\theta$$

The SI unit of magnetic flux is Weber (Wb), thus

$$1Wb = 1Tm^2$$

☞ **Note that,** when the magnetic field cross the area at right angle, the normal line and the magnetic field become parallel and hence, $\theta = 0°$ and the magnetic flux (ϕ) become maximum, $\phi = BA$

Illustrative Examples

26. Consider a circular loop with a 2.5cm radius in a constant magnetic filed of 0.625T. Find the magnetic flux through this loop when its normal makes an angle of

a) $0°$ b) $30°$

c) $60°$ d) $90°$ with the direction of the magnetic field.

Solution: $\phi = BA\cos\theta$, $A = \pi r^2$

a) $\phi = 0°$, $\phi = BA\cos\theta = BA\cos0° = (0.625)(\pi \times 0.025)^2\cos0°$

$\phi = 1.23\times 10^{-3}Wb$, $\cos0° = 1$.

b) $\phi = 30°$, $\phi = BA\cos\theta = (0.625)\pi\,(0.025)^2\cos30°$
$\phi = 1.06 \times 10^{-3}$Wb , $\cos30° = 0.867$.

c) $\phi = 60°$, $\phi = BA\cos\theta = (0.625)\pi\,(0.025)^2\cos60°$
$\phi = 6.14\times 10^{-4}$Wb , $\cos 60° = 0.5$.

d) $\phi = 90°$, $\phi = BA\cos\theta = (0.625)\pi\,(0.025)^2\cos90°$
$\phi = 0$, $\cos 90° = 0$.

Faraday's Law of Induction

The magnitude of the induced emf is determined using faraday's law.

Faraday's law, states that;

"The magnitude of the induced emf is proportional to the rate of change of flux."

$$\text{That is, induced emf} \simeq \frac{\text{change in flux}}{\text{change in time}}$$

$$\Rightarrow \varepsilon \simeq \frac{\Delta\phi}{\Delta t}$$

$$\boxed{\varepsilon = \frac{N\Delta\phi}{\Delta t}}$$

Where, N, is number of turns of the coil

$\Delta\phi = \phi_f - \phi_i = \Delta BA$

$\frac{\Delta\phi}{\Delta t}$, rate of change of flux

☞ The magnitude of the induced emf depends on the following factors.

The speed (V); as the relative movement between the magnet and the coil increases, the induced emf increases.

The magnetic filed (B): as the magnetic field strength increase the induced emf also increases.

The number of turns (N): as the number of turns increase the induced emf also increase.

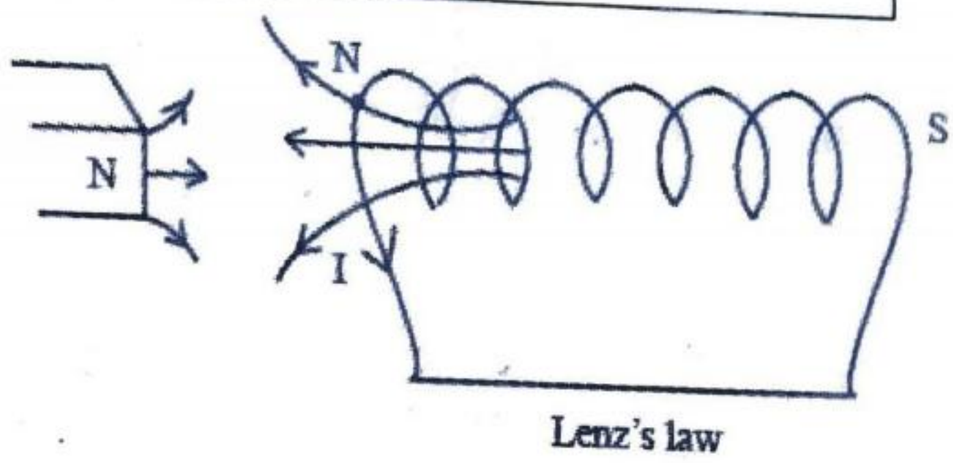
Lenz's Law

The direction of the induced emf or induced current is determined using *Lents law*.

Lent's Law: states that

"The direction of the induced current (emf) is such as to oppose the change that is causing it."

For example, as the north pole of the magnet comes towards a solenoid, according to Lenz's law, the induced current makes the top of the solenoid a North pole so, by the repulsion action, the solenoid opposes the approach of the magnet.



Lenz's law

☞ **Note that,** the "Change" that produce the induced current or the induced emf is the motion between the magnet and the coil.

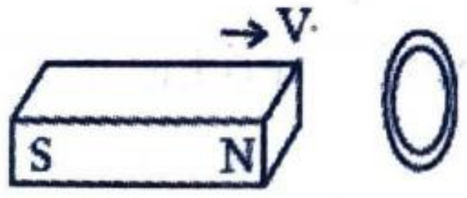
The induced emf in a coil is then generally given by:

$$\varepsilon = -\frac{N\Delta\phi}{\Delta t}$$

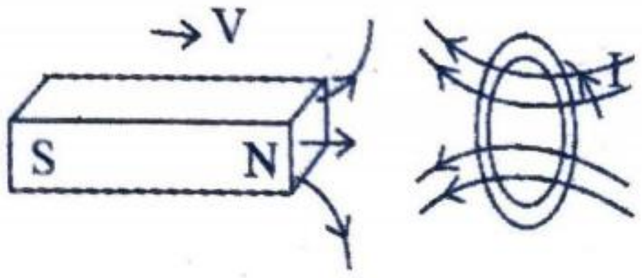
Where, the negative sign comes from Lenz's law and it indicates direction of opposition.

Conceptual Examples

27. A bar magnet is moved towards a conducting ring. What is the direction of the induced current in the ring?



Explanation: When the magnet is moved, towards the stationary conducting loop, a current is induced in the direction shown and produces a magnetic field directed to the left so as to oppose the approaching magnet.



28. Lenz's law is just another way of stating the law of conservation of energy. True or False

Answer: True.

Illustrative Examples

29. A coil of wire having an area of $10^{-3}m^2$ is placed in a flux density (magnetic field) of 1.5T. In a time interval of 0.001sec, the flux

density is reduced to 1T. If the coil consists of 50 turns of wire, what is the induced emf?

Solution: $\varepsilon = -\dfrac{N\Delta\phi}{\Delta t}$, $\Delta\phi = \phi_f - \phi_i$

$$\text{or}\Delta\phi = \Delta BA \ , \ \Delta B = 1.5 - 1 = 0.5T$$

$$\Delta\phi = (0.5)\,(10^{-3}) = 5 \times 10^{-4} wb$$

$$\varepsilon = \frac{50 \times 5 \times 10^{-4}}{0.001} = 25V$$

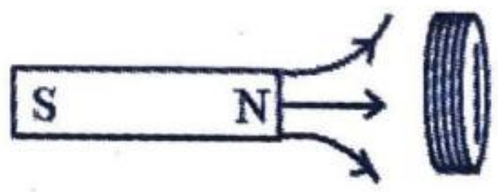
30. The induced emf in a single loop of wire has a magntiude of 1.48V when the magnetic flux is changed from $0.85Tm^2$ to $0.11Tm^2$. How much time is required for this to happen?

Solution: $\varepsilon = -\dfrac{N\Delta\phi}{\Delta t} \Rightarrow \Delta t = \left|\dfrac{N\Delta\phi}{\varepsilon}\right| = N\left|\dfrac{\phi_f - \phi_i}{\varepsilon}\right|$

$$\Delta t = (1)\left|\frac{0.11 - 0.85}{1.48}\right| = 0.5sec$$

31. A bar magent is moved rapidly toward a 40 turn, circular coil of wire. As the magnet moves, the magnetic field increases from 0.0125Tto 0.45T in 0.25sec. If the radius of the coil is 3.05cm, and the resistance of its wire is 3.55Ω. Find the magntidue of the

a) Induced emf
b) Induced current



Solution:

a) $\varepsilon = -\dfrac{N\Delta\phi}{\Delta t}$, $\Delta\phi = \Delta BA$, $A = \pi r^2$

$$\Delta\phi = \phi_f - \phi_i = B_f A - B_i A = (0.45)\pi(0.0305)^2 - (0.0125)\pi(0.0305)$$

$$= 1.32 \times 10^{-3} - 3.65 \times 10^{-5} = 0.0012835\ Wb$$

$$|\varepsilon| = \left|-\frac{N\Delta\phi}{\Delta t}\right| = -\frac{(40)(0.0012835)}{0.25} = 0.205V$$

b) $I = \frac{V}{R} = \frac{0.205}{3.55} = 0.0577A$

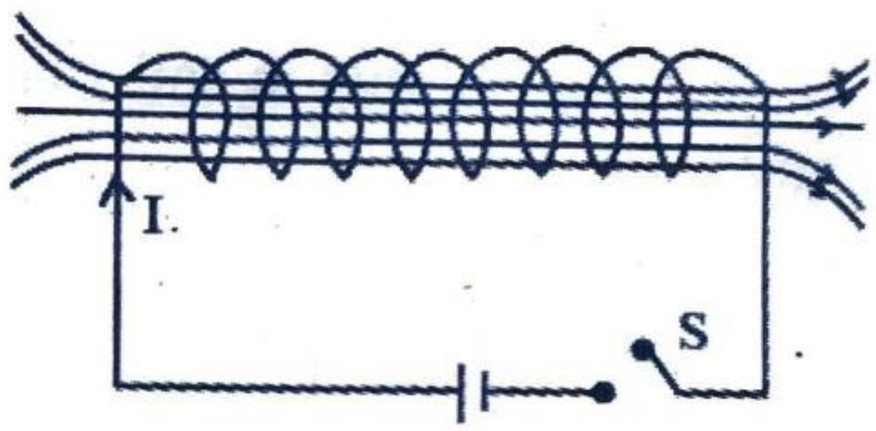
Inductance

Inductance: Is the property in an electric circuit where a change in the electric current through that circuit induces an emf that opposes the change in current.

Induction is of two types, *self induction* and *mutual induction*.

Self Induction: Is the process of producing an induced emf on a coil itself due to a change in its own current.

As the switch is closed, current starts flowing, it causes a change in magnetic field which in turn causes a change in flux. The change in flux again causes an induced emf to develop in the coil itself, which is given by;



$$\boxed{\varepsilon = -\frac{L\Delta I}{\Delta t}}$$

Where, L is called the *self inductance* of the coil and its unit is Henry (H), $\boxed{1H = \frac{1VS}{A}}$

Illustrative Examples

32. Find the induced emf when the current in a 45×10^{-5}H inductor increases from 0 to 515mA in 16.5×10^{-3}sec

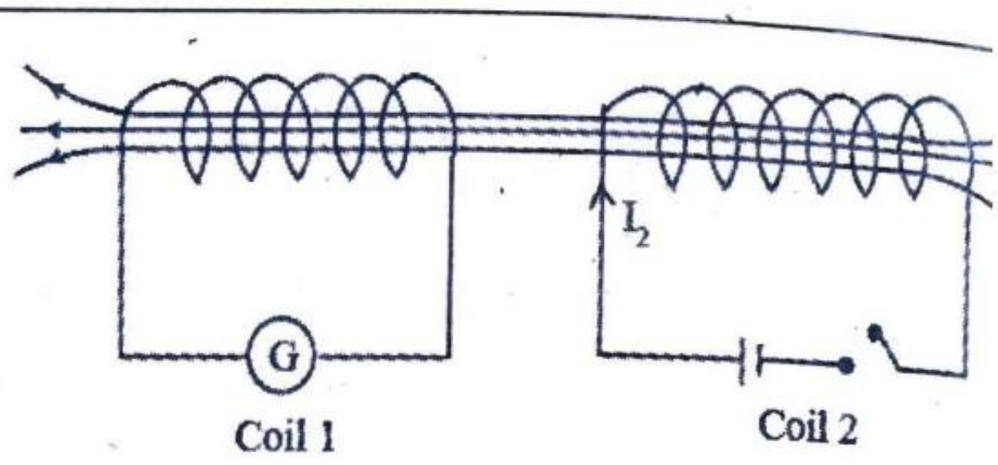
Solution: $E = -\frac{L\Delta I}{\Delta t} = -\frac{\left(-45\times10^{-3}\right)\left(515\times10^{-3}\right)}{16.5\times10^{-3}} = 1.4V$

33. When the current in a solenoid is increased form 0 to 2.5A in 0.35 sec the magnitude of the induced emf is 0.012V. what is the inductance of the solenoid?

$$\textbf{\textit{Solution}}: E = -\frac{L\Delta I}{\Delta t} \Rightarrow L = \left|\frac{E}{\Delta I/\Delta t}\right| = \frac{0.012}{\left(2.5/0.35\right)} = 1.7\times10^{-3}\text{H}$$

Mutual Induction: Is the process of producing an induced emf in one coil due to a changing current in another coil.

As the switch of coil 2 is closed, current passes through it and causes a changing magnetic field to cross coil 1. As a result coil 1 experiences a changing flux



This change in flux in coil 1 causes an induced emf to develop which is given by

$$\varepsilon = -\frac{M\Delta I_2}{\Delta t}$$

Where, M is called *mutual inductance* and its unit is Henry (H).

Illustrative Examples

34. Two coils of wire have a mutual inductance of 0.5H. How large average emf is induced on coil 1 as the current in coil 2 increases from 2A to 3A in 0.01 sec?

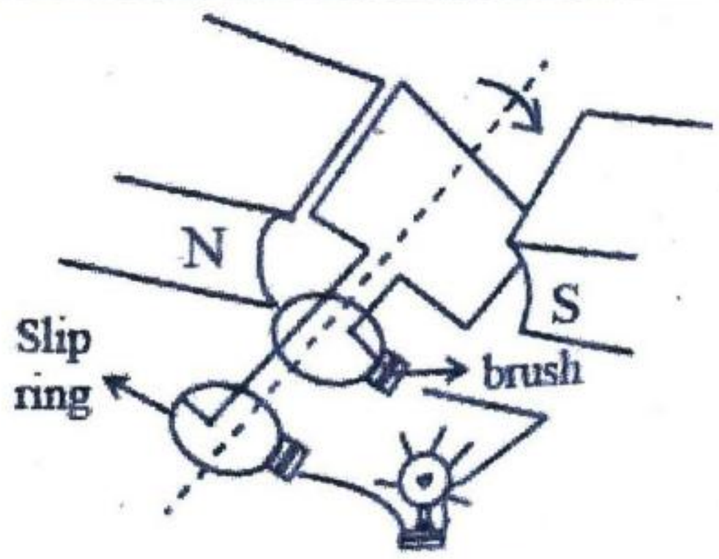
$$\textbf{\textit{Solution}}: |\varepsilon| = \left|-\frac{M\Delta I_2}{\Delta t}\right| = (0.5)\left(\frac{1}{0.01}\right) = 50\text{V}$$

AC Generator

Electromagnetic induction is the bases for the operation of an ac generator.

Generator: Is a device that changes mechanical energy into electrical energy.

A typical generator consists of coil (armature), slip rings, brush, external rotator and magnets. As the coil rotates, the magnetic flux through it changes continually. This changing magnetic flux



AC generator

induces an electromotive force in the coil which inturn causes current to flow in the coil.

The maximum emf induced in the generator is given by

$$\boxed{\varepsilon_{max} = NBA\omega}$$

Where, $\omega = 2\pi f$, which is angular speed of the coil.

Illustrative Examples

35. The armature of a simple ac generator consists of 100 turns of wire, each having an area of $0.2m^2$. If the armature is turned with frequency of 60 rev/s in a constant magnetic field of 10^{-3}T. what is the maximum emf generated?

Solution: $\varepsilon_{max} = NBA\omega$, but $\omega = 2\pi f = (2\pi)(60) = 377$m/s.

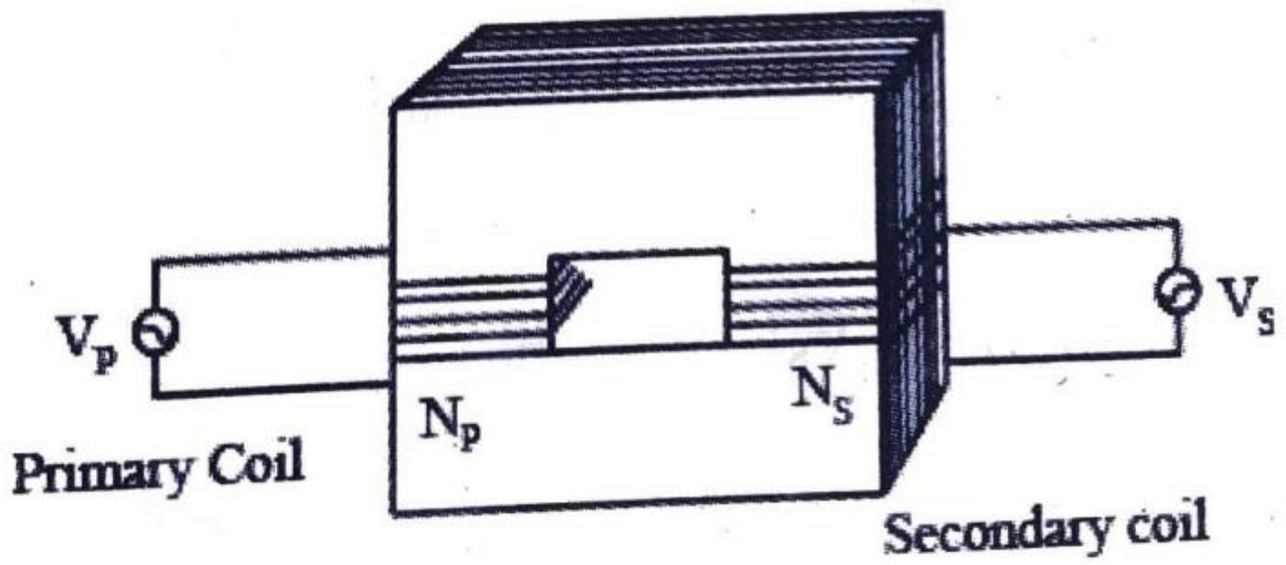
$$= (100)(10^{-3})(0.2)(377)$$

$$\varepsilon_{max} = 7.54V$$

Transformer

Transformer: is a device that transfers electrical energy from one circuit to another usually with a change of voltage, it increase or decrease ac voltage.

A typical transformer consists of an *iron core, primary coil* and *secondary coil*.



Where, N_P is the number of turns in the primary coil

N_S is number of turns in the secondary coil

V_P is the input voltage in the primary coil and

V_S is the output voltage of the secondary coil

A transformer increases (steps up) or decreases (steps down) one ac voltage to another ac voltage to suit the needs of a particular application.

Now, from Faraday's law the induced emf in the primary coil is:

$$V_P = -N_P \frac{\Delta\phi}{\Delta t} \ldots\ldots\ldots (1)$$

Similarly, the emf induced in the secondary coil is

$$V_S = -N_S \frac{\Delta\phi}{\Delta t} \ldots\ldots\ldots (2)$$

Dividing equation (1) by equation (2), we get

$$\boxed{\frac{V_P}{V_S} = \frac{N_P}{N_S}} \leftarrow \text{transformer equation}$$

☞ **Remember, that**

→ For a step down ($V_S < V_P$) transformer, the secondary coil has less number of turns than the primary coil ($N_S > N_P$)

→ For a step up ($V_S > V_P$) transformer, the secondary coil has large number of turns than the primary coil ($N_S > N_P$)

Moreover, assuming a constant power in both coils,

Power input = Power output

$V_P I_P = V_S I_S$, from this,

$$\boxed{\frac{V_P}{V_S} = \frac{I_S}{I_P}} \leftarrow \text{ideal transformer equation}$$

So in general we have,

$$\boxed{\frac{V_P}{V_S} = \frac{N_P}{N_S} = \frac{I_S}{I_P}}$$

Conceptual Example

36. Can a transformer increase or decrease DC voltage? Explain

Explanation: No, a transformer cannot be used on dc voltage because a continually changing flux is the basic for its operation.

Illustrative Examples

37. The input coil of a transformer has 1000 turns, the output coil has 100 turns. If the input voltage is 220V, what is the output voltage of the transformer?

Solution: $\frac{V_P}{V_S} = \frac{N_P}{N_S} \Rightarrow V_S = V_P\left(\frac{N_S}{N_P}\right) = (220)\left(\frac{100}{1000}\right) = 22V$

38. An ac generator what delivers 20A at 6000V is connected to a step up transformer. What is the output current at 120,000V if the transformer efficiency is 100%?

Solution: $\frac{V_P}{V_S} = \frac{I_S}{I_P} \Rightarrow I_S = (I_P)\left(\frac{V_P}{V_S}\right) = (20) = \left(\frac{6000}{120000}\right) = 1A$

General Secondary Education Certificate Exam

39. Magnetic domains are,

A. The possible magnetic pole of a magnet
B. Microscopic set of magnets within a magnet
C. Collections of the same magnetic poles
D. Bar magnets arranged in the same direction.

Answer: B

40. The unit of magnetic flux is

A. Tesla
B. Weber
C. Henery
D. Ġauss

Answer:B

41. Which one of the following is the unit of magnetic flux

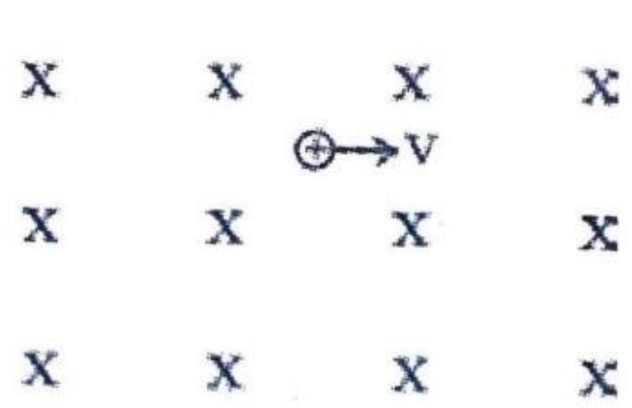
A. T/m^2
B. $\frac{N.m}{A}$
C. $\frac{Tm}{A}$
D. $\frac{NA}{m^2}$

Solution: $1\text{Wb} = 1\text{Tm}^2 = \dfrac{1\text{Nm}}{\text{A}}$.

Answer: B

42. The magnetic field is into the page. If a proton is moving in the field at velocity v, direction of the force on the proton is

A. ←
B. →
C. ↑
D. ↓



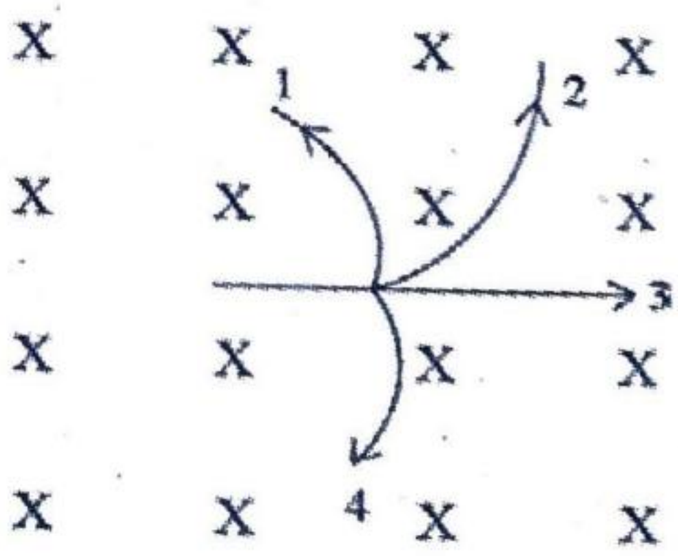
Explanation: Using the right hand rule, pointing the thumb in the current (velocity) direction, first figure in the field, then the second figure points force direction.

Answer: C

43. Positive and negative charges are shot into a magnetic field. Which one is the path followed by the negative charge?

A. Path 2
B. Path 1
C. Path 4
D. Path 3

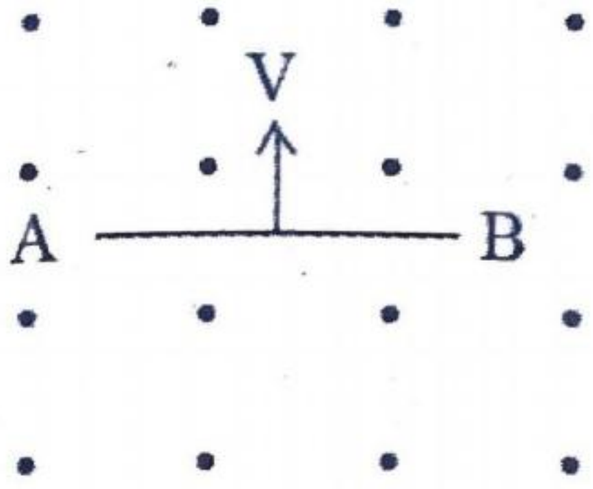
Answer: C



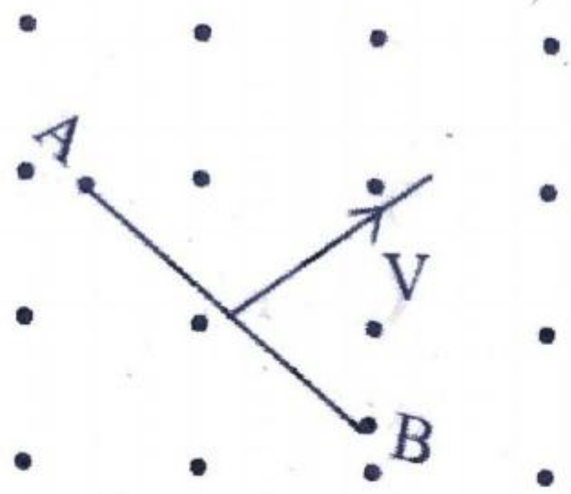
Note: The direction of negative charges is opposite to the direction of conventional current.

44. A copper wire is moving at velocity V in a magnetic field pointing out of the page. All wires have an induced emf across the ends A and B EXCEPT?

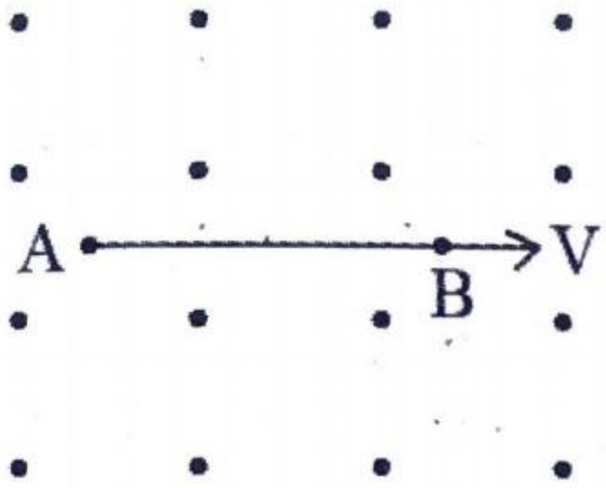
A.



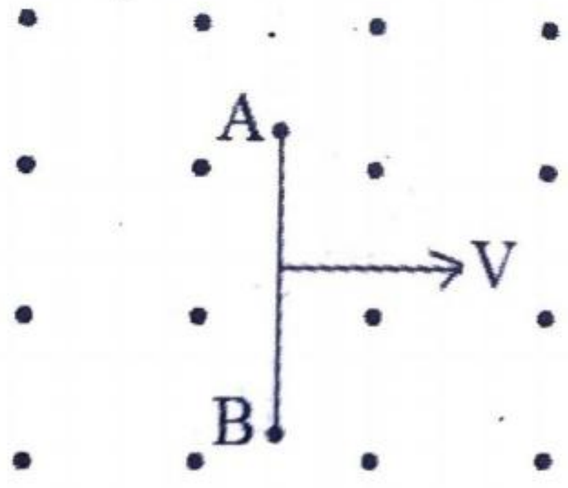
C.



B.



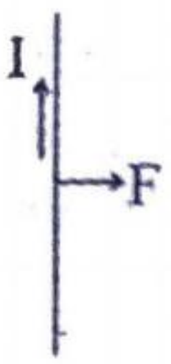
D.



Explanation:induced emf is produced when there is a change in flux, that is when there is a relative motion between the magnetic field and the conductor and the field lines cross the conductor.

Answer: B

45. The direction of the force acting on a current carrying conductor is as shown. What is the direction of the magnetic field



A. Into the plane of this page
B. In the direction of the force
C. In the direction of the current
D. Out of the plane of this force

Answer: A

46. A circular ring has an area of $100cm^2$. If the ring is immersed perpendicular in a uniform magnetic field of 0.5T. What is the flux through the ring?

A. $5\times10^{-4}Wb$
B. $5\times10^{-2}Wb$
C. 50Wb
D. $5\times10^{-3}Wb$

Solution:$\phi = BA = (0.5)\,(0.01) = 5\times10^{-3}Wb$

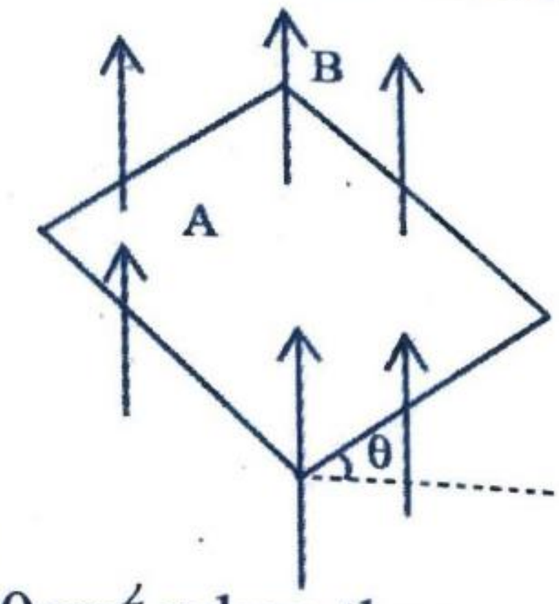
Answer: D

47. What is the direction flux through the square loop of area A tilted at an angle θ from the horizontal level?

A. $BA\sin\theta$

B. BA
C. BA tanθ
D. BA cosθ

Answer: D



Note: $\phi = BA_{\perp}$, $A_{\perp} = A\cos\theta$

48. What is the average p.d across a coil of 100 turns and cross sectional area of $1000cm^2$ when the magnetic field strength changes from $10^{-3}Wb/m^2$ to $10^{-4}Wb/m^2$ in 0.1 sec?

A. 0.9V B. 1V C. 10V D. 0.1V

Solution:

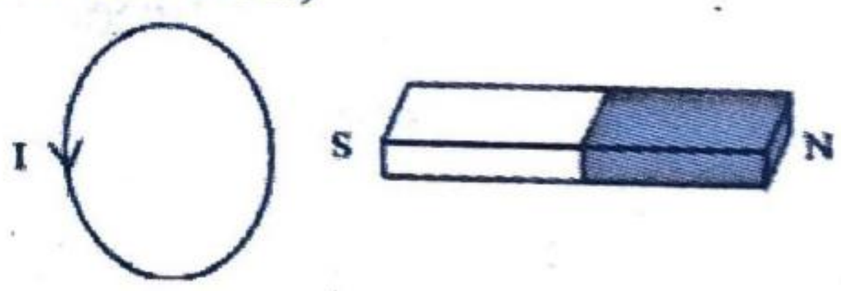
$$\varepsilon = -\frac{N\Delta\phi}{\Delta t} = -\frac{N\Delta BA}{\Delta t} = -100\frac{\left(10^{-4} - 10^{-3}\right)}{0.1}\left(1000\times10^{-4}\right)$$

$\varepsilon = 0.9V$

Answer: A

49. A coil carrying a current and a bar magnet are arranged side by side. As the bar moves towards the coil, the magnetic force on the bar magnet due to the coil is directed,

A. to the right
B. to the left
C. upward
D. downward



Explanation: according to lenz, the induced emf opposes the cause producing it. Hence, the coil repels the magnet

Answer: A

50. Which law states that the induced emf in a closed loop is directly proportional to the rate of the magnetic flux in the loop?

A. Ampere's law
B. Faraday's law
C. Lenz's law
D. Columb's law

Answer: B

51. The following are applications of electromagnetism EXCEPT

A. Transformer
B. Van De Graff Generator
C. Electric Motor
D. Galvanometer.

Explanation: Electromagnetism involves moving charges, but Van De Graff generator works on static charges.

Answer: B

52. Electromangetic induction is?

A. The application of magnetic force on a current Carrying coil
B. The exertion of force on a magnet by current carrying coil
C. The production of electric current by a changing magnetic flux
D. The production of magnetic field by electric current

Answer: C

53. A current carrying wire is put in uniform magnetic field. All the following modifications will change the magnitude of the magnetic force EXCEPT

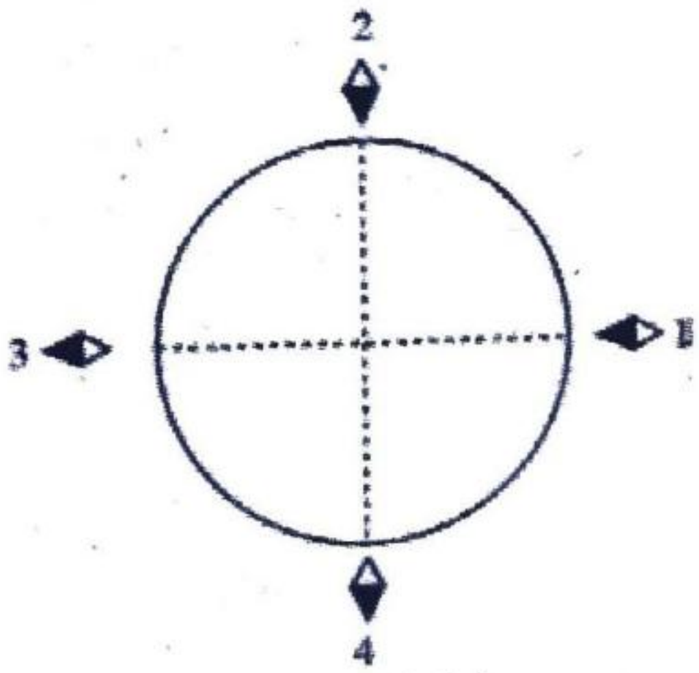
A. Doubling the current through the wire
B. Increasing the magnetic field strength
C. Rotating the wire to change its orientation
D. Reversing the direction of flow of the current

Answer: D

54. Which of the compass needle orientation correctly describe the earth's magnetic field at the point? (The dark side of the compass needle is the North side).

A. 1 B. 2 C. 3 D. 4

Explanation: The magnetic field of the Earth originated from number 4 and enter into number 2. Thus, the north pole of the compass needle point towards the geographic North pole of the earth.



Answer: B

55. A coil is put on the top of another coil with their faces parallel to each other. If the magnetic field in one of the coils is constantly increased, what will happen to the induced emf of the other coil?

A. Increasing constantly
B. Decrease constantly
C. remain constant
D. no induced emf

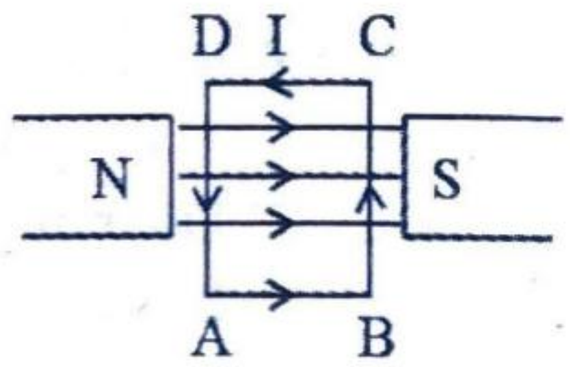
Explanation: According to Faraday, "the induced emf is directly proportional to the rate of change of flux". That is,

$$\varepsilon = \frac{N\Delta\phi}{\Delta t} = \frac{N\Delta BA}{\Delta t}$$ as ΔB increases, εincrease

Answer: A

56. A rectangular coil of wire carrying current is placed between the poles of a permanent magnet. The magnetic field rotates the coil by moving

A. Side AB out of the page and side CD into the page
B. Side AB into the page and side CD out of the page
C. Side BC out of the page and side DA into the page
D. Side BC into the page and side DA out of the page

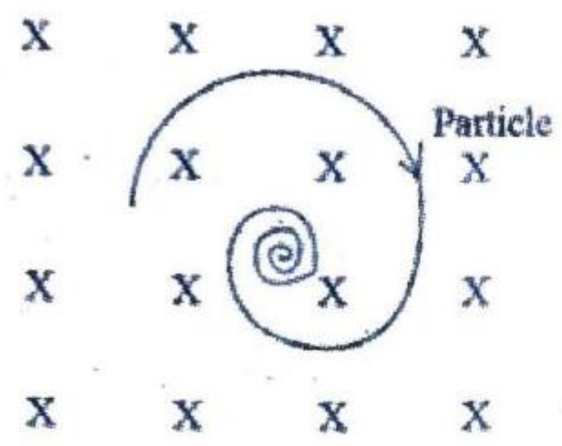


Explanation: the turning effect of force, torque is produced when the field and the current are perpendicular.

Answer: D

57. A uniform magnetic field is directed into the page. A charged particle, moving in the plane of the page follows a clockwise spiral of decreasing radius. Which one of the following is a reasonable explanation?

A. The charge is negative and slowing down
B. The charge is positive and slowing down
C. The charge is positive and speeding up
D. The charge is negative and speeding up



Explanation: $R = \frac{mV}{qB}$

Answer: A

58. The magnetic field strength inside a long ideal solenoid can be doubled by doubling

A. The spacing of winding keeping the current and number of turns constant
B. The area keeping the current and number of turns constant
C. The length keeping the current and number of turns constant
D. The current keeping the area and number of turns constant

Solution: $B = \frac{\mu_o N}{\ell} I$

Answer: D

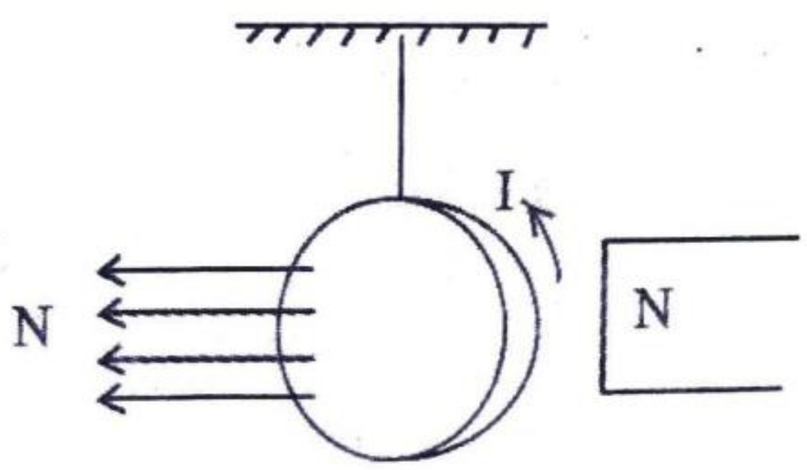
59. Which of the following statement is correct

A. In an electric generator, motion is created by the forces magnets exert on each other.
B. In any generator, a magnetic field rotates around a current carrying coil
C. All electric generators produced a voltage of constant magnitude
D. In an electric generator, kinetic energy is converted into electrical energy.

Answer: D

60. A long wire wound in the form of a ring is freely suspended near the North Pole of a bar magnet. If current I is set in the ring, the ring will

A. move away from the magnet
B. not be affected by the magnet
C. move toward the magnet
D. move up and down



Answer: C

61. A thin copper rod of length 1m has a mass of 50g. What is the minimum current in the rod that would allow into "float" in a magnetic field of 0.1T. (take $g = 10m/s^2$)

A. 5mA
B. 50mA
C. 5A
D. 0.5A

Solution: $F = W \Rightarrow BI\ell = mg \Rightarrow \frac{mg}{B\ell} = \frac{\left(50\times10^{-3}\right)(10)}{(0.1)(1)} = 5A$

Answer: C

62. What is the purpose of using transformer in electric devices?

A. To transform mechanical energy into electrical energy

B. To transform small electrical energy into height electrical energy

C. To convert electrical voltage from one value to another

D. To convert electrical energy to mechanical energy

Answer: C

63. A long solenoid has 1000 turns over a length of 0.4m. What current is required to produced a field strength of $\pi \times 10^{-4}$T ($\mu_o = 4\pi \times 10^{-7}$Tm/A)

A. 0.1A C. 0.01A

B. 1A D. 10A

Solution:

$$B = \mu_o nI = \frac{\mu_o N}{\ell} I \Rightarrow I = \frac{B\ell}{\mu_o N} = \frac{\left(\pi\times10^{-4}\right)(0.4)}{\left(4\pi\times10^{-7}\right)(1000)} = 0.1A$$

Answer: A

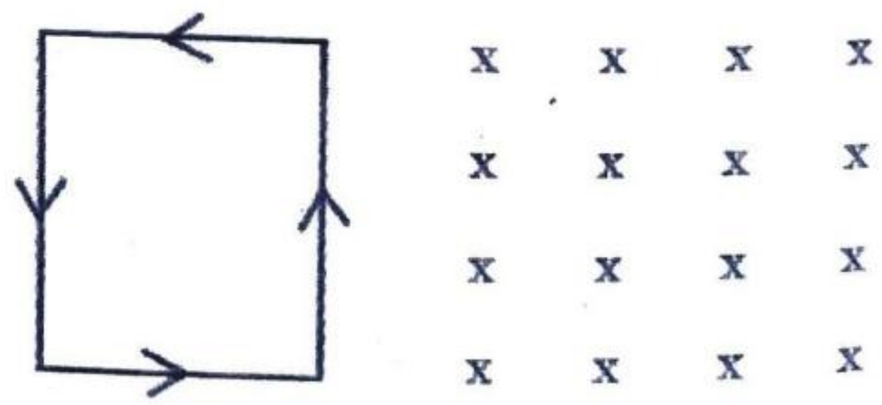
64. What is the unit of magnitude flux density?

A. T B. Tm C. T/m D. Tm^2

Explanation: magnetic flux density is magnetic field strength and its unit is Tesla (T).

Answer: A

65. A rectangular current carrying wire moves to the right with its plane perpendicular to a uniform magnetic field. The net magnetic force acting on the wire is the right



A. When the right side of the wire is within the magnetic field but not the left side

B. When the left side of the wire is within the magnetic field but not the right side

C. When the rectangular wire is completely within the magnetic field

D. When the rectangular wire is completely outside the magnetic field

Answer: B

66. A metal bar magnet has a magnetic field in the region of space around it. The magnetic field is due to

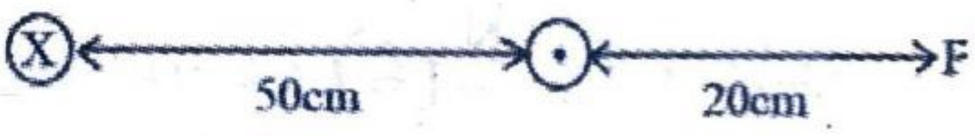
A. A hidden voltage source in the metal
B. An electric current that runs along the length of the magnet
C. Magnetic monopoles embedded in the metal
D. Motion of charged particles in the metal

Answer: D

67. Two long parallel conductors that are 50cm apart carry currents $I_1 = 7A$ and $I_2 = 10A$, in opposite directions. What is the resultant magnetic field at point P?

A. 8×10^{-6}T up
B. 8×10^{-6}T down
C. 1.2×10^{-5}T up
D. 1.2×10^{-5}T down

Solution: $B = \dfrac{\mu_o I}{\pi r^2}$

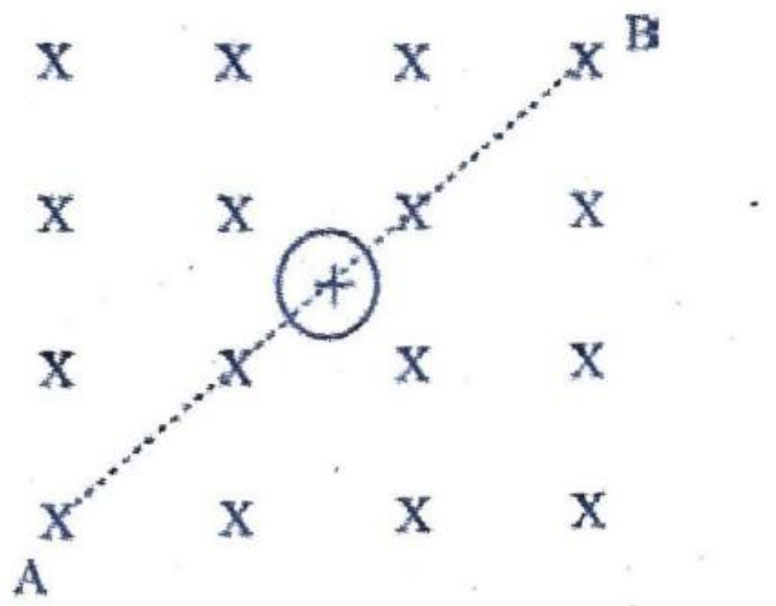


$$B = B_2 - B_1 = \frac{\mu_o I_2}{2\pi r_2} - \frac{\mu_o I_1}{2\pi r_1} = \frac{4\pi \times 10^{-7}}{2\pi}\left(\frac{10}{0.2} - \frac{7}{0.7}\right) = 8 \times 10^{-6}\text{T up}$$

Answer: A

Note: The direction of the magnetic field around each current carrying wire is determined using Right Hand Rule Accordingly, it is downward for wire 1 and upward for wire 2.

68. A positive charge moves from point A to B along the diagonal in a magnetic field pointing into the paper. What is the direction of the magnetic force on the charge?



A. Down and right
B. Down and left
C. Up and right
D. Up and left

Explanation: using right hand rule.

Answer: D

69. If the long, parallel wires 1m apart carry the same current and the force per unit length on each wire is 2×10^{-7}N/m the current in each wire is

A. 2 Amperes
B. 1 Ampere
C. 1 coulomb
D. 2×10^{-7} coulomb

Solution: $F = \frac{\mu_o I_1 I_2 \ell}{2\pi r} = \frac{F}{\ell} = \frac{\mu_o I_1 I_2}{2\pi r} = \frac{\mu_o I^2}{2\pi r}$

$$I^2 = \left(\frac{F}{\ell}\right)\frac{(2\pi r)}{\mu_o} = \frac{\left(2\times10^{-7}\right)(2\pi\times1)}{4\pi\times10^{-7}} = 1\,\text{Ampere}$$

Answer: B

70. What is the magnetic field strength at the center of a solenoid of 200 turns and 4cm long when it is carrying a current of 2A?

A. 1.256×10^{-4}T
B. 1.256×10^{-3}T
C. 1.256×10^{-2}T
D. 1.256T

Solution:

$$B = \mu_o nI = \mu_o \frac{N}{L} I = \left(4\pi\times10^{-7}\right)\frac{(200)}{4\times10^{-2}}(2) = 1.256\times10^{-2}\text{T}$$

Answer: C

71. An emf of 4V is induced in a coil when the current in the changes at the rate of 16A/S. what is the inductance of the coil?

A. 4H
B. 1H
C. 0.5H
D. 0.25H

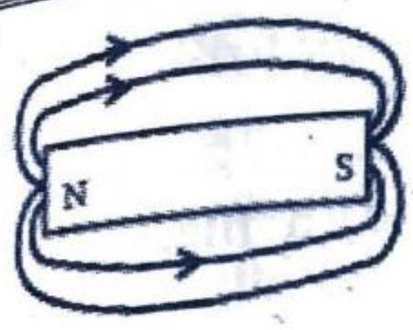
Solution: $\varepsilon = \frac{\Delta I}{\Delta t} \Rightarrow L = \varepsilon \Big/ \left(\frac{\Delta I}{\Delta t}\right) = \frac{4}{16} = 0.25\text{H}$

Answer: D

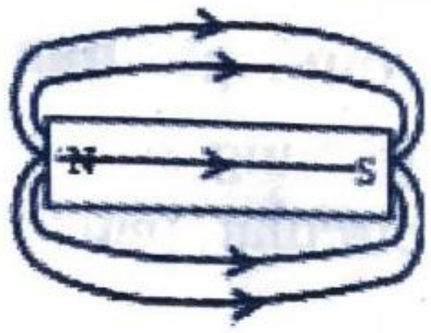
Additional Examples

72. The magnetic field lines due to a bar magnet are correctly shown in

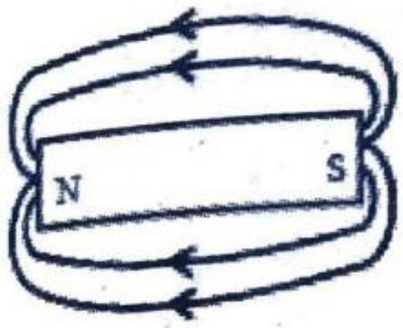
A.



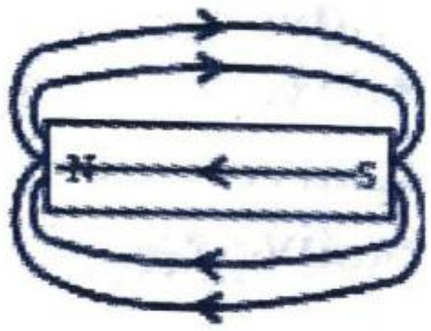
C.



B.



D.



Answer: D

73. Current is flowing in two long parallel conductors in the same direction the conductors

A. Attract each other
B. Repel each other
C. Neither attract nor repel
D. Get perpendicular to each other

Answer: A

74. A solenoid 10 turns per cm carries a current of 5A flowing through it. The magnetic field in side the solenoid

A. $2\pi \times 10^{-3}$T
B. $2\pi \times 10^{-2}$T
C. $2\pi \times 10^{-5}$T
D. $2\pi \times 10^{-4}$T

Solution: $B = \mu_o nI = \left(4\pi \times 10^{-7}\right)\frac{(10)}{\left(10^{-2}\right)}(5) = 2\pi \times 10^{-3}T$

Answer: A

75. A positive charge is moving upward vertically. It enters a magnetic field directed to the north. The force on the charge will be towards.

A. north
B. south
C. east
D. west

Answer: D from right hand rule

76. A current carrying power line carries current form west to east. What will be the direction of the magnetic field at a point above the wire?

A. east to west
B. west to east
C. north to south
D. south to north

Answer: C from right hand rule

77. An electron is moving along positive x axis. To get it moving on anticlockwise circular path in x – y plane, a magnetic field is applied

A. along positive y – axis
B. along positive z – axis
C. along negative y – axis
D. along negative z – axis

Applying the right hand rule,Answer: B

78. An electron continues to move in a straight line while passing through magnetic field

A. It is never possible
B. It is possible if B is perpendicular to the electron motion
C. It is possible if B is parallel to the electron motion
D. It is possible if B is at 45° to the electron motion

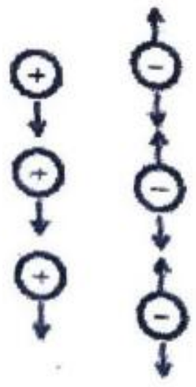
Explanation: when the field and the charge are parallel, the force on the particle is zero and the particle moves in a straight line at constant velocity.

Answer: C

79. A proton beam is going from north to south and an electron beam is going from south to north. The electron beam will be deflected.

A. Towards the proton beam
B. Away from the proton beam
C. Upwards
D. Downwards

Explanation: The two beams act like currents moving in a wire in the same direction.



Answer: A, the conventional current for the electron motion is down ward

80. Protons are shot perpendicular to a magnetic field.

A. Magnetic field will have no influence on the motion of proton

B. Protons will continue to move in the same direction, gain momentum
C. Protons will continue to move in the opposite direction, gain momentum
D. They will bend in an arc of circle.

Explanation: the magnetic force becomes centripetal force.

Answer: D

81. Two beams of protons moving parallel to each other

A. attract
B. repel
C. exert no force
D. deflect perpendicularly

Explanation: the electrostatic force dominates the magnetic force.

Answer: B

82. A positively charged particle projected towards east is deflected towards north by a magnetic field. The field may be

A. towards west
B. towards south
C. upward
D. downward

Answer: D

83. A charged particle enters in a strong magnetic field. Then its kinetic energy

A. increase
B. decrease
C. remains constant
D. first increases then become constant

Explanation: since the work done by magnetic field is zero, the kinetic energy remains constant.

Answer: C

84. A magnetic field

A. always exerts a force on a charged particle
B. never exerts a force on a charged particle
C. exerts a force, if the charged particle is moving across the magnetic field lines
D. exerts a force, if the charged particle is moving along the magnetic field lines.

Answer: C

85. An electron enters a magnetic field whose direction is perpendicular to the velocity of the electron. Then,

A. the speed of the electron will increase
B. the speed of the electron will decrease
C. the speed of the electronwill remain the same
D. the velocity of electron will remain the same.

Answer: C , kinetic energy constant, speed constant

86. the north of earth's magnet is near the geographic

A. south
B. north
C. west
D. east

Answer: A

87. A permanent magnet

A. attracts all substances
B. attracts only magnetic substances
C. attracts magnetic substances and repels non magnetic
D. attracts non magnetic substances and repel magnetics

Answer: B

88. Which of the following demonstrates that earth has a magnetic field?

A. The earth is surrounded by a shell of charged particles
B. A large quantity of iron ore is found in the earth
C. The cosmic rays (stream of charged particles combing form the outer space). Is more at the poles than at the equator
D. There is a naturally placed huge bar magnet in it

Answer: C

89. Out of the following, which is non magnetic?

A. Zinc
B. Steel
C. Copper
D. Cobalt

Answer: A

90. The magnetic field strength at a distance of 4cm from a long current carrying wire is 10^{-3}T. the magnetic field at a distance of 12 cm from the same wire is

A. 9×10^{-3}T
B. 3×10^{-3}T
C. 3.33×10^{-4}T
D. 1.11×10^{-4}T

Solution: $B_1 = \frac{\mu_o I}{2\pi r_1} = 10^{-3}$ and $B_2 = \frac{\mu_o I}{2\pi r_2}$, taking ratio

$$\frac{B_2}{B_1} = \frac{r_2}{r_1} \Rightarrow B_2 = \frac{r_1}{r_2} \times B_1 = \frac{4}{12} \times 10^{-3} = 0.333 \times 10^{-3}$$

$$B_2 = 3.33 \times 10^{-4} T$$

Answer: C

91. A straight wire of length 0.5m carrying a current of 1.2A is placed in a uniform magnetic field of 2T. Perpendicular ly the force on the wire is

A. 3.4N
B. 2.4N
C. 3N
D. 1.2N

Solution: $F = BI\ell = (0.5)(2)(1.5) = 1.2N$

Answer: D

92. A power line lies along the east west direction and carries a current of 10A. The force per meter due to the earth's magnetic field of 10^{-4}T is

A. 10^{-5}N
B. 10^{-4}N
C. 10^{-3}N
D. 10^{-2}N

Solution: $F = BI\ell \Rightarrow \frac{F}{\ell} = BI = (10)(10^{-4}) = 10^{-3}$ N.

Answer: C

93. Which one of the following is the correct unit of magnetic flux

A. testa
B. Wbm^{-1}
C. Wbm^{-2}
D. Tm^2

Solution: $1Wb = 1Tm^2$

Answer: D

94. Whenever magnetic lines move across conductor, there is produced an electric field. This is referred to as

A. Lenz's law
B. Faraday's law
C. Ohm's law
D. Gauss's law

Answer: B

95. Lenz's law is consequence of the law of conservation of

A. Charge
B. Mass
C. Momentum
D. Energy

Answer: D

96. A current increases from zero to 1A in 0.1 sec in a coil of 5mH. The magnitude of the induced emf is

A. 0.005V
B. 0.5V
C. 0.05V
D. 5V

Solution: $\varepsilon = \left|-\frac{L\Delta I}{\Delta t}\right| = \frac{L\Delta I}{\Delta t} = \left(5\times10^{-3}\right)\left(\frac{1}{0.1}\right) = 5\times10^{-2} = 0.05V$

Answer: C

97. When the current in a coil is reduced from 20A to zero in 0.5sec, an emf of 4V is induced in a nearby coil. What is the value of the mutual inductance between the coils?

A. 0.1H
B. 0.4H
C. 2.5H
D. 40H

Solution: $\varepsilon = \frac{M\Delta I}{\Delta t} \Rightarrow M = \varepsilon \Big/ \left(\frac{\Delta I}{\Delta t}\right) = \frac{4}{20/0.5} = 0.1H$

Answer: A

98. "The direction of the induced current in a circuit is always such that it opposes the cause due to which it is produced." This law is named as ?

A. Ohm's law
B. Kirchhoff's law
C. Lenz's law
D. Faraday's law

Answer: C

99. A step down transformer is employed to reduce the main supply of 200V to 11V. If the primary coil draws 5A of current, the secondary coil draws a current of

A. 11A
B. 5A
C. 90A
D. 200A

Solution: $\frac{V_P}{V_S} = \frac{I_S}{I_P} \Rightarrow I_S = \frac{V_P}{V_S} \times I_P = \frac{200}{11} \times 5 = 90A$

Answer: C

100. The mutual inductance between the coils is 1.25H. if the current in the primary changes at the rate of 80A/S, then the induced emf in the secondary is

A. 12.5V
B. 64V
C. 0.016V
D. 100V

Solution: $\varepsilon = \left|\frac{M\Delta I}{\Delta t}\right| = 1.25 \times 80 = 100V$

Answer: D

101. In a circular coil, when current increases from 2A to 18A in 0.05sec, the induced emf is 20V. the self-inductance of the coil is

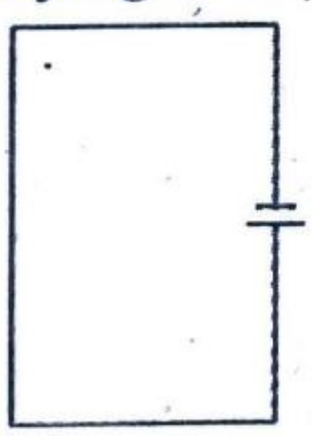
A. 62.5mH
B. 6.25mH
C. 50mH
D. 20mH

Solution: $\varepsilon = \left|\frac{L\Delta I}{\Delta t}\right| \Rightarrow L = \varepsilon \Big/ \frac{\Delta I}{\Delta t} = \frac{20}{\left(\frac{18-2}{0.05}\right)} = 0.0625 = 62.5mH$

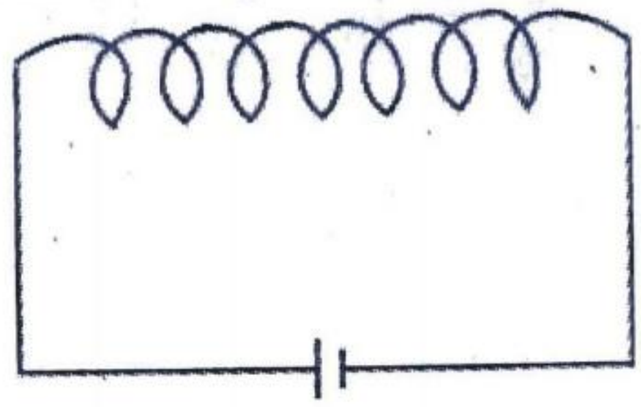
Answer: A

Unit Twelve: Conceptual Questions

1. Explain why a compass will show you which direction is magnetic north.
2. What happens to the domains when a magnetic material is heated?
3. Why would you use a magnetic shield?
4. Copy and complete the diagram to show the direction of the field lines around the current carrying wire



5. Copy the determine the north pole of the solenoid



6. Explain why the magnetic field in a solenoid is greater if the solenoid has an iron core

7. Explain the factors on which the force on a moving charge in a magnetic field depends.

Unit Twelve: Problems

Magnetic field

8. Calculate the magnetic field strength when a current of 6A flows in a wire 3m long and produces a force of 36N

9. Find the magnetic field strength at the center of a solenoid with 500 turns and current of 5A assuming its length is 50cm

Magnetic Force

10. Calculate the force felt by a 20cm wire carrying 1.5A at right angles to a magnetic field of 6×10^{-5}T

11. Work out the radius of a circle an electron revolves in a magnetic field of 0.02T at a speed of 4.5×10^{7}m/s ($m_e = 9.1 \times 10^{-31}$kg, $q = 1.6 \times 10^{-19}$C)

Electromagnetic Induction

12. What is induced emf when the rate of change in magnetic flux is $6Tm^2S^{-1}$

13. What is the induced emf when a current of 3A flows for 2 sec in an inductor of inductance 3×10^{-2}H

14. A uniform magnetic field of 3×10^{-2}T crosses. A 20cm × 20cm square card board perpendicularly. What is the magnetic flux through the board.

15. Suppose a coil has 1000 turns and the flux is changing at the rate of 0.003Wb/s. What is the induced emf.

16. The input coil of a transformer has 100 turns, the output coil has 100 turns. If there input voltage is 220V, what is the output voltage of the transformer?

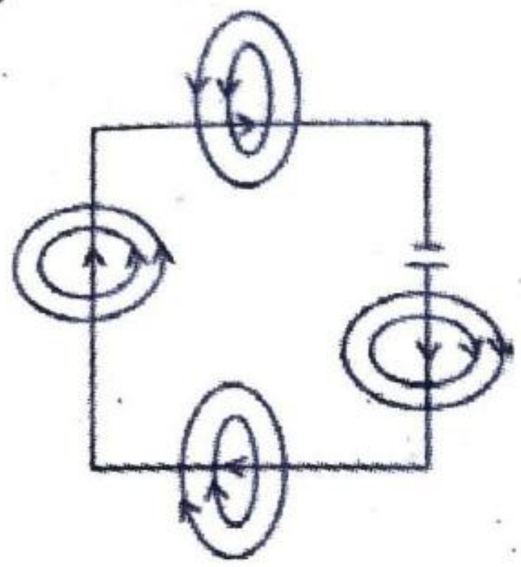
Unit Twelve: Explanation to Questions

1. Since a compass is a piece of magnet, it is attracted by the magnetic field of the earth. Hence its north pole is attracted by the earth's magnetic south pole.

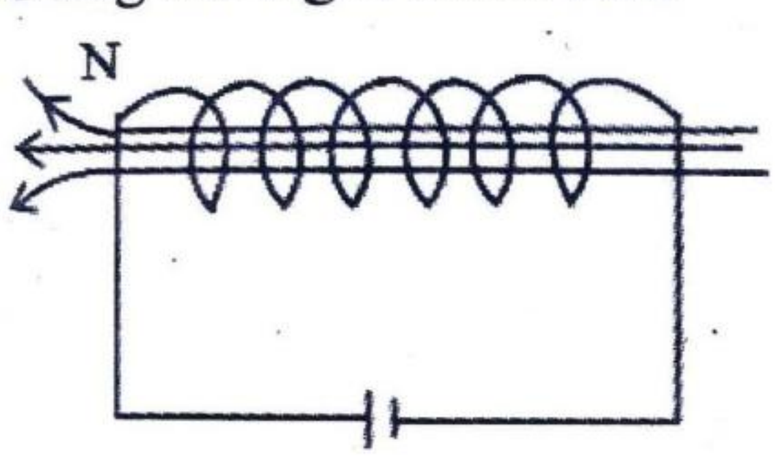
2. They become disordered

3. To prevent the material from the effect of magnetism

4. Using the right hand rule



5. Using the right hand rule



6. Because, the soft iron becomes a magnet itself when the current is flowing and hence a stronger magnetic field

7. The magnetic field, the speed of the particle, and the charged carried by the particle.

Unit Twelve: Solution to problems

8. $$B = \frac{F}{IL} = \frac{36}{(6)(3)} = 2T$$

9. $$B = \mu_o nI = \mu_o \frac{N}{\ell} I = \left(4\pi \times 10^{-7}\right)\frac{(5000)}{0.5}(5) = 6.3 \times 10^{-5}\,T$$

10. $$F = BIL = \left(6 \times 10^{-7}\right)(1.5)(0.2) = 1.8 \times 10^{-5}\,N$$

11. $R = \dfrac{mV}{Bq} = \dfrac{\left(9.1\times10^{-31}\right)\left(4.5\times10^{7}\right)}{\left(0.02\right)\left(1.6\times10^{19}\right)} = 1.2\times10^{-2}\,m$

12. $\varepsilon = -\dfrac{N\Delta\phi}{\Delta t} = \dfrac{\Delta\phi}{\Delta t} = -6V$

13. $\varepsilon = -\dfrac{L\Delta T}{\Delta t} = -3\times10^{-3}\left(\dfrac{3}{2}\right) = -4.5\times10^{-3}\,V$

14. $\phi = BA = \left(3\times10^{-2}\right)\left(0.2\times0.2\right) = 1.2\times10^{-3}\,Wb$

15. $\varepsilon = -\dfrac{N\Delta\phi}{\Delta t} = -1000\left(0.03\right) = -3V$

16. $\dfrac{V_P}{V_S} = \dfrac{N_P}{N_S} \Rightarrow V_S = V_P\dfrac{N_S}{N_P} = 220\left(\dfrac{100}{1000}\right) = 22V$

Unit Thirteen

Introduction to Electronics

The world is being changed by electronics. It is causing a revolution in every sphere of life and it is affecting life in most dramatic ways. Telephone, radio, Satellites Television, microcomputers and even Robots are and are going to have impact on our daily lives. Now adays, using electronic devices is becoming a must to live and cope with modern life.

Electronics: Is the study and design of systems that use the flow of electrons through semiconductors, resistors and capacitors.

13.1 Vaccum tube device

When a metal is heated up, electrons become energetic and escape from the metal surface just like the evaporation of water molecules from a boiling water. This effect is known as *thermionic emission.*

Thermionic Emission: is the escape of electrons from a heated metal surface.

Thermionic emission can be achieved in vacuum tubes.

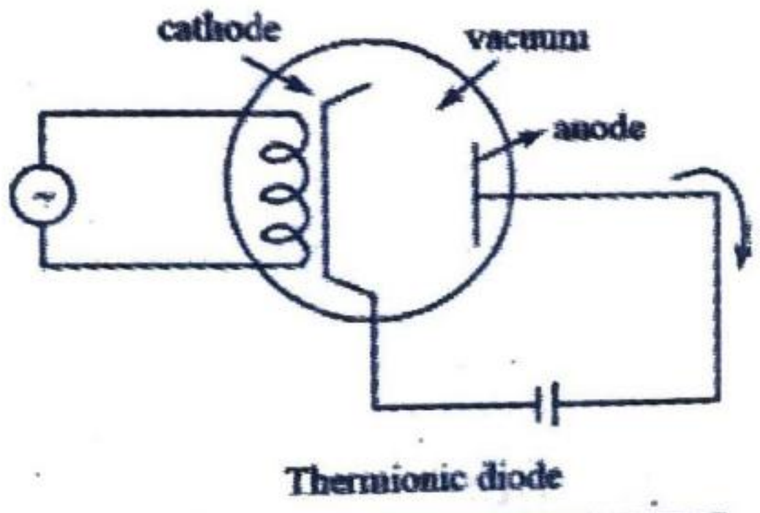
Vacuum tube: is a tube where there is no air or matter.

The thermionic diode.

Diode: is an electrical component having two electrodes; anode (+ve) and Cathode (-ve) and used for rectification, i.e for converting alternating current (a.c) to direct current (d.c)

When the cathode is heated, electrons 'boiled off' from it and attracted to the anode. In this way current flows in the circuit.

The arrow direction (↷) in the figure is the electron flow direction.



Thermionic diode

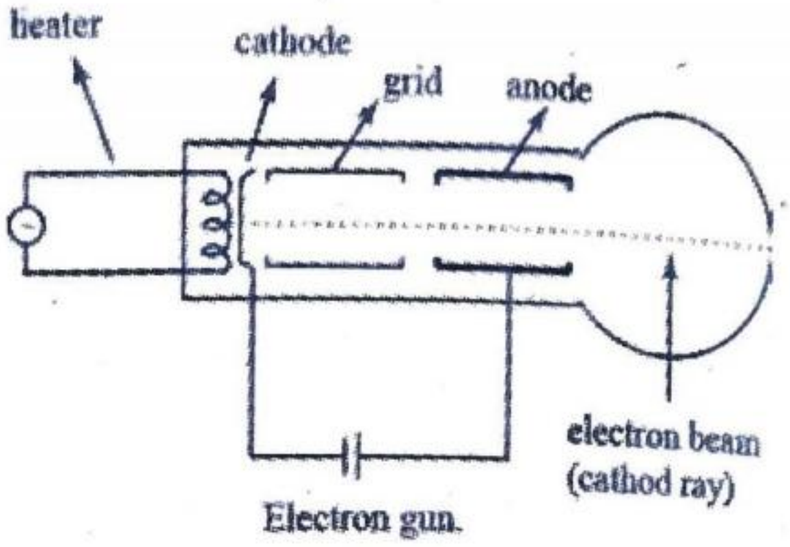
Cathode rays: Are beam of electrons moving through a vacuum at a high speed.

Cathode rays are produced by an *electron gun.*

Electron gun: Is a device that produce and shot a beam of electrons through vacuum at high speed.

The basic elements of an electron gun are *heater, cathode, grid* and *anode.*

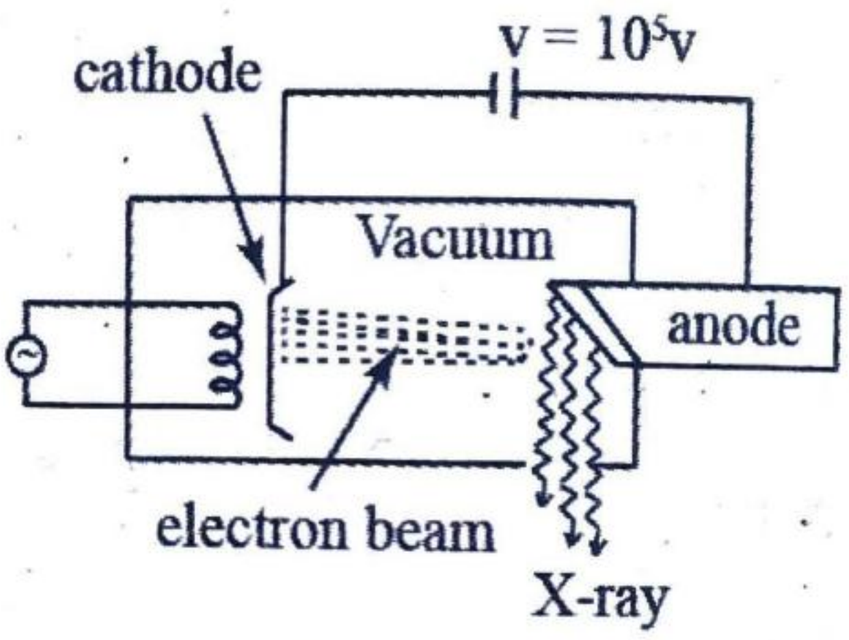
Electrons are produced when the cathode is heated. The grid which is more negative determines the number of electrons reaching the screen. The anode accelerate the electrons and focus them to a narrow beam.



X – ray tube

The x-ray machine is a vacuum tube device.

Electrons are generated from the cathode by thermionic emission and are then accelerated through a p.d of the order of 10^5V so as to hit the anode at an extremely high speed. When such fast moving cathode rays are suddenly stopped, x-rays are produced.

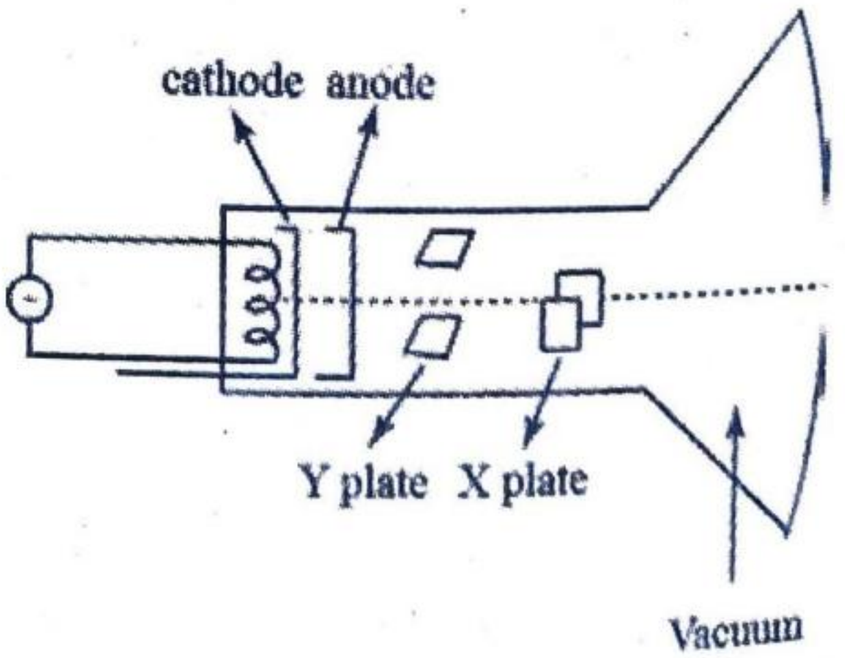


X-ray tube

Cathode Ray Oscilloscope (CRO)

CRO: Is an electronics equipment that provides visual images of electrical singles and oscillation such as *voltage, current* and *frequency.*

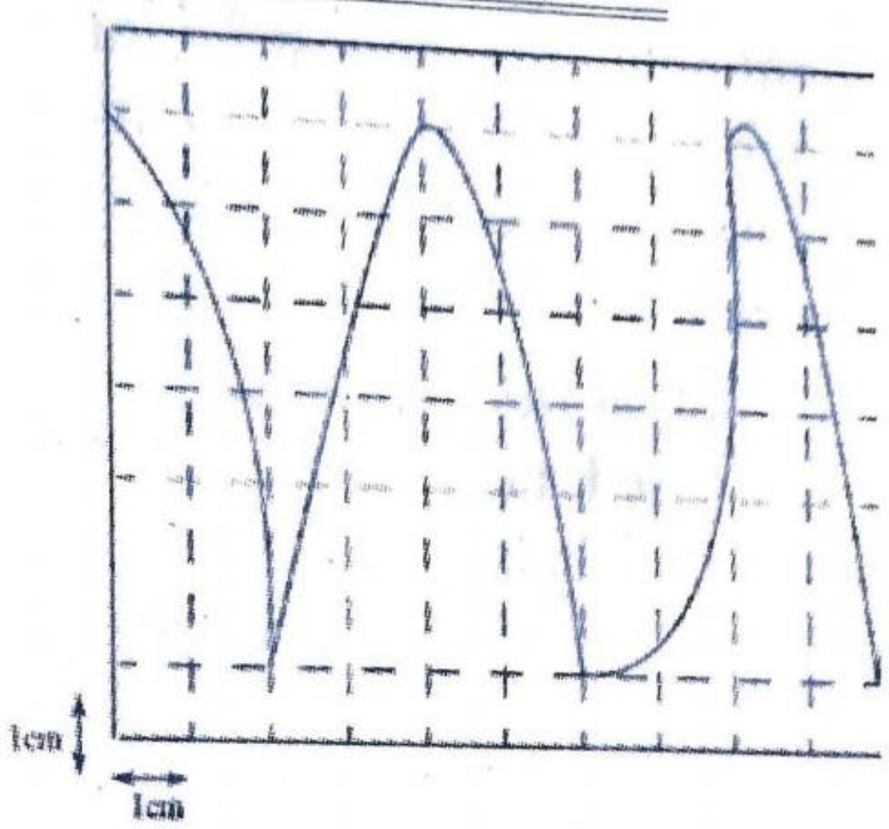
In a cathode ray oscilloscope the beam of electrons produce a spot on a fluorescent screen. Before reaching the screen it passes between two sets of deflecting plates, one to deflect in the y direction and the other in the x direction.



The cathdod ray oscilloscope.

Illustrative Examples

1. In a CRO screen, an AC voltage, the y gain control reads 10v/cm and the time base reads 5 millisecond/cm. Calculate.



a) the period
b) the frequency
c) the peak voltage

Solution:

a) Wavelength = 4cm
Period = $4\times(5\times10^{-3})$
$= 2\times10^{-2}$sec

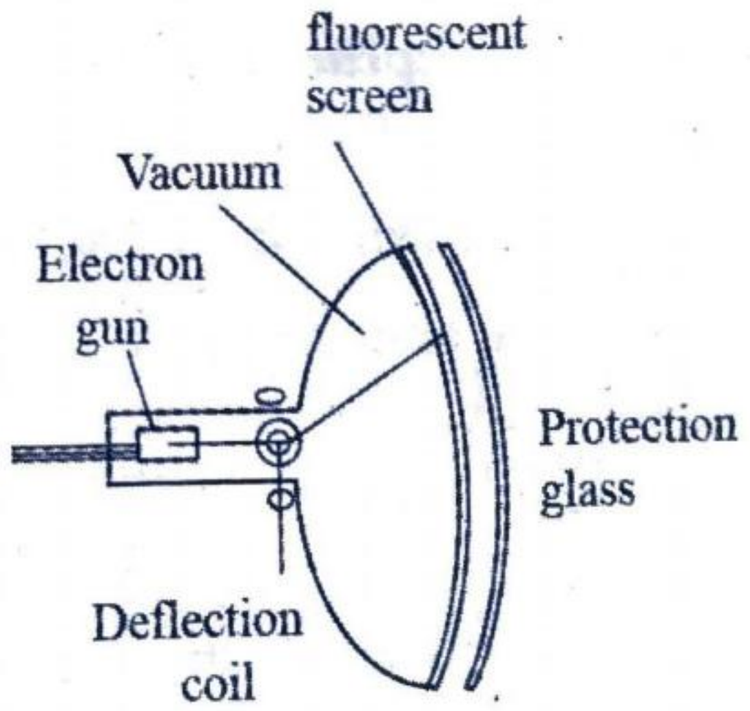
b) $\text{Frequency} = \dfrac{1}{\text{period}} = \dfrac{1}{2\times10^{-2}} = 50\text{Hz}$

c) Peak to peak height of wave = 4cm
Peak to peak voltage = $4\times10 = 40\text{V}$

$$\text{Peak voltage} = \frac{40}{2} = 20\text{V}$$

The Television (TV)

A television set has a cathode Ray tube (CRT) with pair of deflection coils mounted outside the neck of the tube. The magnetic fields of these coils deflect the electron beam by the catapult force.



Cathdod ray tube of TV

13.2 Semiconductors

Semiconductors form the backbone of electronics. Semiconductors affect all walks of life, starting from communications, computers, medicine, power, to aviation, defenses and entertainment. The transistors, integrated circuits, lasers, detectors, sensors and other semiconductor devices affect our daily life.

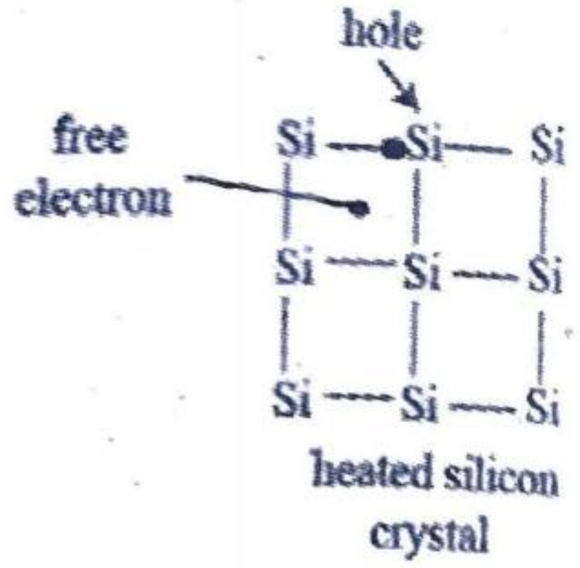
Semiconductor: Is a material that has a conductivity level somewhere between that of conductor and insulator.

Semiconductors can be *pure* or *impure*.

Pure (intrinsic) semiconductors.

Silicon (Si) and germanium (Ge) are the best examples of pure semiconductors. They have four valence electrons and these electrons form covalent bond. At low temperature, a semiconductor acts as an insulator.

When a semiconductor is heated, electrons detached and break free from the covalent bond. When the electron leave the bond a hole that acts like a positive charge is created.



Conceptual Examples

2. When a pure semiconductor is heated, it acts as conductor. What are the causes for this? Electrons or holes.

   Explanation:Both, the electrons and holes move in opposite directions forming the intrinsic conduction

3. The resistance of conductors increase with temperature but the resistance of semiconductors decrease with temperature why?

   Explanation:Because, conductors have positive temperature coefficient of resistance and semiconductors negative coefficient.

Impure (Extrinsic) Semiconductors

Impure semiconductors are formed by adding (doping) other elements to the pure semiconductors.

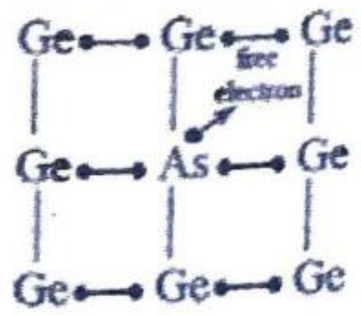
Doping: Is deliberately introducing impurities into a pure semiconductor to change its electrical properties.

Impure semiconductors are of two types. *N-type* and *P-type*.

N-type Semiconductor

When the pure semiconductors Si or Geare doped with pentavalent atoms (atoms of five valence electrons), such as phosphorus (P), arsenic (Ar) and Antimony (Sb), an N-type semiconductor is obtained.

For example, when Germanium (*Ge*) is doped with Arsenic (*As*), four of the five valence electrons of As form covalent bond with the four valence electrons Ge. The fifth electron is



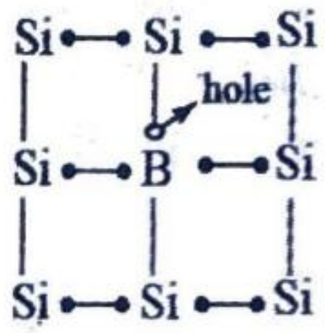
N-type semiconductor

unpaired and so is "free" to move and produce current. Since the majority charge carriers, charges that create the current are electrons (negatives), we call the semiconductor N-type. And we call the pentavalent atom, **Donor** atom since it donates free electron to the crystal.

P-type semiconductor

When the pure semiconductor Si or Ge are doped with trivalent atoms (atoms of three valence electrons) such as Indium (In), Boron (B) and Gallium (Ga), a p-type semiconductor is obtained.

For example, when silicon (Si) is doped with Boron (B), three of the four valence electrons of silicon form covalent bond with the three valence electrons of Boron, but the fourth is unpaired. So there remains a space or hole needing an electron. in this case, the majority charge carries, "Charges" that create current are holes. Since holes attract neighboring electrons, they act like positive and hence we call the semiconductor P-type. And, since the trivalent atom accepts electron to fill the hole, we call it **Acceptor atom**.



P-type semiconductor

Conceptual Examples

4. Why do impure semiconductors are preferred rather than pure semiconductor?

   Explanation:Because, pure semiconductors are not conductors at low temperature and heating it to improve its conductivity will affect the circuit. However, adding impurity of small amount to pure semiconductor improve its conduction property dramatically. Thus, impure semiconductors are often used.

5. Can we conclude that an N-type semiconductor is a negatively charged crystal and a P-type a positively charged? Explain

   Explanation: No. It is good to remember that the material itself is electrically neutral even after doping. The name n-type and p-type only signifies how the charge is carried.

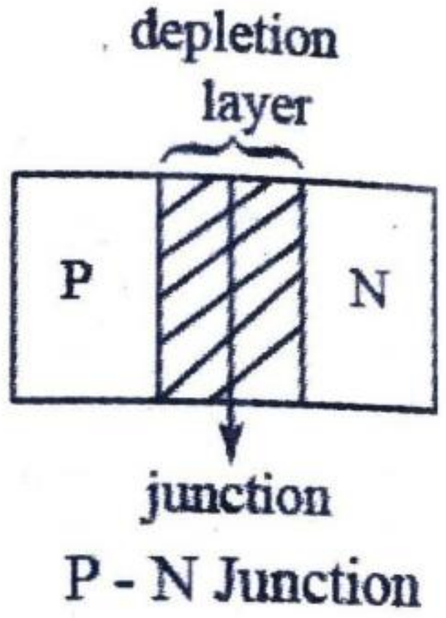
6. What makes a semiconductor material so special than conductors? Explain

   Explanation:The conduction properties of semiconductor materials can be controlled and changed as required which is impossible for conductors.

13.2 The P-N junction diode

When a p-type semiconductor is joined to an N-type semiconductor a PN junction is formed. The surface where the two semiconductors meet is called a ***junction***.

At the junction where they meet, some of the N-type electrons move into the P-type holes and vice versa forming the ***Diffusion current.*** The middle of the P-N junction then act as insulator and form a layercalled***Depletion layer***. The potential difference formed across the deflection layer is called ***Barrier Potential***, since it prevents movement of charge carriers across the depletion layer.



P - N Junction

Conceptual Examples

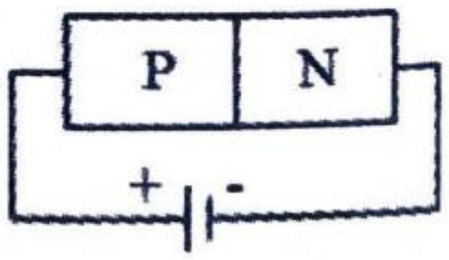
7. Why does the depletion layer act as an insulator? Explain
   Explanation:Because, the electrons and holes cancel each other in this region, so there are no free charge carriers inside it.
8. Why is the potential difference set up in the depletion layer called barrier potential? Explain
   Explanation: It is because the negative charge formed in the region of the p-type oppose the movement of electrons from n-type to this region. Similarity, the positive charges formed in the n-type region oppose the movement of holes from p-type to this region.

Forward and reverse bias

The P-N junction can be connected in a circuit in one of these two ways.

Forward biased:

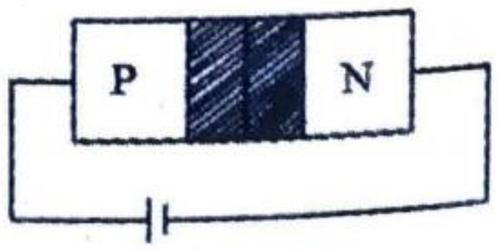
In this mode, the positive terminal of the battery is connected to the p-type and the negative terminal to the N-type. In forward bias, there is *narrow depletion* layer and *large current*.



forward bias

Reverse biased

In this mode, the positive terminal of the battery is connected to the N-type and the negative terminal to the p-type



reverse bias

☞ In reverse bias, there is wide depletion layer and very small current.

Conceptual Examples

9. Why is the depletion layer become thin and the current large in forward bias? Explanation

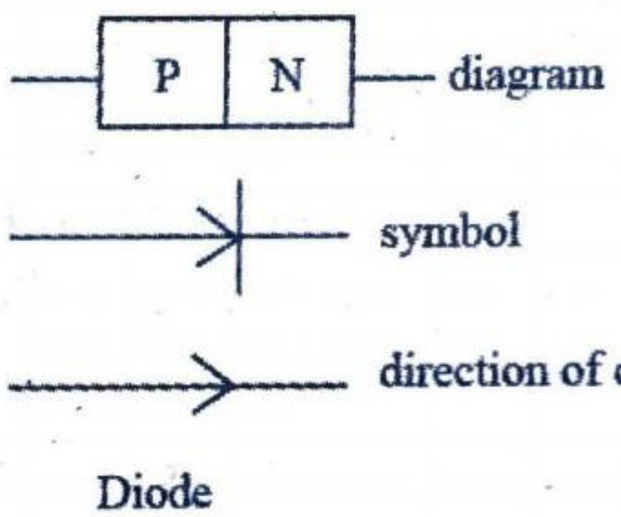
Explanation: Because the holes are repelled by the positive terminal of the battery and the electrons by the negative terminal, result in a thinner depletion layer hence charge carriers can easily diffuse through and thus a larger current.

10. Why is the depletion layer wide and the current very small in reverse bias mode.

Explanation: Because, the holes are attracted towards the negative terminal and the electrons to the positive terminal resulting in a wider depletion layer and larger barrier potential and very small current.

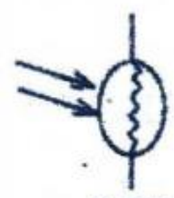
Some semiconductor devices

Diode: It allows current to flow only in one direction. The PN junction is a diode where current flows from P to N side.



Diode

LDR: Light dependent resistor conducts current when light shines on it and has very large resistance in the dark



LDR

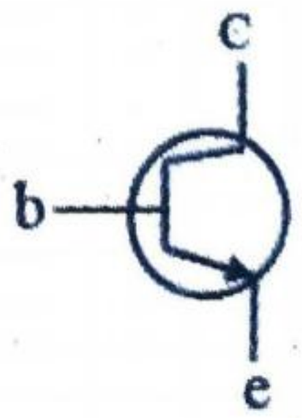
Thermistor: Has a high resistance in the cold, and its resistance drops as it becomes warmer.

Thermistor

LED: Light emitting diode, when a current is passed in the forward direction it emits light.

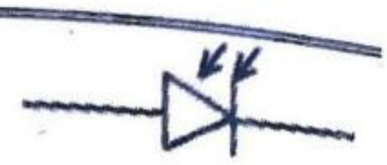
LED

Transistor: Is made of three layers of doped semiconductor and it has three terminals.



npn transistor

Photodiode: Is a light sensitive diode used to detect light or to measure its intensity. They are reverse biased so they do not conduct.



photodiode

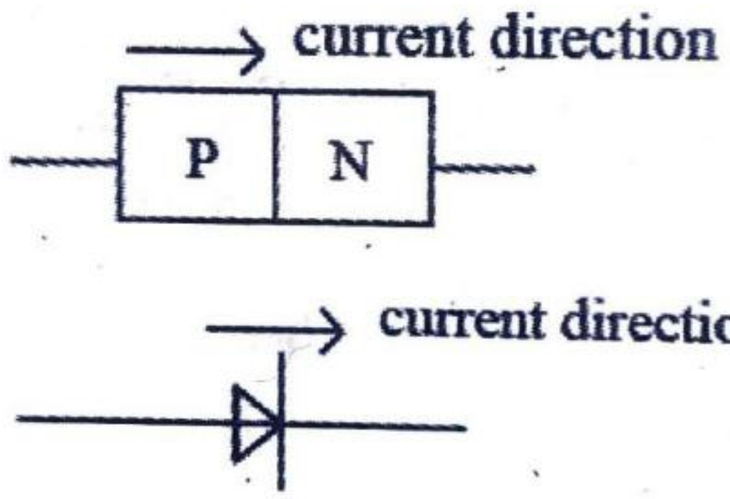
Photovoltaic cell: Is a cell made from P-N semi-conductor and converts solar energy into electrical energy.

photovoltaic cell

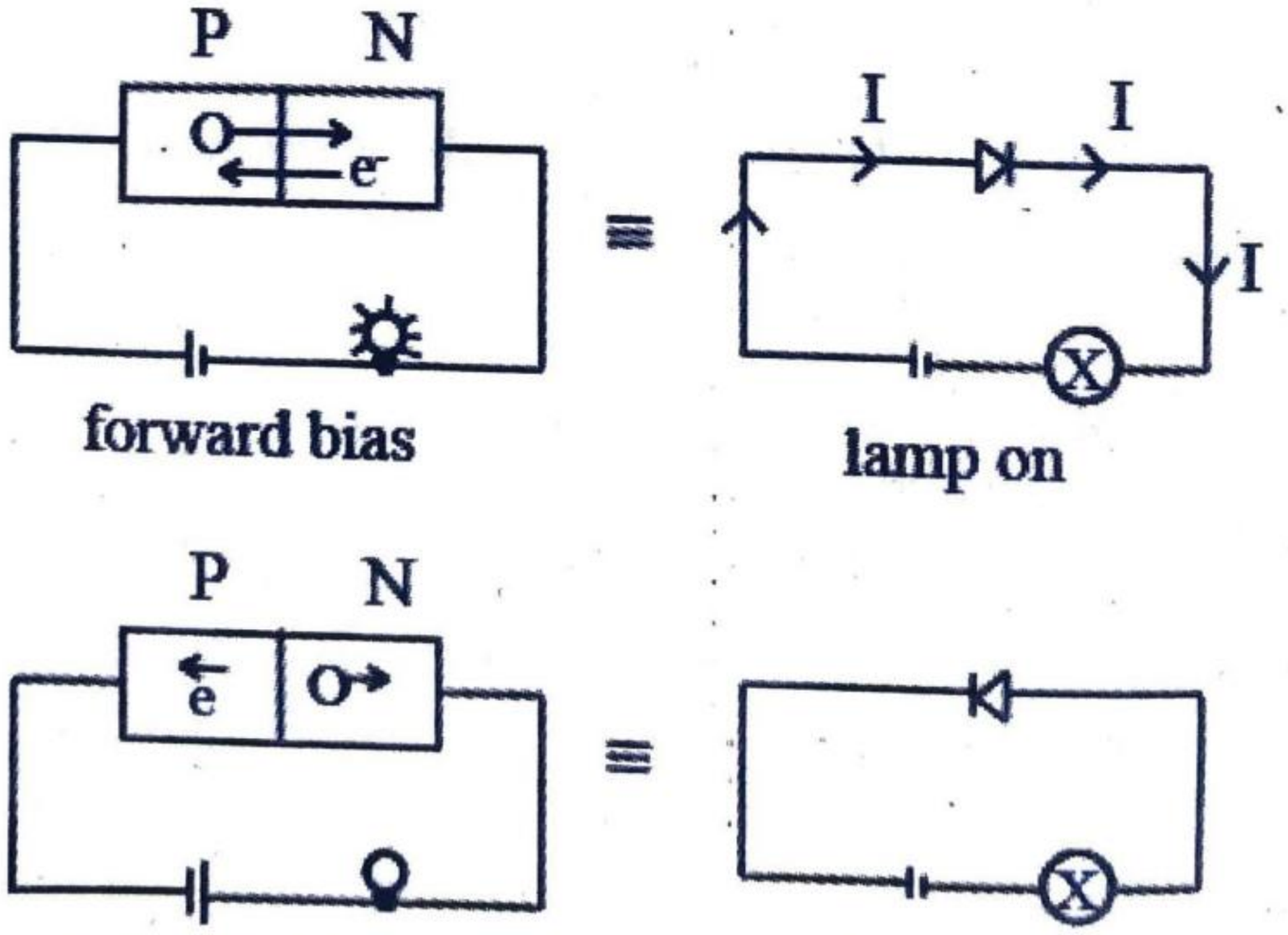
Comparing forward and reverse bias Modes

'Biasing' refers to placing voltage across terminals of a device, when a diode is in the forward connection, current flows in the circuit, but in the reverse bias, it is not. In the P-N junction diode current always flows only in one direction. That is from the P (triangle shape ▷) to the N (bar shape |) only. The other way round, the current is negligible.

In symbols,



In circuit

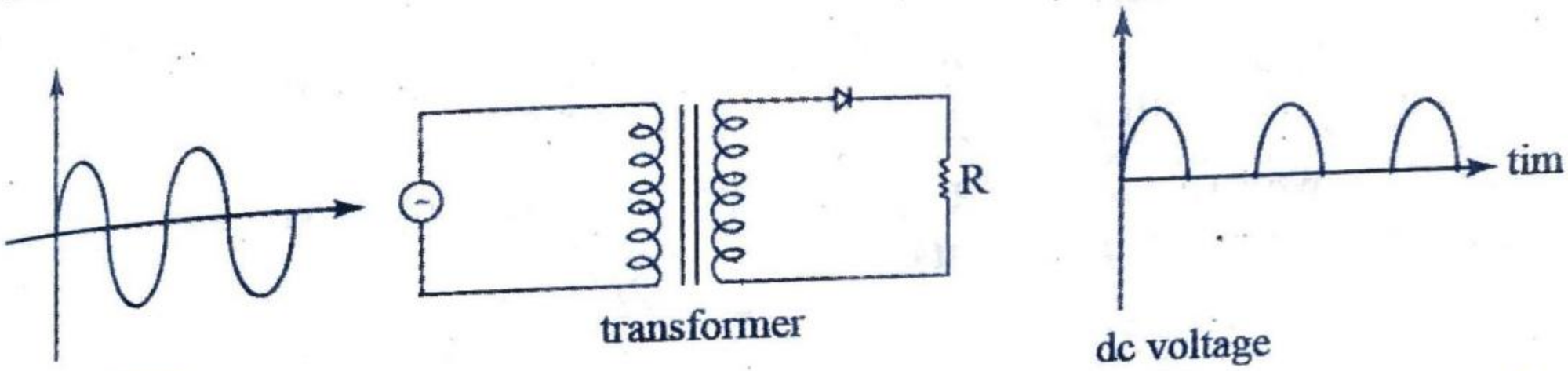


Rectification using the P-N junction diode

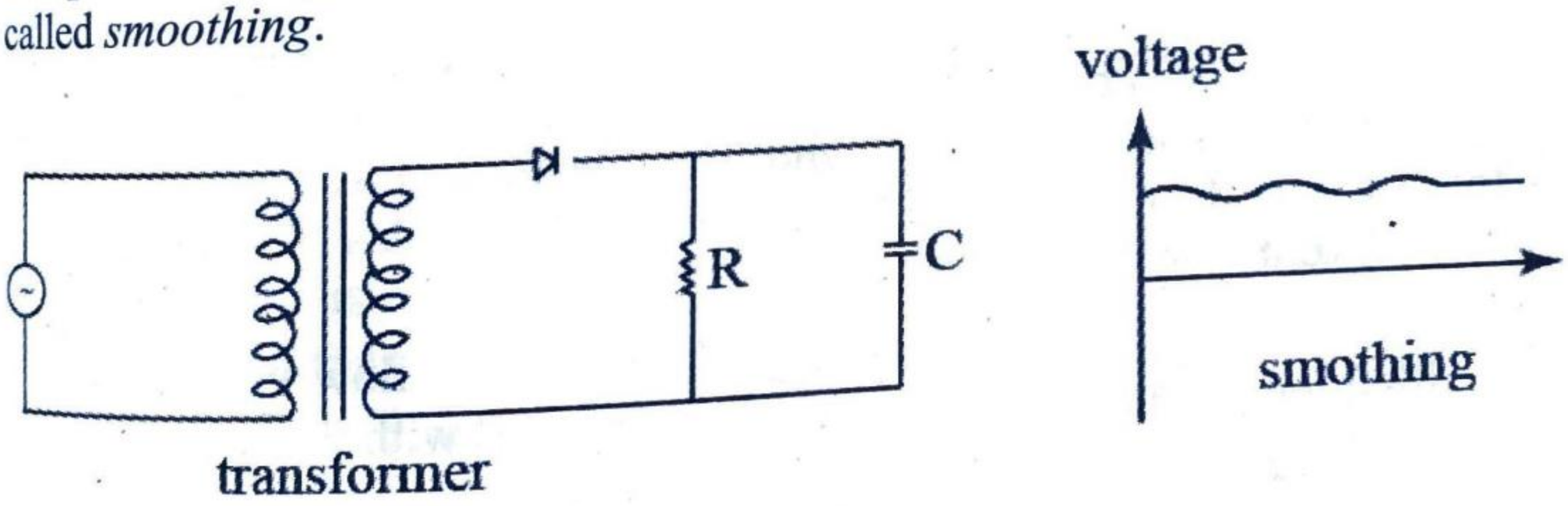
The process of changing an alternating current (*ac*) into a direct current (*dc*) is called ***Rectification.*** This can be in partial, *half wave* or in total, *full wave* rectification.

Half wave rectification

One diode is used in the circuit, and since current flows through the diode in a forward bias only, the *ac* is changed to *dc*by half only.



If you connect a lamp in this circuit, it will be 'on' for half time and 'off' the other half. But if you connect a capacitor in parallel to the resistor, the lamp will be on for a long time. The capacitor removes the humps of the output considerably making it more smooth. This action is called *smoothing*.



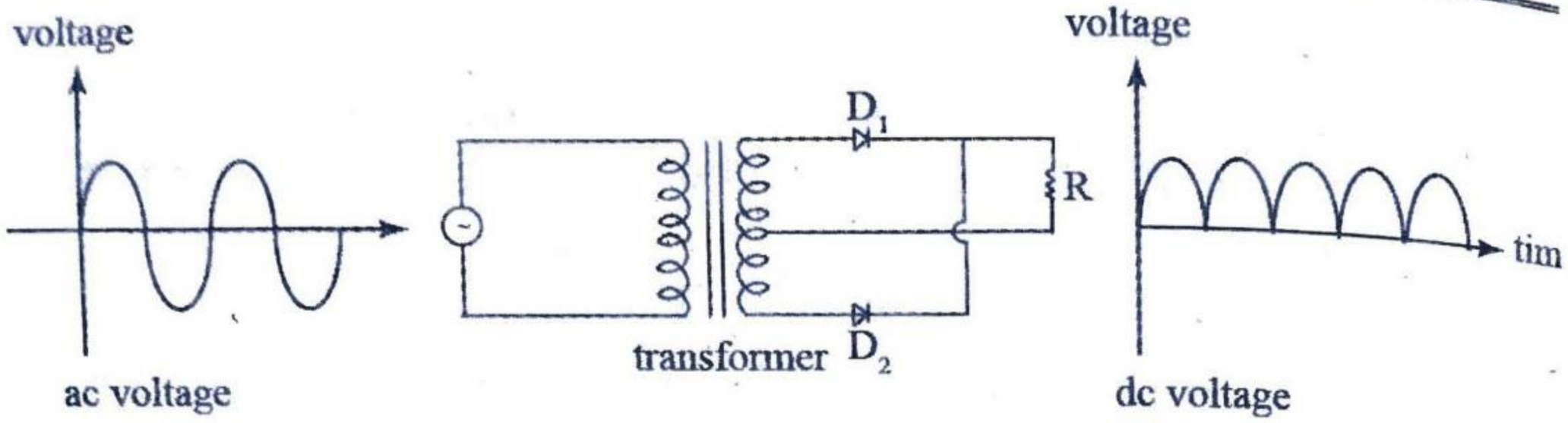
Conceptual Example

11. What is main disadvantage of a half wave rectification?

Explanation:The output voltage is non-zero for only half the time. Hence there is no continues flow of current in the circuit.

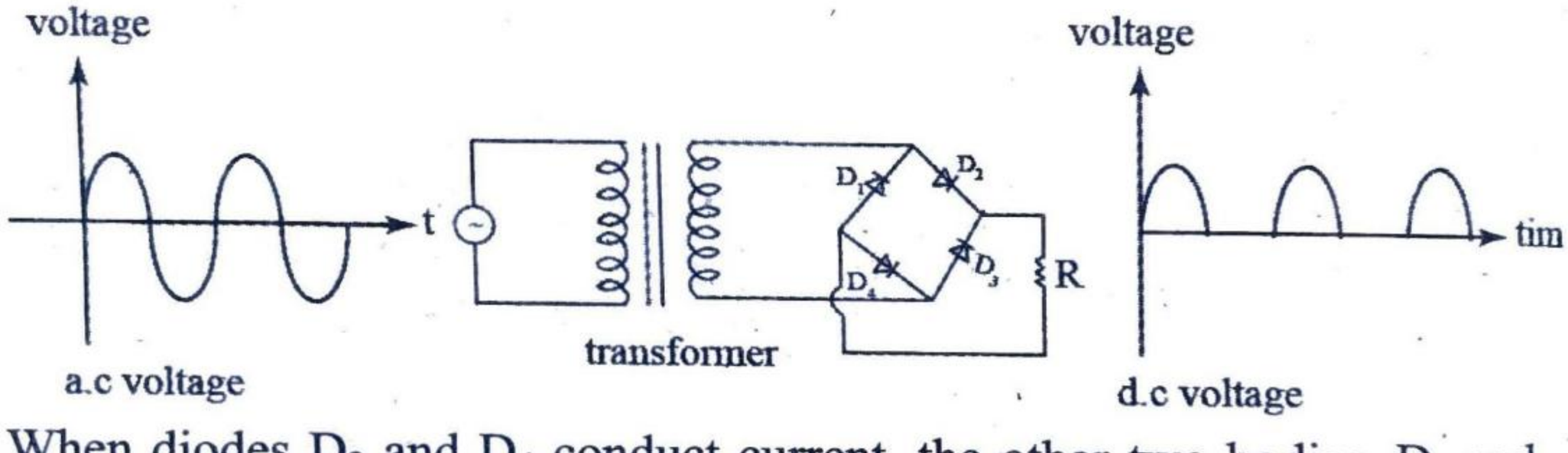
Full wave rectification

It can be achieved used two or four diodes. First let's see full wave rectification using two diodes.



When current flows through D_1, there is no current flow through diode D_2. Similarly, current doesn't flow through D_1 when it flows through D_2 (why?). As a result of this, there is current flow through the resistor throughout the full time.

Another way of obtaining a full wave rectification involves using four diodes (*bridgearrangement*).



When diodes D_2 and D_4 conduct current, the other two bodies, D_1 and D_3 do not conduct. Similarly, when current flows through D_1 and D_3, it doesn't flow through D_2 and D_4. At the end of the day, the circuit end up with current flow for full time.

Note that: For further smoothing of the current flow in the full wave rectification, a capacitor is connected in parallel with the resistor.

13.4 Transistors

A bipolar or junction transistor is an extremely important electronic element formed by sandwitching a semiconductor material of one type between two semiconductors that are opposite to it.

A junction transistor has three layers and hence three terminals. Namely, the base (b), the emitter (e) and the collector (c).

Base (b): Is the middle of the transistor, which is very thin and is doped very lightly. It is used as a bridge for the charge carriers movement between the two terminals or layers.

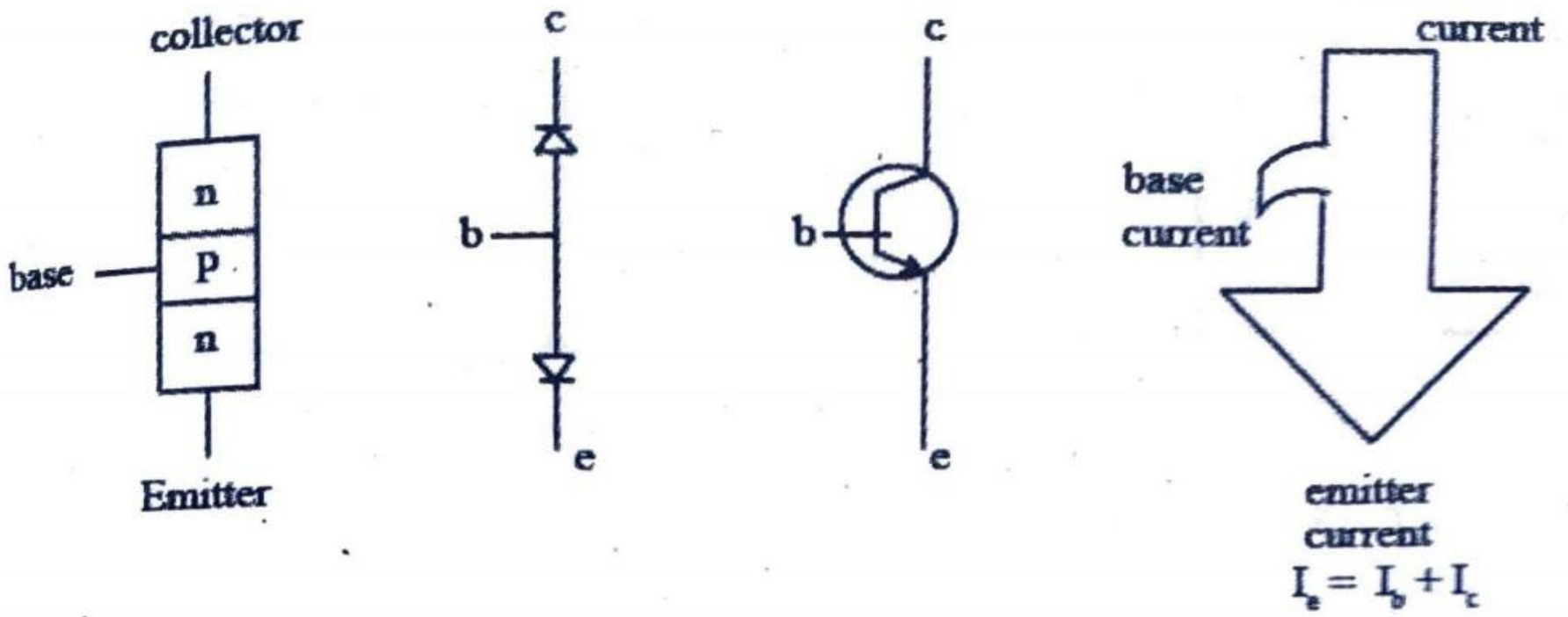
Emitter (e): is one of the outer layer, which is doped heavily since it is the source of charge carriers.

Collector (c): is the other outer layer, which is designed to have larger surface area so as to store much charge carriers.

Broadly speaking, there are two different types of bipolar transistors. Namely, NPN transistor and PNP transistor.

NPN Transistor

An NPN transistor is formed when a thin layer of P-type semiconductor is sandwitched between two thicker layers of N-type semiconductors.

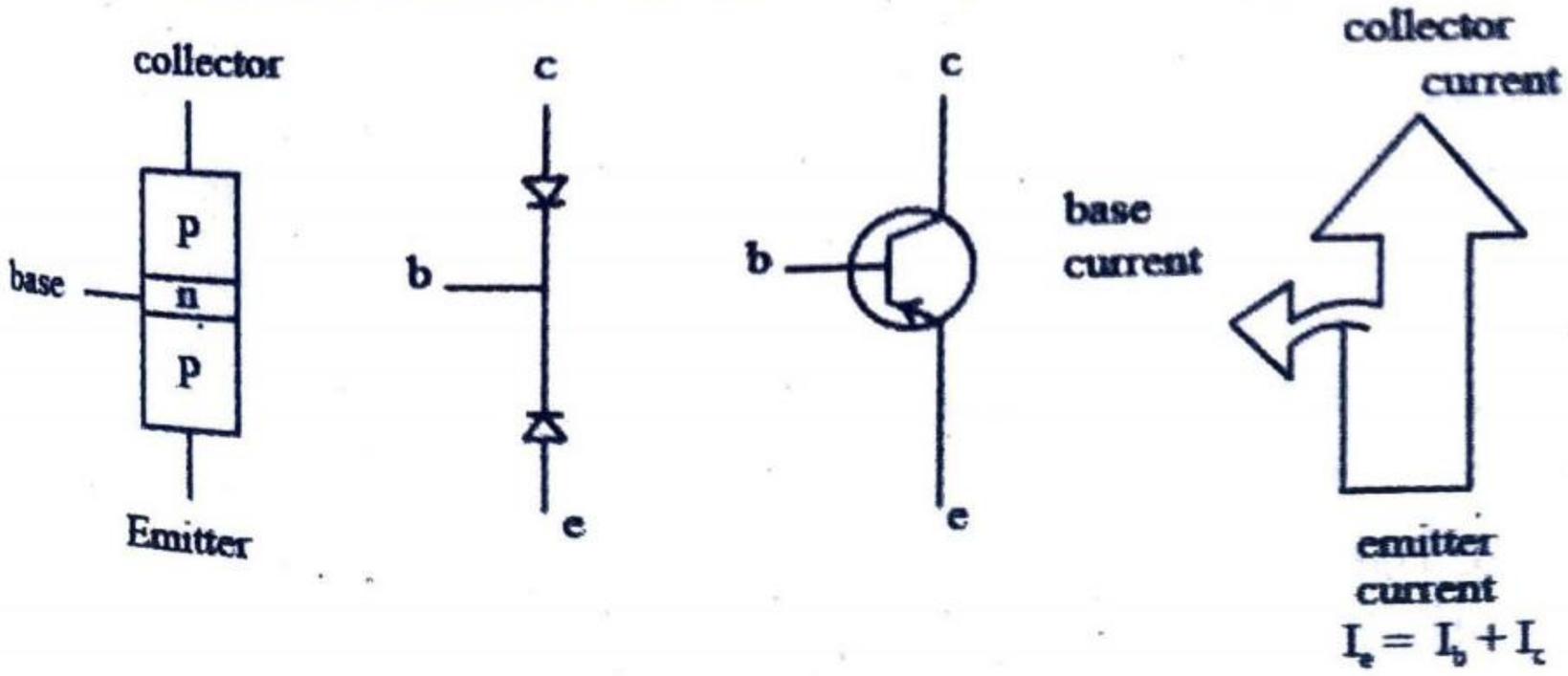


n-p-n transistor symbols

☞ **Remember,** the arrows in the symbols indicate the direction of flow of conventional current (holes). Therefore the arrow always points to the N-type material.

PNP Transistor

A PNP transistor is formed when a thin layer of N-type semiconductor is sandwitched between two thicker layers of P-type semiconductors.



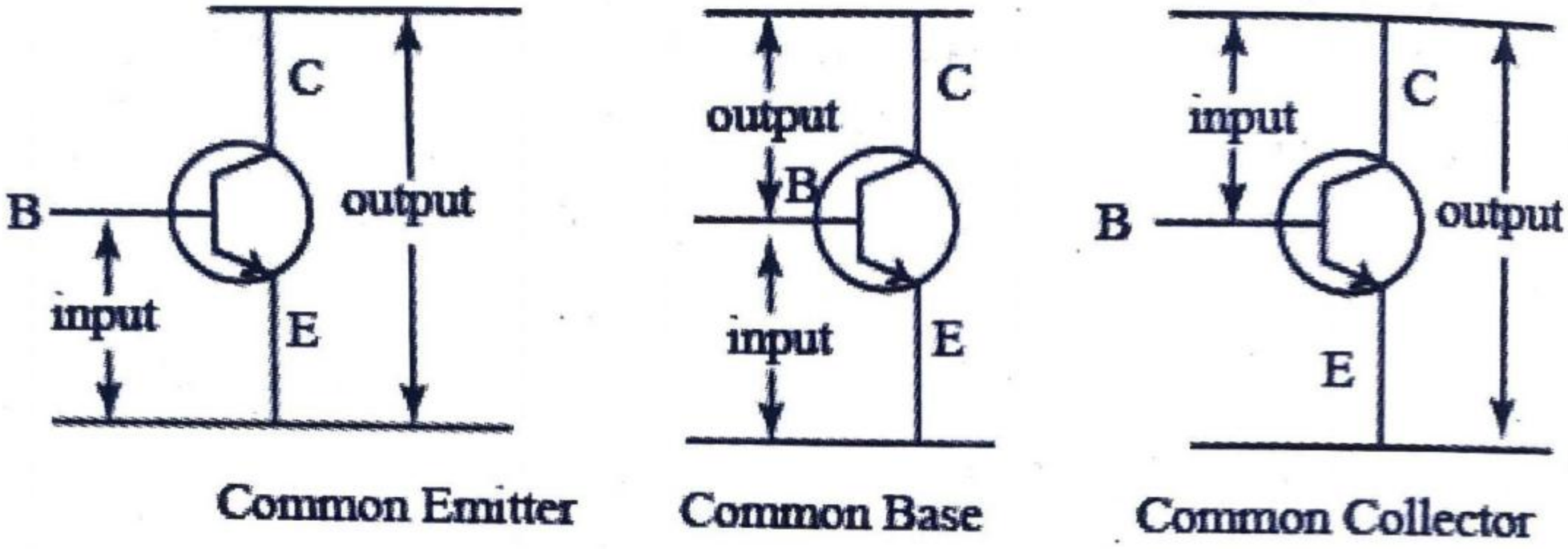
p - n - p transistor symbols

☞ **Tip:** An easy way of identifying a transistor from its circuit symbol is using the direction of the arrow. For a p-n-p transistor, the arrow

points into the transistor. So we can read p-n (for p-n-p) as *pointing –in*. For the n-p-n transistor, the arrow points out, so we can read n-p-n as *notpointing-in*

Transistor characteristics

Depending on the common terminal to the input and the output we can have common emitter, common base or common collector circuits.



Common Emitter Common Base Common Collector

Transistor as switch and as amplifier.

Transistors use the input of relatively small signals to control circuits carrying large currents. This makes them very important as switches and amplifiers.

Amplification: is the process of increasing the strength of a signal such as current, voltage or power in a circuit.

Transistors as a current Amplifier

A small change in the base current can control a large current through the collector. The current amplification factor (A_I) or the current gain is given by

$$A_I = \frac{\Delta I_c}{\Delta I_b}$$, it is unitless.

Illustrative Example

12. In a circuit, the collector current changes form 5mA to 30mA when the base current changes form 10μA to 40μA. What is the current amplicfication factor?

Solution: $A_I = \frac{\Delta I_c}{\Delta I_b} = \frac{(30-5)10^{-3}}{(40-10)10^{-6}} = 833.3$

☞ **Note:** the amplification factor is a number without unit. It tells us how much larger the change in the output signal is than the change in the input signal.

Transistor as a voltage Amplifier

The voltage amplification factor (A_V) often called the voltage gain of the amplifier is given by

$$A_V = \frac{\Delta V_{out}}{\Delta V_{in}}$$

Illustrative Example

13. If the output voltage changes by 4V when the input voltage changes by 0.1V. What is the voltage gain?

Solution: $A_V = \frac{\Delta V_{out}}{\Delta V_{in}} = \frac{4}{0.1} = 40$

13.5 Logic gates

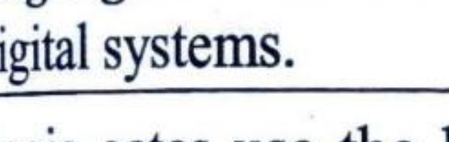
Logic gates: are electronic devices that work like switches or gates in digital systems.

Logic gates use the binary digits 0 and 1for their operation. Where 0 represents 'low', 'off', 'false' or 'no' and 1 represents 'high', 'on, 'true' or 'yes' signals.

There are five types of basic logic gates form which all the more complicated logic gates are constructed. Those are NOT, OR, NOR, AND and NAND gates

NOT (Inverter) gate

It inverts the input signal fed into it. Thus, it changes 0 to 1 and 1 to 0.



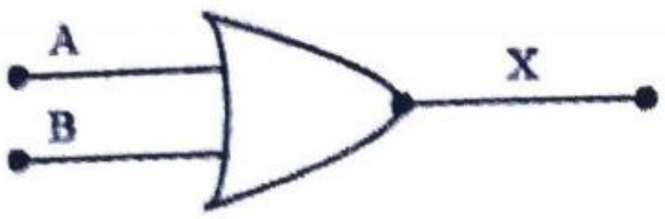
NOT gate symbol

Input (A)	Output (X)
0	1
1	0

Not gate truth table

OR gate

It gives a high (1) output if input A or input B or both are high (1).



OR gate symbol

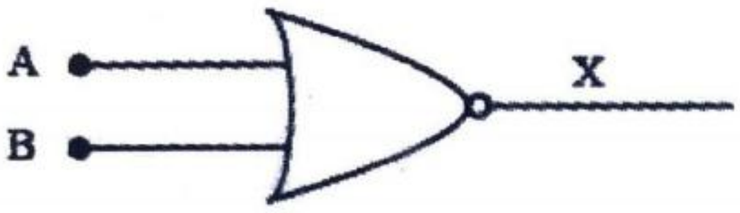
Inputs		Output
A	B	X
0	0	0
1	0	1
0	1	1
1	1	1

OR gate truth table

☞ You can think of an OR gate as two switches in parallel.

NOR (NOT OR) gate

It gives an output which is exactly the opposite of the OR gate



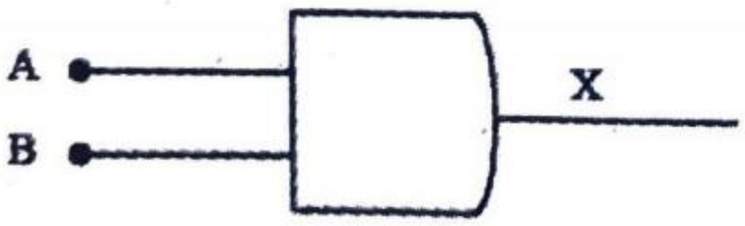
NOR gate symbol

Inputs		Output
A	B	X
0	0	1
1	0	0
0	1	0
1	1	0

NOR gate truth table

AND gate

It gives a high (1) output when both inputs are high



AND gate symbol

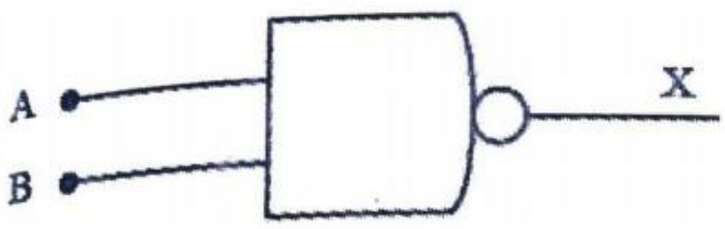
Inputs		Output
A	B	X
0	0	0
1	0	0
0	1	0
1	1	1

AND gate truth table

☞ You can think of an AND gate as two switches in series.

NAND (NOT AND) gate

It gives an output which is exactly the opposite of the AND gate.



NAND gate symbol

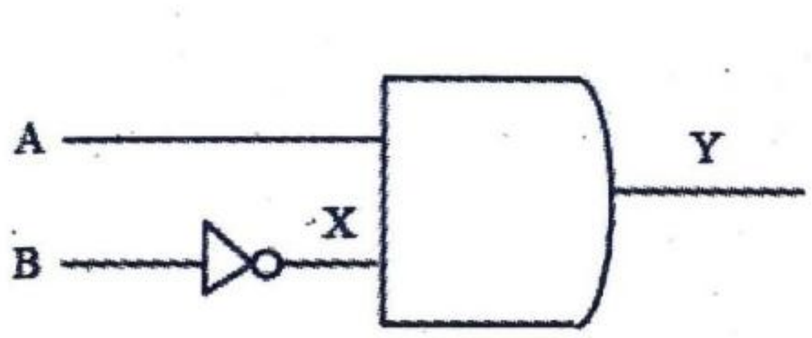
Inputs		Output
A	B	X
0	0	1
1	0	1
0	1	1
1	1	0

NAND gate truth table

Truth table: is a table showing all possible inputs and the resulting output produced by a particular logic gate.

Illustrative Examples

14. Complete the truth table for the system of logic gates shown.

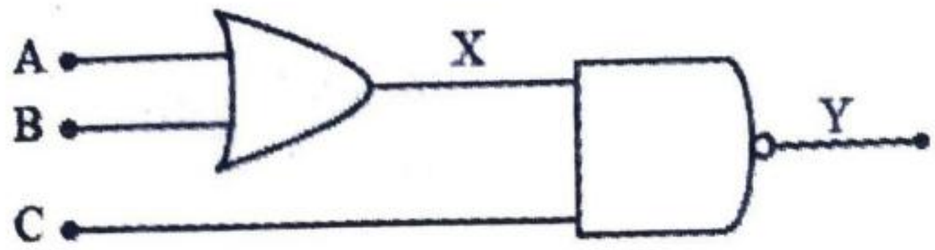


A	B	X	Y
0	0		
1	0		
0	1		
1	1		

Solution:X is the output of input B through the NOT gate and Y is the output of the inputs A and X through the AND gate. So we have

A	B	X	Y
0	0	1	0
1	0	1	1
0	1	0	0
1	1	0	0

15. Complete the truth table for the system of logic gates shown



A	B	C	X	Y
0	0	0		
0	0	1		
0	1	1		
1	0	0		
1	0	1		
1	1	1		

Solution: X is the output of input A and B through the OR gate and Y is the output of the inputs X and C through the NAND gate.

A	B	C	X	Y
0	0	0	0	1
0	0	1	0	1
0	1	1	1	0
1	0	0	1	1
1	0	1	1	0
1	1	1	1	0

General Secondary Education Certificate Exams

16. The truth table for AND gate is shown. What are the missing outputs

A. M = 0, N = 1
B. M = 0, N = 0
C. M = 1, N = 0
D. M = 1, N = 1

Inputs		Output
X	Y	Z
0	0	0
0	1	M
1	0	N
1	1	1

Solution: The output of AND gate is 1 when both inputs are 1 otherwise 0.

Answer: B

17. Which logic gates has the following truth table?

Inputs	Output
0	O
1	1

A. AND
B. NOR
C.
D. NOT

Answer: D

18. Digital systems are referred as binary systems because

A. They are made from two parts
B. They have two terminals
C. The can have only two values
D. Electrons and holes exist in them.

Explanation: digital systems use only 0 and 1

Answer: C

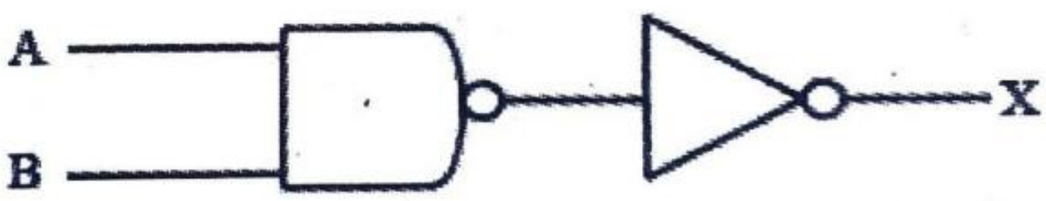
19. The truth table represents

A	B	?
0	0	1
0	1	1
1	0	1
1	1	0

A. AND gate
B. NAND gate
C. OR gate
D. NOR gate

Answer: B

20. For the digital circuit shown, output is found to be x = 1. What are the values of inputs A and B?



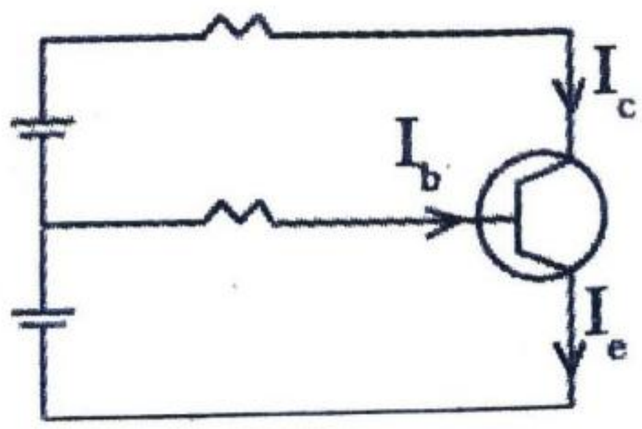
A. A = 0, B = 0
B. A = 1, B = 0
C. A = 0, B = 1
D. A = 1, B = 1

Solution:

A	B	AND	NAND	NOT(X)
0	0	0	1	0
0	1	0	1	0
1	0	0	1	0
1	1	1	0	1

Answer: D

21. Which one of the following is correct about the current in the circuit shown



A. $I_b = I_e + I_c$
B. $I_e = I_b + I_c$
C. $I_b = I_e + I_c = 0$

D. $I_c = I_b + I_e = 0$

Answer: B

22. The collector current of a transistor used as an amplifier changes from 6mA to 24mA as the base current changes from 10μA to 50μA. What is the current amplification factor of the transistor?

A. 600
B. 450
C. 5
D. 4

Solution: Current amplification $= \frac{\Delta I_c}{\Delta I_b} = \frac{240\times10^{-3} - 6\times10^{-3}}{50\times10^{-6} - 10\times10^{-6}}$

Current amplification = 450.

Answer: B

23. Which one of the following electronic components can best be used as a switch?

A. Capacitor
B. Diode
C. Resistor
D. Transistor

Answer: D

24. What are the constituents of the beam in cathode ray tube?

A. Light
B. Ions
C. Protons
D. Electrons

Answer: D

25. The source of electrons in thermionic diode is

A. a hot metal
B. a p type semiconductor
C. an n type semiconductor
D. an impurity atom

Answer: A

26. A small amount of impurity is added to an intrinsic semiconductor to increases its

A. mechanical strength
B. resistance to heat
C. electrical conductivity
D. reliability

Answer: C

27. A semi-conductor diode is made of

A. Doping elements
B. N type material
C. p type material
D. p-n junction

Answer: D

28. When does a NOR gate have a high output?

A. When either input is low
B. When either input is high
C. When both input are low
D. When both input are high

Answer: C

29. In a thermionic diode, once the current reaches saturation, what can be done to increase the anode current?

A. Increase the heater voltage
B. Decrease the cathode voltage
C. Increase the anode voltage
D. Decrease the anode temperature

Answer: A

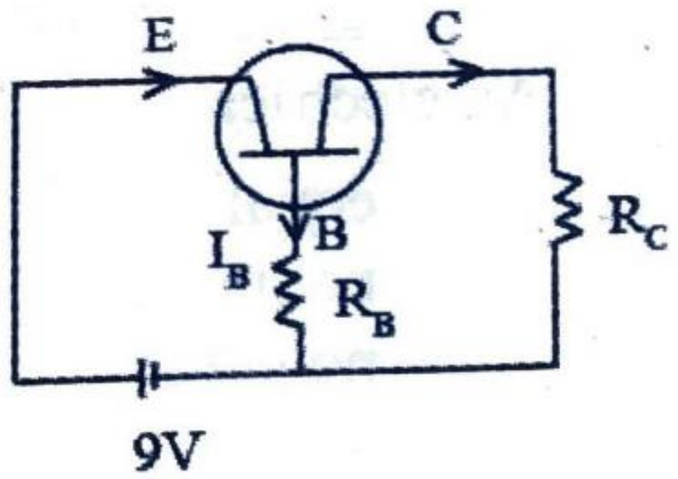
30. In thermionic diode, the voltage supplied to the heater is

A. anode voltage
B. cathode voltage
C. base voltage
D. triode voltage

Answer: B

31. For the pnp transistor circuit, the depletion voltage is 0.6V and the base resistance is 280kΩ what is the value of the base current I_B?

A. 3×10^{-3}A
B. 3.4×10^{-5}A
C. 3×10^{-5}A
D. 3.2×10^{-2}A

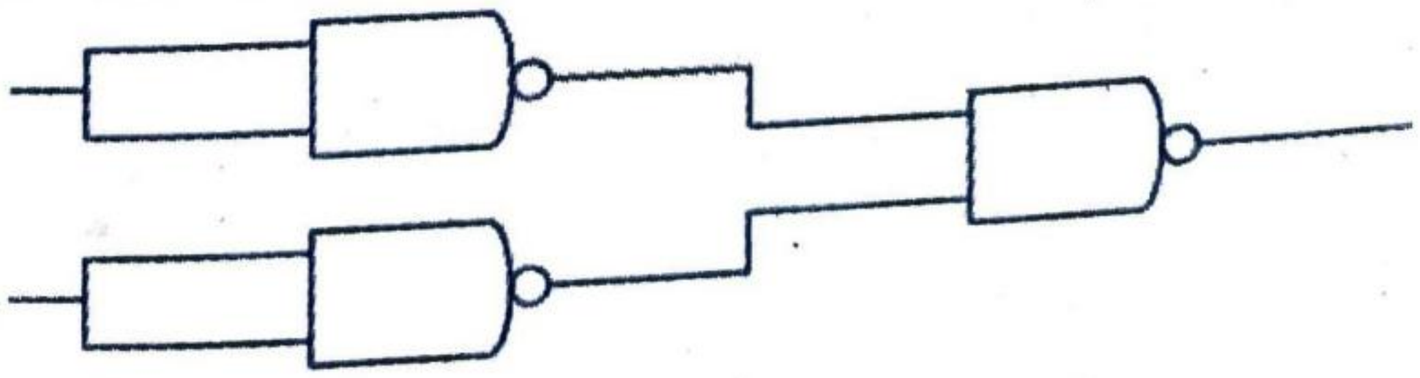


Solution:

$$I_B = \frac{V_B}{R_B} = \frac{9-0.6}{280,000} = 3\times10^{-5}\text{A}$$

Answer: C

32. The diagram below represents a combination of logic gates. Which one of the following gates is equivalent to the combination



A. OR
B. NOR
C. AND
D. NOT

Explanation: The inputs in the first two NAND gates are the same, and the final result is as an OR gate.

Answer: A

33. When a p-n junction is forward biased

A. Electrons flow from p-type to n-type
B. Current cannot flow through the dioxide
C. The depletion layer size increases
D. The depletion layer size decrease

Answer: D

34. …on a cathode Ray Oscilloscope, the distance between crests is 4cm. if the time base is set at $10mscm^{-1}$, what is the period of the wave.

A. 0.04sec
B. 0.4sec
C. 2.5sec
D. 25sec

Solution:

$$V = Sf = \frac{S}{T} \Rightarrow \frac{1}{V} = \frac{T}{S} \Rightarrow T = \left(10 * 10^{-3}\right)(4) = T = 0.04\,\text{sec}$$

Answer: A

Additional Examples

35. An electrically neutral semiconductor has

A. equal amounts of negative and positive charges
B. no minority charge carriers
C. no majority charge carriers
D. no free charges

Answer: A

36. The barrier potential in a semiconductor diode opposes.

A. Majority carriers in N region
B. Majority carriers in P region
C. Majority carriers in both region
D. Majority carriers in both region

Answer: C

37. The reverse saturation current in a PN junction diode is due to only the
A. minority carriers
B. acceptor ions
C. majority carriers
D. donor ions
Answer: A

38. A hole diffuses from the p-side to the n-side in a p-n junction. This means that
A. A bond is broken on the n-side and electrons jump on it
B. A bond is broken on the p-side and electrons jump from it
C. A bond is broken on the n-side and electrons jump to the p-side
D. A conduction electron on the p-side jumps to n-side
Answer: C

39. With rise in temperature, resistance of semiconductors
A. increases
B. decreases
C. remains unchanged
D. first decreases and then increases
Answer: B

40. When a PN junction diode is forward biased, the flow of current across the junction is mainly due to
A. drift of charges
B. diffusion of charges
C. both drift and diffusion
D. nature of material
Answer: B

41. A PN junction diode cannot be used
A. as rectifier
B. for converting light energy to electrical energy
C. For getting light radiation
signal

42. To use a transistor as an amplifier
A. emitter-base junction is forward biased and collector base junction is reverse biased.

B. both junctions are forward biased
C. both junctions are reverse biased
D. it always works as amplifier regardless of being biased.

Answer: A

43. When an impurity is doped into an intrinsic semiconductor the conductivity of the semiconductor

A. increases
B. decreases
C. remains the same
D. become zero

Answer: A

44. An NPN transistor is biased to work as an amplifier. Which of the following statement is not correct?

A. The electrons go from base region to collector region
B. The electron go from emitter region to base region
C. The electrons go from the collector region to base region
D. The holes go from base region to emitter region

Answer: C

45. Diffusion current in a p-n junction is greater than the drift current in magnitude

A. If the junction is forward biased
B. If the junction is reverse biased
C. If the junction is unbiased
D. In no case

Answer: A

46. Which of the following is based on photovoltaic effect?

A. Solar cell
B. Zener diode
C. light emitting diode
D. PNP transistor

Answer: A

47. Electric conduction in a semiconductor takes place due to

A. Electrons only
B. Holes only
C. Both electrons and holes
D. Neither electrons nor holes

Answer: C

48. Silicon is a semiconductor. If a small amount of (Arsenic) is added to it, then its electrical conductivity

A. Decrease
C. Remains Uncharged

B. Increases D. Becomes zero

Answer: B

49. On increasing the reverse biase to a large value in a PN junction, current

A. increase slowly C. suddenly increases
B. remains fixed D. decrease slowly

Answer: C

50. PN junction is said to be forward biased when

A. The positive terminal of the battery is joined to the p-side and the negative terminal to the N-side
B. The positive terminal of the battery is joined to the N-side and the negative terminal to the p-side
C. The positive terminal of the battery is connected to the N or P side
D. A mechanical force is applied in the forward direction.

Answer: A

51. The valency of an impurity element added to germanium crystal to convert it into p-type semiconductor is

A. 6 B. 4 C. 5 D. 3

Answer: D

52. The majority and minority charge carriers in N-type semiconductors are respectively

A. electrons and electrons C. holes and electrons
B. electrons and holes D. holes and holes

Answer: B

53. An intrinsic semiconductor at absolute zero temperature behaves as

A. Insulator C. Metallic conductor
B. Super conductor D. Semiconductor

Answer: A

54. Iron and silicon wire are heated form 30°C to 50°C. The correct statement is that

A. Resistance of both wires increases
B. Resistance of both wires decreases

C. Resistance of iron wire increases and that of silicon wire decreases
D. Resistance of iron wire decreases and that of silicon wire increases

Answer: C

55. The electrical resistance of the following decreases with rise in temperature

A. Metals
B. Gold
C. Semiconductors
D. constantan **Answer: C**

56. The following truth table is for

		A	B	Y
A.	OR gate	0	0	1
B.	AND gate	1	0	0
C.	NOT gate	0	1	0
D.	NOR gate	1	1	0

Answer: D

57. The depletion layer in the P-N junction region is caused by

A. Diffusion of carriers
B. Drift of holes
C. Drift of electrons
D. Migration of impurity ions **Answer: A**

58. An example of a semiconductor is

A. Platinum
B. Germanium
C. Quartz
D. Mica **Answer: B**

59. A semiconductor device is connected in a circuit with a battery and a resistor and a current is found to pass through the circuit. If the polarity of the battery is reversed, the current drops almost to zero. The device may be

A. P type semiconductor
B. N type semiconductor
C. P-N junction
D. Intrinsic semiconductor

Answer: C

60. When the conductivity of a semiconductor is only due to breaking of covalent bonds, the semiconductor is

A. extrinsic
B. intrinsic
C. n-type
D. p-type

Answer: B

61. The impurity atom that should be added to germanium to make it n-type is

A. Arsenic
B. Iodine
C. Aluminium
D. Indium

Answer: A

62. Reverse bias applied to a junction diode

A. Lowers the potential barrier
B. Raises the potential barrier
C. Increases the majority carrier current
D. Increases the minority carrier current

Answer: B

63. To obtain P-type Si semiconductor, we need to dope pure Si with

A. Aluminium
B. Phosphorus
C. Arsenic
D. Germanium

Answer: A

64. The part of a transistor which is most heavily doped to produce large number of majority carriers is

A. emitter
B. base
C. collector
D. any one of the three terminals

Answer: A

65. A pure semiconductor has

A. Infinite resistance
B. No resistance
C. Finite resistance that decrease with temperature
D. Finite resistance that increase with temperature

Answer: C

66. To obtain electrons as majority charge carriers in a semiconductor, the impurity mixed is

A. Monovalent
B. Divalent
C. Trivalent
D. Pentavalent

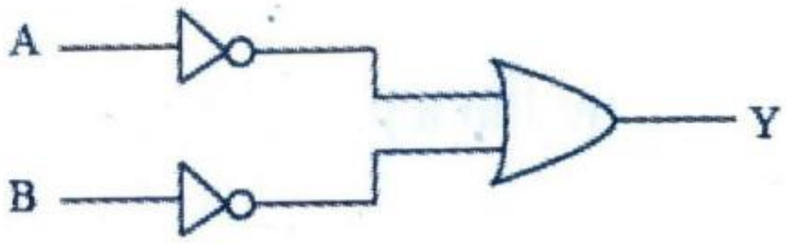
Answer: D

67. Bonding in a germanium crystal is

A. Metallic
B. Ionic
C. Vander Waal's type
D. Covalent

Answer: D

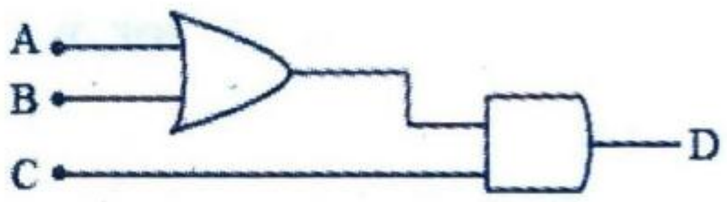
68. The gate represented by the combination is



A. OR
B. AND
C. NAND
D. NOR

Answer: C

69. If A = B = C = 0 and then A = B = 1 and C = 0 then the outputs are respectively



A. 0, 0
B. 0, 1
C. 1, 0
D. 1, 1

Answer: A

70. An AND gate is equivalent to

A. A series switching circuit
B. A parallel switching circuit
C. A single switch
D. A single transistor

Answer: A

71. In the depletion region of unbiased P-N junction there are

A. electrons
B. holes
C. both
D. fixed ions

Answer: D

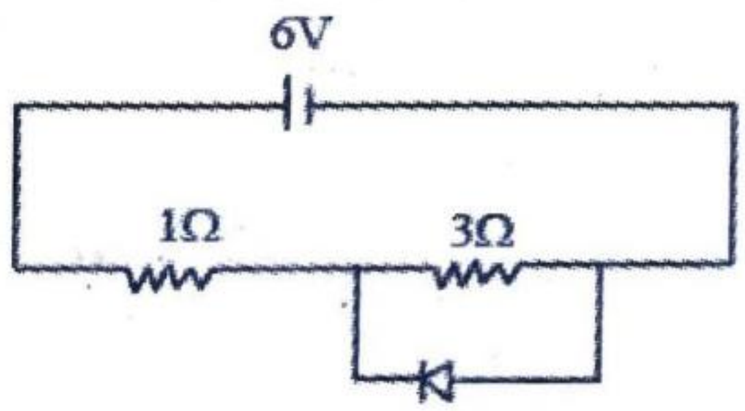
Unit Thirteen: Conceptual Questions

1. If you heat a metal, what will happen to its conductivity? What if you heat semiconductor?
2. Why do impure semiconductors become popular in the electronic industry?
3. Explain why pure semiconductors act as insulator at low temperature.
4. How is the current flow in a PN junction diode? Explain

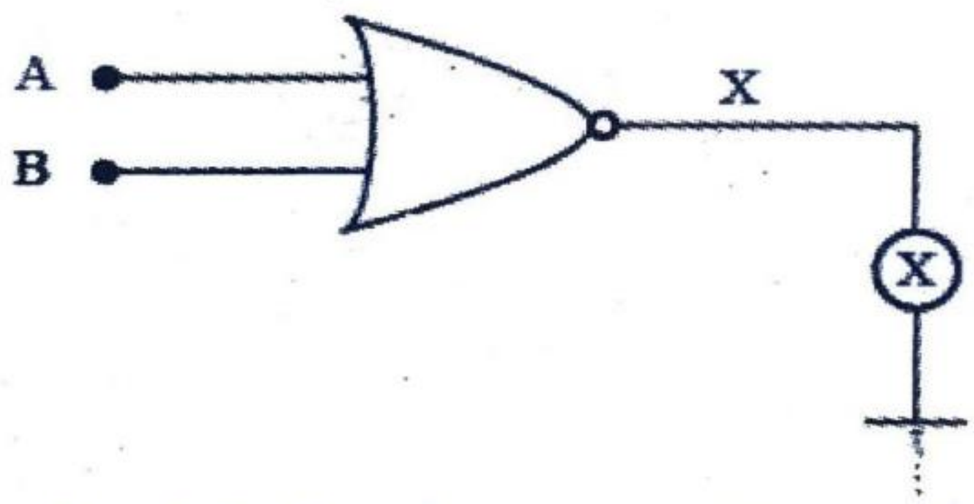
5. Explain the advantage of a full wave rectification over a half wave rectification
6. What is the purpose of connecting a capacitor in both a half wave and full wave rectification
7. Explain why a diode acts as a rectifier
8. What is the difference between d.c obtained from a battery and that obtained from an alternating supply when you add a diode in series.

Unit Thirteen: Problems

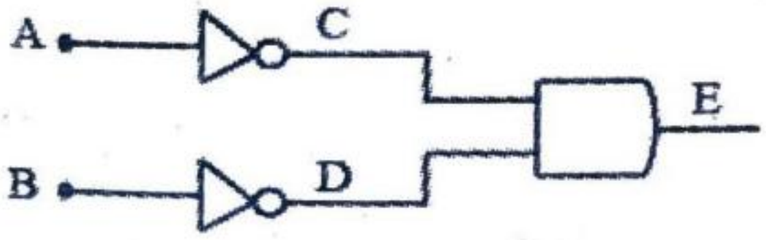
9. a) What size current will be drawn from the battery in the figure shown. Give your reasoning.
   b) If the battery is turned round, what size current will it now supply? Explain



10. A transistor has a current gain of 200. If a current of 3μA is sent into the base, what size collector current would you expect?
11. What inputs will be needed at A and B to light a tiny bulb shown in the figure



12. Complete the truth table for the arrangement of gates shown in the figure



A	B	C	D	E
0	0			
1	0			
0	1			
1	1			

13. A battery is connected to the y terminal of an oscilloscope and the spot deflects 3cm vertically. The sensitivity of the oscilloscope is set at 1v/cm. what is the p.d across the battery.

Unit Thirteen: Explanation to Questions

1. Decreases for a metal and increases for a semiconductor.
2. Because, they can be designed in such a way that the current can be controlled as required
3. Because, at normal condition, there is no free electrons. The valence electrons form covalent bond with in the lattice.
4. It flows from the p-side to the N-side in the forward bias.
5. It supplys continues current in one direction throughout.
6. To smoothen the current and reduce interruption (on, off)
7. Because, it allows current to flow in one direction only
8. The voltage from a battery is steady but when the voltage is obtained from rectification, it is not smooth. It will show a little fluctuation even in the full wave.

Unit Thirteen: Solution to Problems

9. a) In this case, current do not flow through the diode, hence,

$$I = \frac{V}{R} = \frac{16}{1+3} \Rightarrow 1.5A$$

b) Now current flows through the diode and not through the 3Ω resistor. Thus, $I = \frac{V}{R} = \frac{6}{1} \Rightarrow 6A$

10. $A_I = \frac{\Delta I_c}{\Delta I_b} \Rightarrow \Delta I_c = (A_I)(\Delta I_b)(200)(3\mu A) = 600\mu A$

11. To light the bulb, the out put, i,e x should be 1. For this to happen, since the gate is NOR gate, the inputs A and B both should be zero (0).

12. C and D are the outputs of A and B respectively through the NOT gate. And, E is the output of C and D through the AND gate so we have the following.

A	B	C	D	E
0	0	1	1	1
1	0	0	1	0
0	1	1	0	0
1	1	0	0	0

13. V = (3cm) (1v/cm) = 3v

Unit Fourteen

Electromagnetic Waves and Geometrical Optics

Starting from helping in vision, communication and cooking, electromagnetic waves do have so much more practical importance in medicine and defense. Using the laws of reflection and refraction images are produced on optical devices such as the eyeglass, magnifying glass, camera, projector, microscope and telescope, which help us to see the very small and the very far, to see into outer space and into the world of subatomic particles.

14.1 Electromagnetic waves

Electromagnetic waves do not need a medium through which to travel-they can travel through vacuum.

Electromagnetic waves: Are transverse waves produced when a magnetic field and an electric field are at right angle to each other.

Electromagnetic spectrum

The sun emits electromagnetic waves with a wide range of frequencies and wavelengths. This is referred as the *electromagnetic spectrum*.

Electromagnetic Spectrum: is the entire frequency range of electromagnetic waves.

The following is the electromagnetic spectrum.

R-rays	x-rays	ultraviolet rays	visible light	infrared rays	micro waves	radio waves
Wave length 10^{-12}m	10^{-10}m	10^{-8}m	10^{-7}m	10^{-6}m	10^{-2}m	1m

- In vacuum, all electromagnetic waves travel at the same speed of 3×10^{8}m/s.

Some uses of electromagnetic radiation

Radio Waves: They have the longest wavelength in the electromagnetic spectrum. They are used in communication systems.

Micro Waves: They are used in microwave ovens for cooking food, in communication systems, in supermarkets to detect customers, in radar system to detect ships, aircraft and missiles.

Infrared Waves: They are used in infrared cameras, which are used in rescue operations to detect the presence of bodies.

Visible Light: It is most important in vision, that is to see objects in our everyday life.

Ultraviolet rays: They are generated from the sun and electric arcs used for welding. They are sources of vitamin D but too much of them cause skin cancer and blindness. They are also used in sterilization and acceleration of chemical reactions.

X – rays: Are used to take picture of inside of our bodies, such as broken bones. They are also used to kill cancer cells. They are dangerous if not limited.

γ-rays: They have the shortest wavelength in the electromagnetic spectrum and they are most penetrating and most dangerous than x-rays. They are used to kill cancer cells without surgery. They are also used to kill microorganisms in food.

Illustrative Examples

1. A radar wave is transmittted and later refelcted off an aircraft, and received 1.2×10^{3}sec after being sent out. How far is the aircraft?

   Solution: $s = \frac{Vt}{2} = \frac{\left(3\times10^{8}\right)\left(1.2\times10^{-3}\right)}{2} = 180\text{km}$

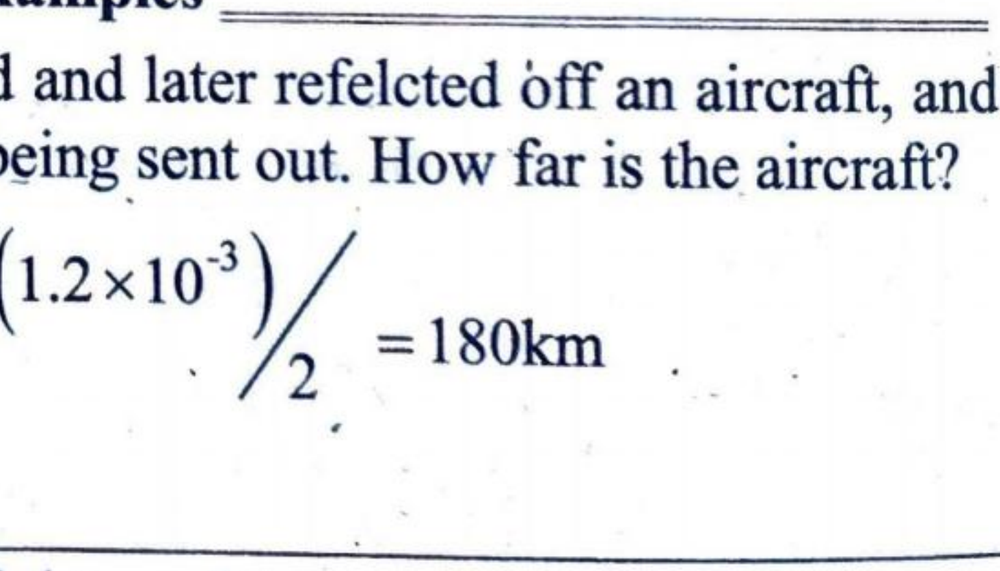
14.2 Reflection of light

Reflection: Is the bouncing off a light ray from a reflection surface.

Laws of reflection

There are *two* laws of reflection. Those are;

1. The angle of incidence equals the angle of reflection $\hat{i} = \hat{r}$
2. The incident ray, the reflected ray and the



normal all lie in the same plane.

Images formed by Mirrors

The laws of reflection are used to form images in plane or curved mirrors.

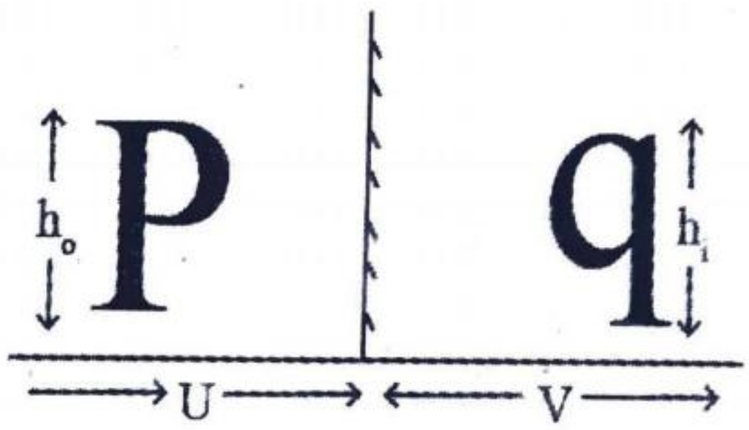
Image: Is an optically formed reproduction of an object

Images formed by a Plane Mirror

Plane mirror: Is a flat and smooth reflecting surface.

Images formed by a Plane Mirror is always

- At the same distance from the mirror as the object (v = u)
- The same size as the object (hi = ho)
- Virtual, cannot be seen on a screen
- Upright or erect not inverted, but
- Laterally inverted, that means, the right side of an object becomes the left side of its image, and vice versa. Similarly, front to back inversion also takes place.



☞ The mirror you looked at yourself in a bathroom this morning is a plane mirror.

Conceptual Examples

2. Does a plane mirror converge or diverge parallel light rays?

Explanation:Parallel light rays remain parallel in plane mirror.

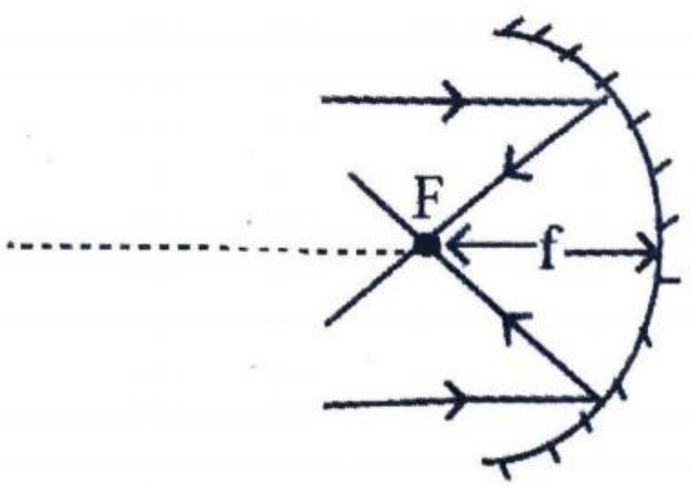
Spherical (curved) mirrors.

Curved mirrors such as *concave mirror* and *convex mirror* are made from a spherical surface.

Concave (Converging) mirror

Concave mirror: Is a mirror with a reflecting surface that curvesinwards away from the light source.

A concave mirror converges (brings together) parallel rays to a point called ***focal point (F)***, that is why we call, concave mirror a *convergingmirror*.



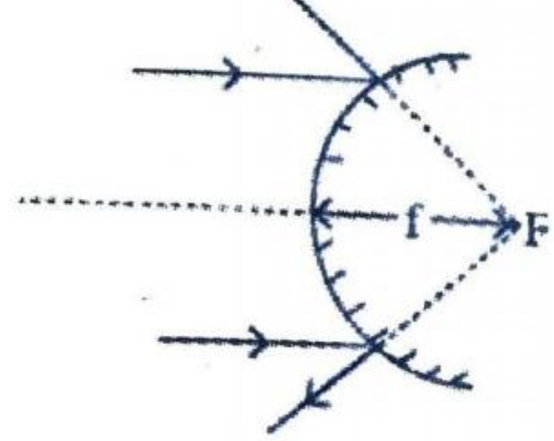
Convex mirror

☞ Concave mirrors are usedin telescope, in torch light, in reflector and as a shaving mirror.

Convex (diverging) mirror

Convex mirror: Is a mirror with a reflecting surface that bulges outwards towards the light source.

A convex mirror diverges (spreads out) parallel rays. For this reason, we call convex mirror a diverging mirror.



☞ Convex mirrors are used as side mirrors of a car (spokio), which helps the driver to control the car from any possible collision from behind,they are also used in stores and supermarkets to control customers. Convex mirror

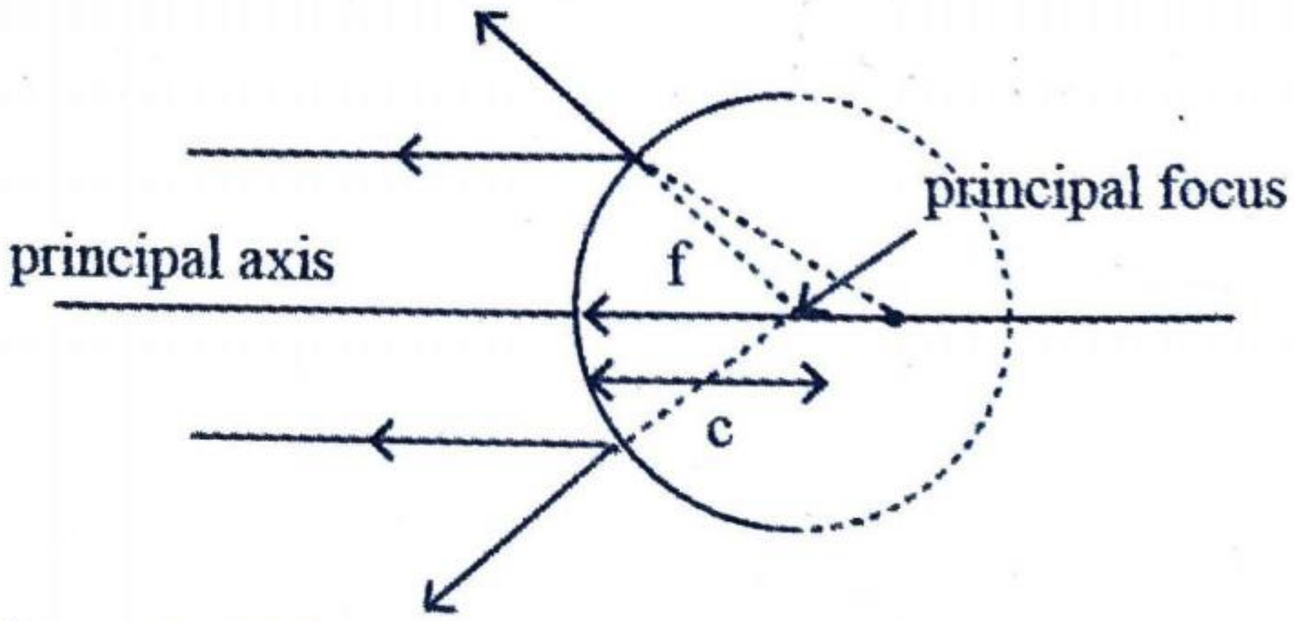
☞ **Terms used in concave and convex mirrors**

Principal axis: Is the line passing through the center of curvature of a curved mirror.

Radius of curvature (C): Is the radius of the sphere from which the curved mirror is formed.

Principal focus (F): Is the point to which reflected rays converge.

Focal length (f): Is the distance from the vertex of the mirror to the principal focus.



→ The focal length (f) is half of the radius of curvature (c)

$$f = \frac{c}{2}$$

Images formed by a given mirror (lens) may be real or virtual.

Real Image: Is an image that is formed when actual light rays meet, and it can be projected on to a screen.

Virtual image: Is an image that is formed without the meeting of actual rays but from the extension of the reflected rays, and it cannot be projected onto a screen.

The mirror equation

The mirror equation enables us to find out where and how the image is formed. It is given by

$$\boxed{\frac{1}{f} = \frac{1}{u} + \frac{1}{v}} \leftarrow \text{mirror equation}$$

Where f, is the focal length

u, is the object distance

v, is the image distance

Magnification (m): is the ratio of image height to object height or image distance to object distance.

$$\text{magnification} = \frac{\text{image height}}{\text{object height}} = \frac{\text{-image distnace}}{\text{object distance}}$$

$$\boxed{m = \frac{h_i}{h_o} = \frac{-v}{u}}$$

☞ **Sign convention**

When we use the mirror equation, we need to make use of the following sign conventions.

Focal length

f is positive for concave mirror and negative for convex mirror.

Image distance

V is positive for real image and negative for virtual image.

Object distance

U is positive for real object and negative for virtual object

Magnification

M is positive for upright (erect) image and negative for inverted image

☞ **Comparing images formed by concave and convex mirrors.**

- The type, size and position of the image formed by a concave mirror depends on the object distance. So, the image can be real or virtual, erect or inverted, smaller, larger or the same size as the object. But,
- A convex mirror always forms virtual, erect and diminished image which is formed between the mirror and the focal point.

Conceptual Example

3. Suppose you want to get fire from the sun. Which one concave or convex mirror will help? Explain

   Explanation:Concave. Because it focuses the sun rays to a point and hence can start fire.

Illustrative Examples

4. a) what is the focal length of converging mirror whose radius of curvature is 20cm?

   b) what is the nature and location of an image if an object is placed 15cm from this mirrors.

   Solution:a) $f = \frac{c}{2} = \frac{20}{2} = 10\text{cm}$

   b) $\frac{1}{f} = \frac{1}{u} + \frac{1}{v} \Rightarrow \frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{10} - \frac{1}{15} = \frac{3-2}{30} = \frac{1}{30}$

   $\frac{1}{v} = \frac{1}{30} \Rightarrow v = 30\text{cm}$, real image

5. Find the position of the image if an object is located 4cm from a convex mirror whose focal length is 6cm.

   Solution: $\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \Rightarrow \frac{1}{f} - \frac{1}{u} = -\frac{1}{6} - \frac{1}{4} = \frac{-2-3}{12}$

   $\frac{1}{v} = \frac{-5}{12} \Rightarrow v = -\frac{12}{5} = -2.4\text{cm}$, virtual image

6. A candle 6cm high is located 60 cm from a concave mirror whose focal length is 20cm. find the postion, nature and size of the image.

Solution: $\frac{1}{f}=\frac{1}{u}+\frac{1}{v}\Rightarrow\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=\frac{1}{20}-\frac{1}{60}=\frac{3-1}{60}=\frac{2}{60}$

$$\frac{1}{v}=\frac{1}{30}\Rightarrow v=30\text{cm, real image}$$

and $m=\frac{h_i}{h_o}=\frac{-v}{u}\Rightarrow h_i=\left(\frac{v}{u}\right)(h_o)=-\left(\frac{30}{60}\right)(6)$

$$h_i=-3\text{cm, inverted}$$

7. How far should a pencil be held form a convex mirror of rasius 40cm to form an image one half the size of the pencil?

Solution: $\frac{1}{f}=\frac{1}{u}+\frac{1}{v}$, but $v=\frac{u}{2}, f=\frac{c}{2}=\frac{40}{2}=20\text{cm}$

$$\frac{1}{f}=\frac{1}{u}+\frac{1}{\frac{u}{2}}\Rightarrow\frac{1}{f}=\frac{1}{u}-\frac{1}{u}=\frac{-1}{u}$$

$$\Rightarrow-\frac{1}{20}=\frac{1}{u}\Rightarrow u=20\text{cm}$$

14.3 Refraction of Light

Refraction: Is the change in direction of light due to a change in speed as it moves from one medium to another.

Snell's law and the refractive index

Snell's law, states that;

"the ratio of the sine of the angle of incidence to the sine of the angle of refraction is constant."

Mathematically, Snell's law is generally given by

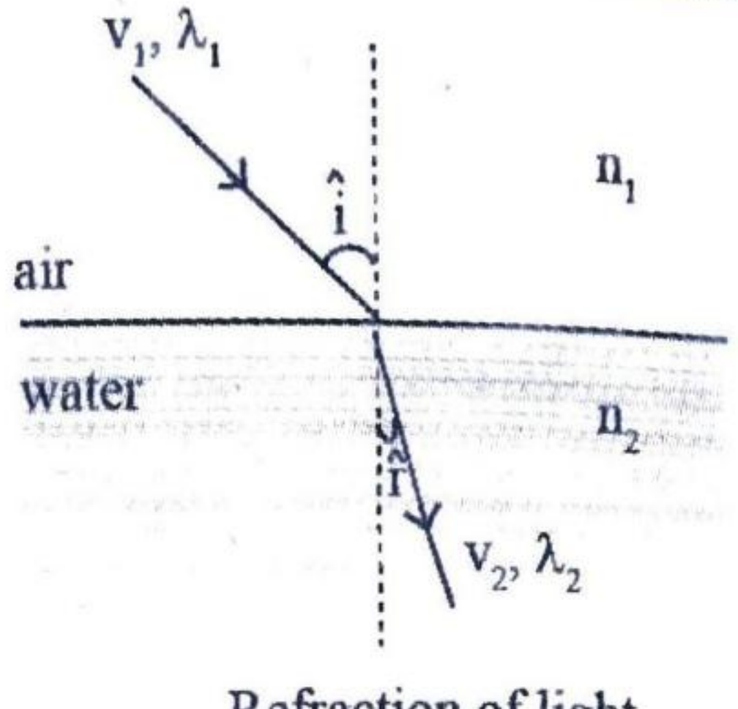
$$\frac{\sin \hat{i}}{\sin \hat{r}} = \frac{n_2}{n_1} = \frac{v_1}{v_2} = \frac{\lambda_1}{\lambda_2}$$

Where 1 and 2 indicates medium one and medium two

n, refractive index of each medium

v, speed of light in each medium

λ, wavelength of light in each medium



Refraction of light

Refractive index (n): Is a measure of the extent to which a medium bends light as it passes through it.

Illustrative Examples

8. If the refractive index of water is a $4/3$ and the angel of incidence is 30°. Find.

   a) The speed of light in water
   b) The angle of refraction

Solution:

a) $$\frac{n_2}{n_1} = \frac{v_1}{v_2} \Rightarrow v_2 = (v_1)\left(\frac{n_2}{n_1}\right) = \left(30\times10^8\right)\left(\frac{1}{4/3}\right) = 2.25\times10^8 \text{ m/s}$$

b) $$\frac{\sin i}{\sin r} = \frac{n_2}{n_1} \Rightarrow \sin r = (\sin i)\left(\frac{n_1}{n_2}\right) = \frac{1}{2}\times\frac{3}{4} = 0.375$$

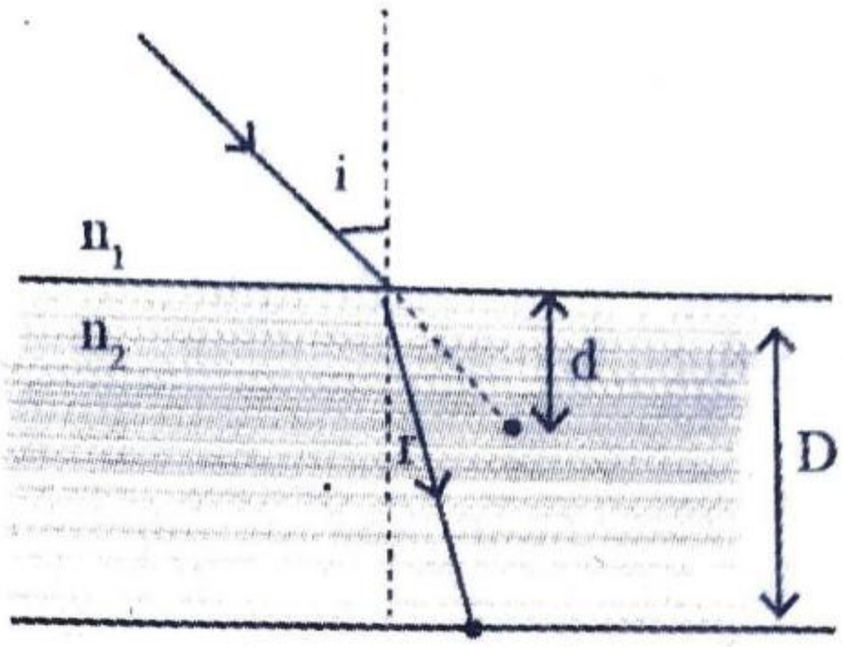
$$\Rightarrow \hat{r} = 22°$$

9. A light of wavelenght 640nm passes form air into glass of index of refraction 1.5. What is the wavelegnth of the light.

Solution: $$\frac{n_2}{n_1} = \frac{\lambda_1}{\lambda_2} \Rightarrow \lambda_2 = (\lambda_1)\left(\frac{n_2}{n_1}\right) = (640)\left(\frac{1}{1.5}\right) = 427\text{nm}.$$

Apparent Depth

Refraction causes an object submerged in a liquid of higher index of refraction to appear closer to the surface than it actually is. We call this depth *apparent depth*.



Apparent depth (d)

Mathematically, it is given by

$$\frac{\sin i}{\sin r} = \frac{n_1}{n_2} = \frac{d}{D}$$

Where d, is the apparent depth

D, is the actual (real) depth

Illustrative Example

10. A coin rests on the bottom of a container filled with water of n = 1.33. The apparent depth of the coin is 9cm. How deep is the container?

Solution: $\frac{n_1}{n_2} = \frac{d}{D} \Rightarrow D = (d)\left(\frac{n_1}{n_2}\right) = (9)\left(\frac{1.33}{1}\right) = 12\text{cm}.$

Total Internal Reflection

☞ Total internal reflection to occur, two conditions must be full filled.

1. Light must travel from denser medium such as water or glass into less dense medium such as air.
2. The angle of incidence must be greater than the critical angle (θ_c)

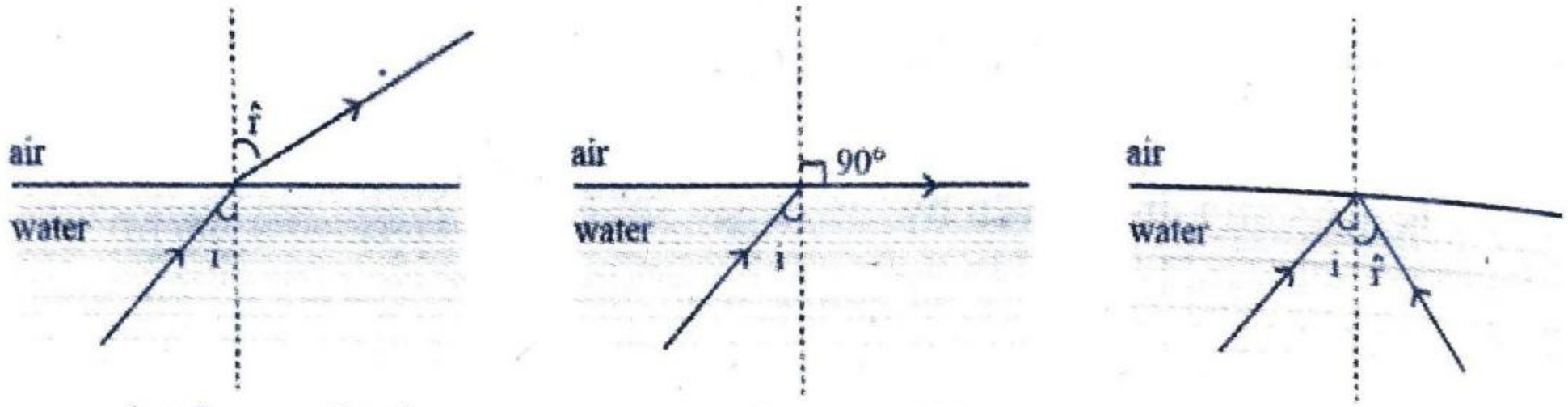
Critical angle(θ_c): Is the angle of incidence for which the angle of refraction is 90°.

Here, at the critical angle, Snell's law modifies to;

$$\frac{\sin i}{\sin r} = \frac{n_2}{n_1} \Rightarrow \frac{\sin \theta_c}{\sin 90^\circ} = \frac{n_2}{n_1}, \text{ but } \sin 90^\circ = 1$$

$$\Rightarrow \boxed{\sin\theta_c \Rightarrow \frac{n_2}{n_1}}$$

The following diagram shows us how refraction and total internal reflection occurs.



$i < \theta_c \rightarrow$ refraction

$i = \theta_c$ and $r = 90°$ → refraction

$i > \theta_c \rightarrow$ total interal refraction

Illustrative Example

11. What is the critical angle for a glass to air surface if the refrative index of glass is 1.5.

Solution: $\sin\theta_c = \frac{n_2}{n_1} = \frac{1}{1.5} = 0.667$

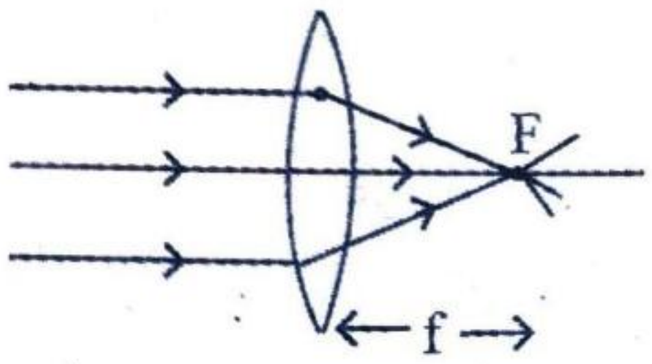
$\theta_c = 42°$

Convex and concave lenses

Mirrors form images due to reflection of light while lenses form images due to refraction of light

Convex (converging) lens

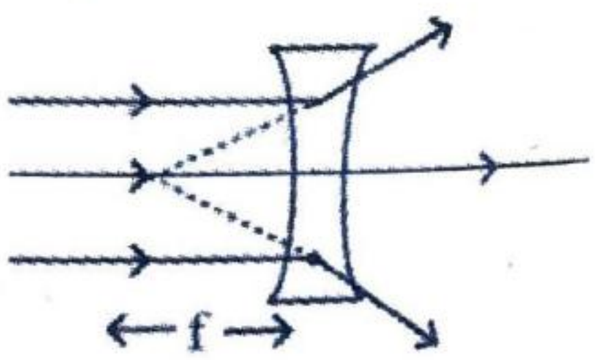
Convex lens is thicker at its centre than at its edges and converges parallel light rays and hence it is called*converging*lens



convex lens

☞ Convex lenses are used in magnifying glass, camera, microscope, telescope etc

Concave(diverging) lens is thicker at its edges than at its center and diverges parallel light rays and hence it is called *Diverging* lens.



concave lens

☞ Concave lens is used when large field of view is needed such as to correct short sightedness.

The power of a lens: is the measure of the strength of the lens to converge or diverge light.

Mathematically, it is given by

$$\Rightarrow \boxed{\text{Power of lens} = \frac{1}{\text{focal length}} = \frac{1}{f}}$$

The unit of power of lens is dioptres (D), thus

$$\boxed{1D = \frac{1}{m} = m^{-1}}$$

☞ **Note that,** the shorter the focal length, the powerful the lens.

Illustrative Example

12. What is the power of a lens whose focal length is 10cm.

Solution: Power of lens $= \frac{1}{f} = \frac{1}{0.1} = 10$ dioptres.

The lens formula.

The lens formula is the same as the curved mirror formula

$$\boxed{\frac{1}{f} = \frac{1}{u} + \frac{1}{v}} \leftarrow \text{lens formula}$$

and magnification, $\boxed{m = \frac{h_i}{h_o} = \frac{-v}{u}}$

☞ **Sign convention**

When we use the lens equation, we use the same sign convention as the mirror equation except that:

Focal length

f is positive for convex lens and negative for concave lens.

☞ **Comparing image formed by convex and concave lens.**

- The nature, position and size of the image formed by a convex lens depends on the position of the objects from the lens. Hence it can be real or virtual, erect or inverted, larger, smaller or the same size as the object.

- The image formed by a concave lens is always erect, virtual and diminished for all positions of the object.

Illustrative Example

13. An object 4cm high is located 10cm from a converging lens of focal length 20cm. what is the nature, size and location of the image?

Solution: $\frac{1}{f}=\frac{1}{u}+\frac{1}{v}\Rightarrow\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=\frac{1}{20}-\frac{1}{10}=\frac{1-2}{20}=-\frac{1}{20}$

$$\frac{1}{v}=-\frac{1}{20}\Rightarrow v=-20\text{cm, virtual}$$

$$\frac{h_i}{h_o}=\frac{-v}{u}\Rightarrow h_i=\left(-\frac{v}{u}\right)(h_o)=\frac{-(-20)}{(10)}(4)=8$$

$h_i = 8\text{cm, erect}$

14. A concave lense has a focal length of 10cm. if an object is held 10 cm from the lens.
    a) Where is the image located?
    b) What is the magnification of the lens?

Solution:a) $\frac{1}{f}=\frac{1}{u}+\frac{1}{v}\Rightarrow\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=-\frac{1}{10}-\frac{1}{10}=\frac{-2}{10}=\frac{-5}{5}$

$$\frac{1}{v}=-\frac{1}{5}\Rightarrow v=-5\text{cm, vertual.}$$

b) $m=\frac{-v}{u}=\frac{-(-5)}{10}=\frac{1}{2}=0.5$, erect

15. An object is placed 32cm from a convex lens that has a focal length of 8cm?
    a) Where is the image formed?
    b) If the object is 3cm high, how high is the image?
    c) Is the image inverted or upright?

Solution: a) $\frac{1}{f}=\frac{1}{u}+\frac{1}{v}\Rightarrow\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=\frac{1}{8}-\frac{1}{32}=\frac{4-1}{32}=\frac{3}{32}$

$$\frac{1}{v}=\frac{3}{32}\Rightarrow v=10.7=11\text{cm, real}$$

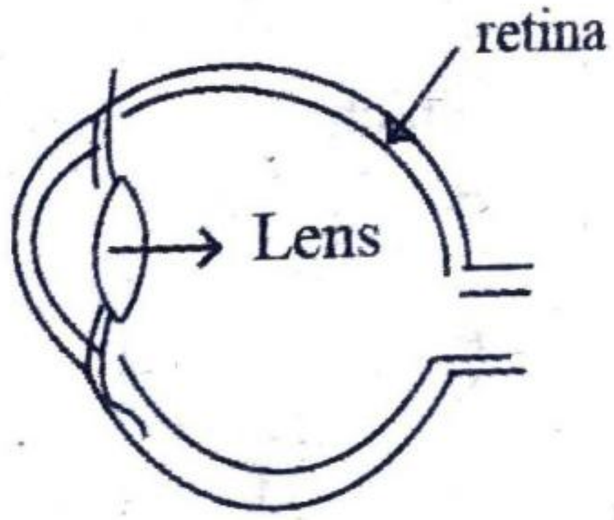
b) $\frac{-v}{u}=\frac{h_i}{h_o}\Rightarrow h_i=h_o\left(-\frac{v}{u}\right)=(3)\left(\frac{-11}{32}\right)=-1\text{cm}$

c) $m=\frac{h_i}{h_o}=-\frac{1}{3}=-\left(\frac{1}{3}\right)$, since it is negative, the image is inverted.

The Human eye

The eye is basically a *converging lens* made of living tissue, and forms an inverted image on the retina.

The normal human eye can focus objects comfortably form about 25cm to infinity and it may be defected for many reasons.



The eye

☞ **Defects of vision and their corrections**

Common defects of the eye are the following

A) **Short sight (myopia)**

A short-sighted person can only see nearby objects clearly

→ This defect is corrected by placing a diverging lens inform of the eye.

B) **Long sight (Hypermetropia)**

A long-sighted person can only see distant objects clearly.

→ This defect is corrected by spectacles of covering lens.

C) **Old sight (Presbyopia)**

Is the inability of the eye to focus as people get aged.

→ It can be corrected by two pair spectacles, diverging and covering.

D) **Astigmatism**

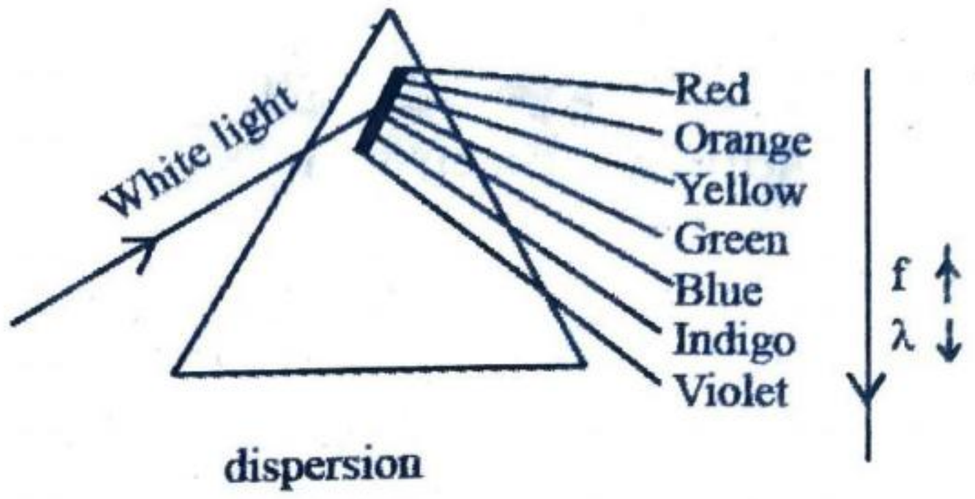
It is caused when the eye has two different focal lengths

→ The remedy is a pair of spectacles fitted with cylindrical lenses.

Dispersion of light.

Dispersion: Is the the process of splitting white light into components by refracting it through prism.

As white light pass through a prism, seven colours are seen on a screen. The frequency *increases* from red to violet, and at the same time the wavelength *decreases*. The refractive index of the prism increases as the frequency of light increases.



dispersion

General Secondary Education Certificate Exam

16. Which one of the following statements is correct?
    A. Visible light consists of an electromagnetic wave of a single frequency
    B. For all electromagnetic waves, the product of the wavelength and frequency in vacuum is constant
    C. Short radio wave have a higher frequency than visible light
    D. X-rays have a larger wave length than visible light.

    Explanation: In vacuum, $\lambda f = c$ (constant $= 3\times10^8$m/s)

    Answer: B

17. Which of the following statements will be true for Green, Blue, and Red light rays in Vacuum?
    A. They will have the same wavelength
    B. They will the same frequency
    C. They will have the same speed
    D. They cannot propagate

    Explanation: They are electromagnetic wave, and can travel in vacuum with the same speed $c = 3\times10^8$m/s.

    Answer: C

18. A certain body is placed 30cm infront of a convex mirror of focal length 15cm. what is the ratio of the image size to the object size
    A. 1:6 B. 1:15 C. 1:12 D. 1:3

    Solution:

$$\frac{1}{f}=\frac{1}{u}+\frac{1}{v}\Rightarrow\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=\frac{1}{30}-\left(-\frac{1}{15}\right)=\frac{3}{30}\Rightarrow v=10\text{cm}$$

$$m=\frac{v}{u}=\frac{h_i}{h_o}\Rightarrow\frac{h_i}{h_o}=\frac{10}{30}=\frac{1}{3}=1:3$$

Answer: D

19. A candle is placed 10cm away from a concave mirror. A virtual image is formed $\frac{50}{3}$ cm cm away from the mirror. What is the focal length of the mirror?

A. $-\frac{50}{3}$ cm

B. $\frac{50}{7}$ cm

C. $\frac{50}{2}$ cm

D. $-\frac{50}{2}$ cm

Solution: $\frac{1}{f}=\frac{1}{v}+\frac{1}{u}\Rightarrow\frac{1}{\frac{50}{3}}+\frac{1}{10}=-\frac{3}{50}+\frac{1}{10}=\frac{-3+5}{50}=\frac{2}{50}$

$$\frac{1}{f}=\frac{2}{50}\Rightarrow f=\frac{50}{2}\text{cm.}$$

Answer: C

20. A 10cm nail is placed 20cm away from a concave mirror of 25cm focal length. Where is the image formed and how big is it?

A. 33.4cm away and 16.7cm tall
B. 16.7 cm away and 33.4cm tall
C. 50cm away and 100cm tall
D. 100cm away and 50cm tall

Solution: $\frac{1}{f}=\frac{1}{u}+\frac{1}{v}\Rightarrow\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=\frac{1}{25}-\frac{1}{20}=\frac{4-5}{15}=\frac{-1}{100}$

$\frac{1}{v}=\frac{1}{100}\Rightarrow v=100$ and

$$m=\frac{v}{u}=\frac{h_i}{h_o}\Rightarrow h_i\left(\frac{v}{u}\right)(h_o)=\left(\frac{100}{20}\right)(10)=50\text{cm}$$

Answer: D

21. A spherical mirror of focal length 20cm forms an erect image where the image size is five times the object size. Where is the object placed?

A. 16cm
B. 120cm
C. 24cm
D. -80cm

Solution: $\frac{1}{f}=\frac{1}{u}+\frac{1}{v}\Rightarrow\frac{1}{u}=\frac{1}{f}-\frac{1}{v}$.

but, $h_i=5h_o,\frac{v}{u}=\frac{h_i}{h_o}\Rightarrow\frac{v}{u}=\frac{5h_o}{h_o}\Rightarrow v=5u$, so we have,

$$\frac{1}{u}=\frac{1}{f}-\frac{1}{5u}\Rightarrow\frac{1}{u}+\frac{1}{5u}=\frac{1}{f}\Rightarrow\frac{6}{5u}=\frac{1}{f}=\frac{1}{20}$$

Criss-crossing, $5u=120\Rightarrow u=24cm$

Answer: C

22. An object is placed at a distance of 24cm from a diverging lens of focal length 12cm. where is the image formed?

A. 8cm
B. 24cm
C. -8cm
D. -24cm

Solution: $\frac{1}{f}=\frac{1}{u}+\frac{1}{v}\Rightarrow\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=-\frac{1}{12}+\frac{1}{24}=\frac{-2+1}{24}=\frac{-1}{24}$

$$\frac{1}{v}=-\frac{1}{24}\Rightarrow v=-24cm$$

Answer: D

Note: f is negative for diverging lens

23. An object is 30cm in front of a converging lens of focal length 10cm. the image is

A. Virtual and smaller than the object
B. Real and smaller than the object
C. Real and larger than the object
D. Real and same size as the object

Solution: $\frac{1}{f}=\frac{1}{v}+\frac{1}{u}\Rightarrow\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=\frac{1}{10}-\frac{1}{30}=\frac{3-1}{30}=\frac{2}{30}=\frac{1}{15}$

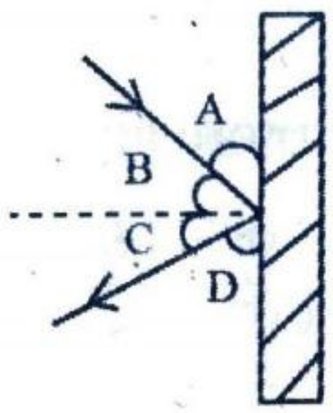
$$\frac{1}{v}=\frac{1}{15}\Rightarrow v=15\text{cm (real)}$$

$$m=\frac{v}{u}=\frac{15}{30}=\frac{1}{2}<1\text{ (small image)}$$

Answer: B

24. Which of the angles shown are compared in the law of reflection

A. B and C
B. A and B
C. B and D
D. A and C



Explanation: the angle of incidence and the angle of reflection are measured from the normal.

Answer: A

25. Light is incident from air into glass of index of refraction $8/5$. If the speed of light in air is 3×10^8m/s, what will be its speed in the glass medium?

A. 1.9×10^8m/s
B. 2.1×10^7m/s
C. 4.8×10^8m/s
D. 3.0×10^7m/s

Solution: $\frac{v_1}{v_2}=\frac{n_2}{n_1}\Rightarrow v_2=\left(\frac{n_1}{n_2}\right)(v_1)=\frac{1}{\left(\frac{8}{5}\right)}\left(3\times10^8\right)=\frac{5}{8}\times3\times10^8$

$$v_2=1.9\times10^8\text{m/s}$$

Answer: A

26. Which of the following will happen as a ray of light passes from water to air?

A. The angle of incidence is less than the angle of reflection
B. The angle of incidence is less than the angle of refraction
C. The angle of incidence is greater than the angle of reflection
D. The angle of incidence is greater than the angle of refraction

Answer: B

27. A monochromatic light source emits a wavelength of 480nm in vacuum. When passing through a liquid, the wavelength reduces to 420nm. What is the liquid's index of refraction?

A. 1.52 B. 1.26 C. 1.33 D. 1.14

Solution: $\frac{\lambda_1}{\lambda_2}=\frac{n_2}{n_1}\Rightarrow n_2=\left(\frac{\lambda_1}{\lambda_2}\right)(n_1)=\frac{480}{420}\times 1=1.14$

Answer: D

28. Which of the following is more appropriate for explaining mirages

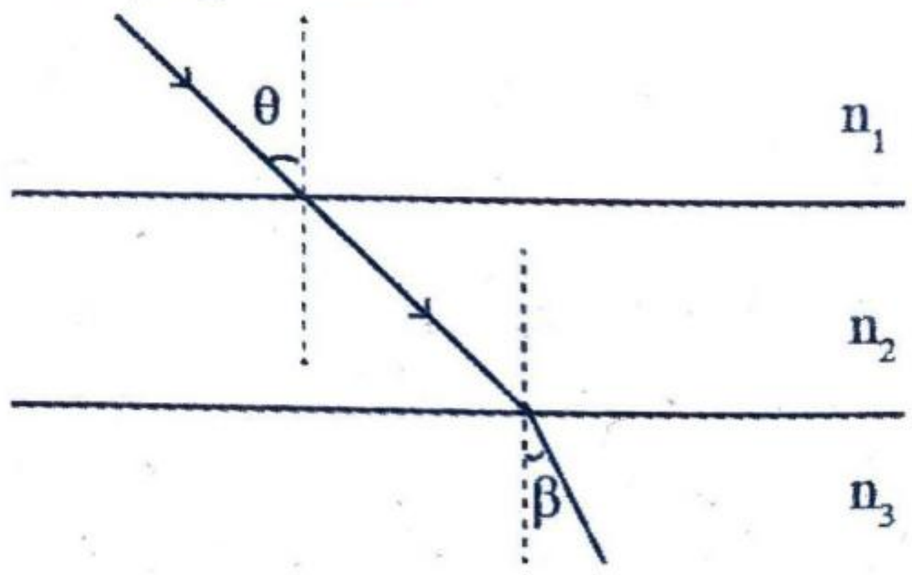
A. Total internal refraction C. diffraction
B. Total internal reflection D. interference

Explanation: A mirage is a phenomena which is caused by variation in the density of air. The air near the earth is warm and less dense. The air above that is cold and denser when incidence ray approaches the warm air, total internal reflection occurs. For an observer, it seems that there is a pool of water on the road in front

Answer: B

29. A ray of light passes through three indices of refractions n_1, n_2 and n_3, if $\theta>\beta$, which one of the following is true

A. $n_1> n_2> n_3$
B. $n_2> n_3>n_1$
C. $n_3>n_2>n_1$
D. $n_3>n_1>n_2$



Explanation: The ray bends away in the second medium hence $n_1> n_2$ and it bends into the normal in the third medium $n_3 > n_2$, hence $n_3 > n_1> n_2$.

Answer: D

30. What is the purpose of the receiver antenna in radio communication?

A. Increasing the amplitude of the radio signal at reception
B. Converting sound wave to electromagnetic wave

C. Converting the electromagnetic wave to electrical oscillation.

D. Separating the carrier wave from the modulation wave

Answer: C

31. What is the unit of a lens?

A. Weber
B. Diopter
C. Rad
D. Rem

Answer: B

32. An electromagnetic wave propagates in vacuum at speed C and wavelength λ. As the wave enters some unknown medium, its speed decrease to 0.6c. What is the wavelength of the wave in the unknown medium?

A. 1.25λ B. λ C. 0.8λ D. 0.6λ

Solution: $\frac{v_1}{v_2}=\frac{\lambda_1}{\lambda_2}\Rightarrow\lambda_2=\left(\frac{v_1}{v_2}\right)\lambda_1=\frac{0.6c}{c}(\lambda)=0.6\lambda$

Answer: D

33. A sinusoidal water waves are generated in a ripple tank. The waves travel at 20cm/s and their adjacent crests are 5cm apart. The period for each cycle is

A. 4sec
B. 0.5sec
C. 0.25sec
D. 100sec

Solution: $v=\frac{\lambda}{T}\Rightarrow T=\frac{\lambda}{v}=\frac{5}{20}=\frac{1}{4}=0.25\text{sec}$

Answer: C

34. A certain wave is characterized by the wavelength λ, frequency f and have speed v. As this wave travels from one medium to another, it gets refracted. Which of these characteristics of the wave are changed due to refraction?

A. λ, f and v
B. λ and f
C. f and v
D. λ and v

Explanation: frequency is not changed during refraction

Answer: D

35. Which one of the following statements describes a convex mirror?

A. The radius of curvature of the mirror is on the reflecting side
B. The principal focus is on the center of curvature of the mirror
C. Rays along the radius of the mirror converge to its principal focus
D. Parallel rays of light diverge when reflected from the mirror

Answer: D

36. Light ravels from medium X to medium Y as shown. Which one of the following is correct

A. Both wavelength and speed are unchanged
B. Speed decreases, wavelength increases
C. Both speed and wavelength decreases
D. Both speed and wavelength increases.

Explanation: the refracted ray is bent away from the normal, thus medium Y is less dense than medium X, which leads the speed and wavelength to increase. **Answer: D**

37. All electromagnetic waves travel with the same speed in

A. Water
B. Air
C. Vacuum
D. Mirror

Explanation: all electromagnetic waves travel at the same speed ($c = 3\times 10^8 m/s$) in vacuum. **Answer: C**

38. If a person is diagnosed to have short sightedness, which of the solutions are appropriate to correct the defect?

A. A diverging lens to focus the distant object at a closer distance
B. A diverging lens to focus the nearly object at a large distance.
C. A diverging lens to focus the distant object at a closer distance
D. A diverging lens to focus the nearly object at a larger distance

Answer: B

39. A pin stands erect 10cm from a converging mirror of focal length 20cm. How far apart are the pin and its image?

A. 10cm
B. 20cm
C. 30cm
D. 7cm

Solution: $\frac{1}{f}=\frac{1}{v}+\frac{1}{u}\Rightarrow\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=\frac{1}{20}-\frac{1}{10}=\frac{1-2}{20}=-\frac{1}{20}$

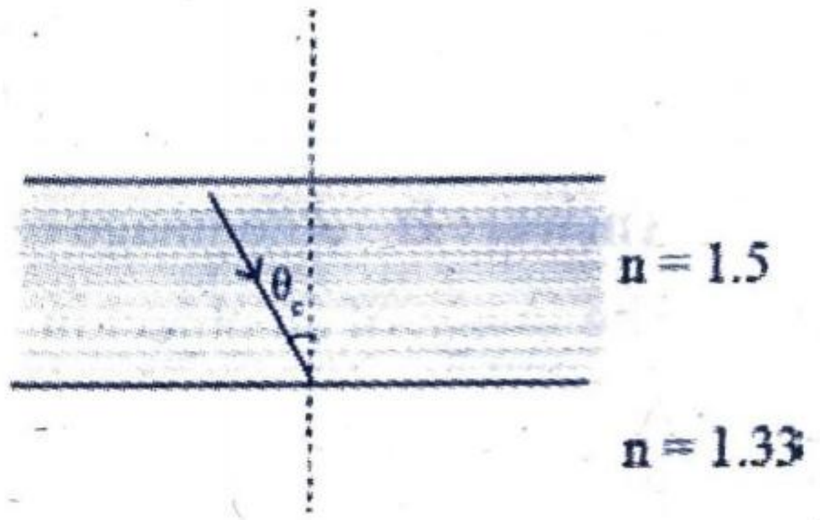
$$\frac{1}{v}=-\frac{1}{20}\Rightarrow v=-20\text{cm (at the back of the mirror)}$$

$$5=u+v=10+20=30\text{cm}$$

Answer: C

40. A beam of light travelling through a medium of refractive. Index 1.5 encounters a boundary leading to another medium of refractive index 1.33. The critical angle is 62.7°. What is the value of θin the second medium.

A. 80°
B. 90°
C. 62.7°
D. 40°



Explanation: the angle of refraction at the critical angel is 90°.

Answer: B

41. If a lens produces a real image that is four times as large as the object and is located 24cm from the lens, What is the focal length of the lens?

A. 4.8cm
B. 8cm
C. 4cm
D. 6cm

Solution: $\frac{1}{f}=\frac{1}{v}+\frac{1}{u}\Rightarrow\frac{v}{u}=\frac{hi}{ho}=\frac{4ho}{ho}=v=4u\Rightarrow u=\frac{v}{4}=6$

$$\frac{1}{f}=\frac{1}{4u}+\frac{1}{u}=\frac{5}{4u}\Rightarrow\frac{1}{f}=\frac{5}{4u}=\frac{5}{4(6)}=\frac{5}{24}$$

$$\frac{1}{f}=\frac{5}{24}\Rightarrow f=4.8\text{cm}$$

Answer: A

Additional Examples

42. Which one of the following statements is correct?

A. The dispersion of light is due to refection

B. The dispersion of light that is produced by a prism is due to diffraction
C. The decomposition of white light in the atmosphere such as rainbow is due to refraction
D. During decomposition of white light by diffraction, violet is diffracted most and red light is diffracted least.

Explanation: dispersion of light is due to refraction

Answer: C

43. The image formed by plane mirror is

A. Larger than the object
B. Smaller than the object
C. Inverted
D. virtual

Answer: D , the image cannot be displayed on a screen

44. The image of an object in a convex mirror is always

A. real
B. virtual
C. Larger
D. Inverted

Answer: B

45. A kind of image that can be captured on a screen is

A. real image
B. virtual image
C. both
D. mirror image

Answer: A

46. "Whenever the light crosses a boundary between two media the sines of the angles of each side of the boundary bear a constant ratio to each other" This law is

A. Newton's
B. Snell's
C. Len's
D. Faraday's

Answer: B

47. A measure of the extent to which a medium refracts light is

A. Reflection index
B. Refraction index
C. transpiration index
D. diffraction index

Answer: B

The angle of incidence above which total internal reflection occurs is

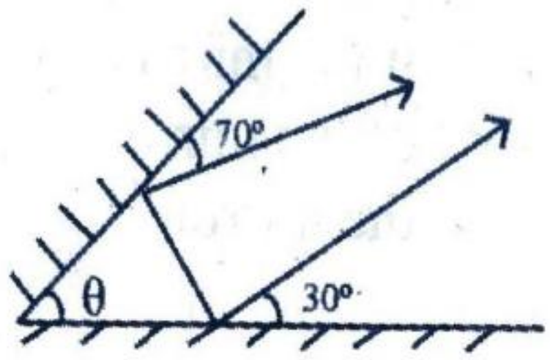
A. actual angle
C. critical angle

B. obtuse angle D. right angle

Answer: C

48. A ray of light is reflected by two mirrors. What is the value of θ

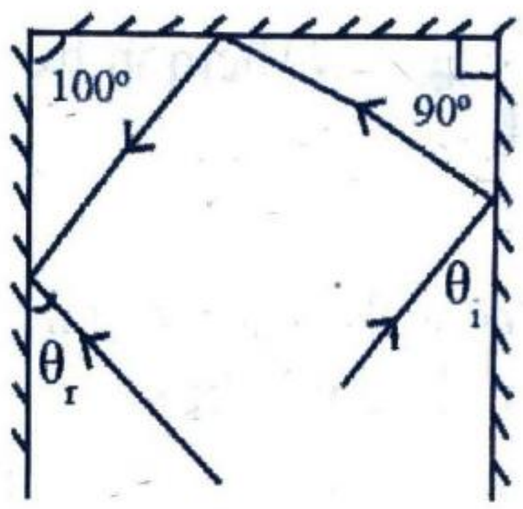
A. 80°
B. 90°
C. 70°
D. 60°



Answer: A

49. A ray of light is reflected by three mirrors. What should be θi if $= \theta r = 35°$?

A. 65°
B. 45°
C. 35°
D. 55°



Answer: B

50. A body is placed 30cm from a concave mirror which has a focal length of 5cm. At what distance from the mirror is the image formed.

A. 5cm C. 6cm
B. 25cm D. 35cm

Solution: $\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \Rightarrow \frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{5} = \frac{1}{30} = \frac{6-1}{30} = \frac{5}{30} = \frac{1}{6}$

$\frac{1}{f} = \frac{1}{6} \Rightarrow f = 6cm$

Answer: C

51. Which of the following statement is false?

A. The wavelength of infrared is smaller than that of radio wave
B. X-rays have a higher frequency than ultraviolet rays
C. Visible light waves have a higher frequency than infrared and a lower frequency than ultraviolet
D. Visible light waves have a smaller wavelength than ultraviolet and larger frequency than infrared waves.

Answer: D

52. An object is placed 10cm in front of a concave mirror which has a focal length of 5cm. where is the image formed?

A. 4cm in front of the mirror
B. 5cm in front of the mirror
C. 10cmin front of the mirror
D. 10cm behind the mirror

Solution: $\frac{1}{f}=\frac{1}{v}+\frac{1}{u}\Rightarrow\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=\frac{1}{5}-\frac{1}{10}=\frac{2-1}{10}=\frac{1}{10}$

$\frac{1}{v}=\frac{1}{10}\Rightarrow v=10\text{cm}$ in front of the mirror

Answer: C

53. A man run towards a plane mirror at 2m/s. The relative speed of his image w.r.t him will be

A. 2m/s
B. 4m/s
C. 8m/s
D. 10m/s

Solution: $v=v_o+v_i=2+2=4\text{m/s}$

Answer: B

54. In a plane mirror, an object is 0.5m in front of the mirror. The distance between the object and its image is

A. 0.5m B. 1m C. 0.25m D. 0.75m

Solution: $s=v+u=0.5+0.5=1\text{m}$

Answer: B

55. An object 0.5m tall is in front of a plane mirror at a distance of 0.2m. The size of the image formed is

A. 0.2m B. 0.5m C. 0.1m D. 1m

Solution: $h_o=h_i=0.5\text{m}$

Answer: C

56. A plane mirror is approaching you at 10cm/s. Your image shall approach you with a speed of

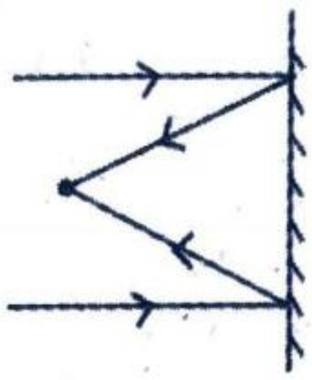
A. 10cm/s
B. -10cm/s
C. 20cm/s
D. -20cm/s

Answer: C

57. When a convergent beam of light is incident on a plane mirror, the image formed is

A. upright and real
B. upright and virtual
C. inverted and virtual
D. inverted and real

Answer: A



58. A convex mirror is used to form an image of a real object which of the following is wrong?

A. The image lies between the vertex and the focus
B. The image is diminished in size
C. The image is erect
D. The image is real

Answer: D

59. A person standing in front of a mirror finds his image larger than himself. This implies that the mirror is

A. Plane
B. Concave
C. Convex
D. Cylindrical

Answer: B

60. The largest distance of the image from a convex mirror of focal length 10cm can be

A. 20cm
B. infinite
C. 10cm
D. depends on the object distance

Answer: C

61. For a real object, a convex mirror always forms an image which is

A. real and inverted
B. virtual and inverted
C. virtual and erect
D. real and magnified

Answer: C

62. A convex mirror forms an image one-fourth the size of the object. If the object is 0.5m from the mirror, the focal length of the mirror is

A. 0.17cm
B. 0.4m
C. -1.5m
D. -0.4m

Solution: $\frac{v}{u}=\frac{h_i}{h_o}=\frac{1}{4}\Rightarrow u=4v\Rightarrow v=\frac{u}{4}$

$$\frac{1}{f}=\frac{1}{v}+\frac{1}{u}=-\frac{1}{v}+\frac{1}{4v}=\frac{-4+1}{4v}=\frac{-3}{4\left(\frac{u}{4}\right)}=\frac{-3}{u}$$

$$\frac{1}{f}=\frac{-3}{u}=\frac{-3}{0.5}\Rightarrow f=\frac{0.5}{3}=0.17m$$

Answer: A

63. A dentist has a small mirror of focal length 16mm. He views the tooth of a patient by holding the mirror at a distance of 8mm form the tooth. The magnification of the mirror is.

A. 1 B. 1.5 C. 2 D. 3

Solution: $\frac{1}{f}=\frac{1}{u}+\frac{1}{v}\Rightarrow\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=\frac{1}{16}-\frac{1}{8}=\frac{1-2}{16}=-\frac{1}{16}$

$\Rightarrow v=16cm$

$$m=\frac{v}{u}=\frac{16}{8}=2$$

Answer: C

64. A concave mirror has radius of curvature of 1m. Light from a distant star is incident on the mirror. The distance of the image of the star from the mirror is

A. 0.5m B. 1m C. 2m D. 0.25m

Solution: $f=\frac{c}{2}=\frac{1}{2}=0.5m$

The image of an object at infinity is formed at the focal length.

Answer: A

65. A concave mirror of focal length f forms an image of the same size as the object. The distance of the object from the mirror is

A. f B. $\frac{f}{2}$ C. 2f D. 4f

Solution: $\frac{1}{f}=\frac{1}{u}+\frac{1}{v}$ but $\frac{v}{u}=1\Rightarrow v=u$

$$\frac{1}{f}=\frac{1}{v}+\frac{2}{v}\Rightarrow v=\frac{1}{f}=\frac{2}{v}\Rightarrow v=2f$$

Answer: C

66. A converging lens has a focal length of 1m. The minimum distance between a real object and its real image formed by this lens is

A. 0.5m B. 1m C. 2m D. 4m

Solution: Minimum distance between an object and its real image is 4f i.e $4\times 1=4$m.

Answer: D

67. An object is placed at a distance of 20cm from a convex lens of focal length 10cm. The image is formed on the other side of the lens at a distance

A. 20cm C. 40cm
B. 10cm D. 30cm

Solution: $\frac{1}{f}=\frac{1}{v}+\frac{1}{u}\Rightarrow\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=\frac{1}{10}-\frac{1}{20}=\frac{2-1}{20}=\frac{1}{20}$

$$\frac{1}{v}=\frac{1}{20}\Rightarrow v=20\text{cm}$$

Answer: A

68. The minimum distance between an object and its real image using a convex lens of focal length f is

A. zero B. 2f C. 4f D. 6f

Answer: C

69. A concave lens has a focal length of 20cm. A beam of light converges to a point 20cm behind the concave lens on the principal axis. the image is formed at

A. infinity C. 10cm
B. 20cm D. 40cm

Solution: $\frac{1}{f}=\frac{-1}{v}+\frac{1}{u}\Rightarrow\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=\frac{1}{20}-\frac{1}{20}=0$

$\frac{1}{v}=0\Rightarrow v=\infty$ **Answer: A**

70. A convex lens of focal length $\frac{1}{3}$m forms a real, inverted image twice in size of the object. The distance of the object from the lens is

A. 0.5m C. 0.33m
B. 0.166m D. 1m

Solution: $\frac{1}{f}=\frac{1}{v}+\frac{1}{u},\frac{v}{u}=2\Rightarrow v=2u$

$\frac{1}{f}=\frac{1}{2u}+\frac{1}{u}=\frac{1+2}{2u}=\frac{3}{2u}\Rightarrow 2u=3f\Rightarrow u=\frac{3}{2}f$

$u=\frac{3}{2}=\left(\frac{1}{3}\right)=\frac{1}{2}m=0.5m$ **Answer: A**

71. The image of an object formed by a device is always virtual and small. The device may be

A. Convex lens C. A glass plate
B. Concave lens D. concave mirror

Answer: B

72. When white light passes through a glass prism, one gets spectrum on the other side of the prism. In the emergent beam, the ray which is deviating least is

A. Violet ray C. Red ray
B. Green ray D. Yellow ray

Answer: C

73. The refractive index of a given piece of transparent quartz is greatest for

A. Red light C. Green light
B. Violet light D. Yellow light

Answer: B

74. When white light passes through a hollow prism, then
A. There is no dispersion and no deviation
B. Dispersion but no deviation
C. Deviation but no dispersion
D. There is dispersion sand deviation both

Answer: A

75. The number of different colours, according to frequency in a spectrum is
A. 5 B. 6 C. 7 D. infinite

Answer: D

76. Colour of light having maximum speed in air is
A. red C. violet
B. yellow D. blue

Answer: A

77. What will be the colour of sky as seen from the earth if there were no atmosphere?
A. Black C. Orange
B. Blue D. Red

Answer: A

78. One important similarity between sound and light waves is that both
A. can pass through even in the absence of any medium
B. are transverse waves
C. travel at the same speed in air
D. can show interference effects.

Answer: D

79. Which one of the following waves is longitudinal wave?
A. Radio wave C. X-ray
B. Sound wave D. light wave

Answer: B

80. Which of the following is an ascending order of frequency
A. Red, blue, yellow, green
B. Blue, green, yellow, red
C. Red, yellow, green, blue
D. Red, green, yellow, blue

Answer: C

81. Which of the following is related to telecommunication

A. Microwave
B. X-ray
C. r-ray
D. sound wave

Answer: A

82. Which of the following electromagnetic radiation has the smallest wavelength?

A. UV ray
B. X ray
C. Microwave
D. γ-ray

Answer: D

83. The frequency is minimum for?

A. Gamma ray
B. Ultraviolet
C. Infrared rays
D. Visible light

Answer: D

84. Which of the following has the maximum frequency?

A. Gamma ray
B. Blue light
C. Infrared light
D. Ultraviolet ray

Answer: A

85. Electromagnetic waves do not transport

A. Energy
B. Charge
C. Momentum
D. Information

Answer: B

86. Which of the following gives electromagnetic spectrum in order of increasing wavelength?

A. Gamma rays, microwaves, visible radiation
B. Radio waves, ultraviolet, x-rays
C. Ultraviolet, infra-red, microwaves
D. Visible radiation, gamma rays, radio waves

Answer: C

87. Pick out the shortest wavelength from the following type of radiations

A. blue light
B. gamma rays
C. x-rays
D. red light

Answer: B

88. The nature of electromagnetic waves is

A. longitude
B. transverse
C. longitudinal stationary
D. transverse stationary

Answer: B

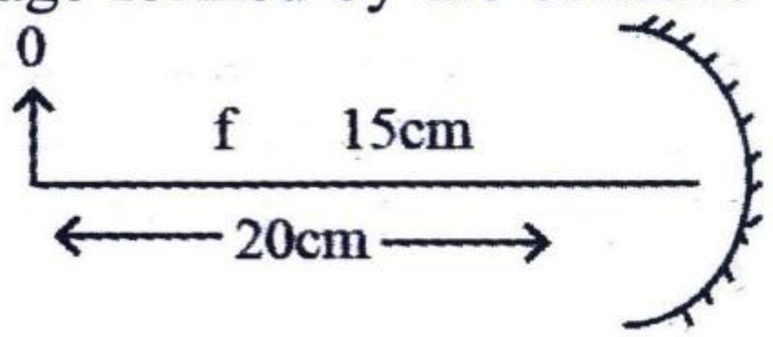
Unit Fourteen: Conceptual Questions

1. which one do have the longest wavelength and the largest frequency in the electromagnetic spectrum
2. what is the advantage of a convex mirror over a plane mirror.
3. What is the physical interpretation of image type when the magnification is, $m > 1$, $m < 1$ and $m = 1$.
4. Explain the difference between the working principle of mirrors and lens.

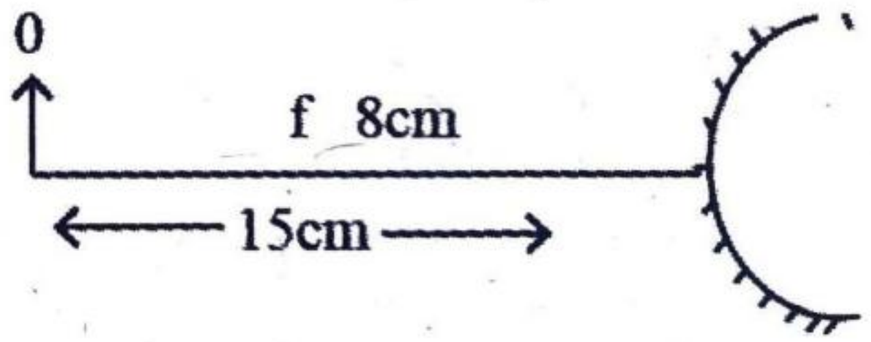
Unit Fourteen: Problems

Reflection of light and images formed by mirrors.

5. What is the frequency of x-ray with a wavelength of 10^{-10}m?
6. Find the position and nature of the image formed by the concave mirror



7. Find the position and nature of the image formed by the convex mirror



Refraction of light and Images formed by Lenses.

8. Find the refractive index of a liquid in which the real depth was 5cm and the apparent depth was 3cm
9. What is the power of a lens of focal length 2m
10. Calculate the critical angel for diamond of refractive index 2.42.
11. What is the critical angle for total internal reflection to occur when the refractive index is 1.52 and incident angle is 32°
12. Find the focal length of convex lens where the object is 4cm from the lens and the image is 5cm from the lens

13. A transparent material A has a higher refractive index than another material B. Through which one does light travel more slowly?

Unit Fourteen: Explanation to Questions

1. γ-ray has the highest frequency because, from the wave equation,

$$v = f\lambda = f = \frac{v}{\lambda} = \frac{c}{\lambda} = \frac{3\times10^{8}}{10^{-12}} = 3\times10^{20}\text{Hz}$$

2. convex mirror helps us to form a smaller image, and to view a wide field, such as supermarket and stores.
3. When $m > 1$, the image is larger than the object, and when $m < 1$, the image is smaller than the object and when $m = 1$ the image is the same size as the object.
4. Mirrors produce image due to reflection of light and lenses produce image due to refraction of light

Unit Fourteen: Solution to Problems

5. $$v = f\lambda = f = \frac{v}{\lambda} = \frac{c}{\lambda} = \frac{3\times10^{8}}{10^{-10}} = 3\times10^{18}\text{Hz}$$

6. $$\frac{1}{f} = \frac{1}{u} + \frac{1}{v} \Rightarrow \frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{15} - \frac{1}{20} = \frac{4-3}{60} = \frac{1}{60}$$

$$\frac{1}{v} = \frac{1}{60} \Rightarrow v = 60\text{cm, real image}$$

7. $$\frac{1}{f} = \frac{1}{u} + \frac{1}{v} \Rightarrow \frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{-1}{8} - \frac{1}{15} = \frac{-15-8}{120} = \frac{-23}{120}$$

$$\frac{1}{v} = \frac{-23}{120} \Rightarrow v = \frac{-120}{23} = -5.2\text{cm, virtual image}$$

8. $$\frac{n_1}{n_2} = \frac{d}{D} \Rightarrow n_2 = \left(\frac{D}{d}\right)(n_1) = \left(\frac{5}{3}\right)(1) = \frac{5}{3}$$

9. $$\text{Power of lens} = \frac{1}{f} = \frac{1}{2} = 0.5 \text{ Dioptres}$$

10. $$\sin\theta_c = \frac{n_2}{n_1} = \frac{1}{2.42} \Rightarrow \theta_c = \sin^{-1}\left(\frac{1}{2.42}\right) = 24°$$

11. $\sin\theta_c = \frac{n_2}{n_1} = \frac{1}{1.52} \Rightarrow \theta_c = \sin^{-1}\left(\frac{1}{1.52}\right) = 41°$

12. $\frac{1}{f} = \frac{1}{u} + \frac{1}{v} = \frac{1}{4} + \frac{1}{6} = \frac{3-2}{12} = \frac{1}{12} \Rightarrow \frac{1}{f} = \frac{1}{12}$

$\Rightarrow f = 12cm$

13. From Snell's law, $\frac{n_2}{n_1} = \frac{v_1}{v_2}$, speed and refractive index are inversely related. Thus the speed is less in material A.

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